

Supplementary Materials for

Ancient genomic changes associated with domestication of the horse

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Materials and Methods

DNA extraction and genome sequencing

DNA was extracted from 120-700 mg of bone powder, starting from the supernatant either of the first or the second powder lysis (Data File S1 and (25)). We followed three different protocols, referred to as S, D and Y (25). Method S performs extraction using silica pellets in solution (26), as modified by (27). Method D is tailored to recover ultra-short DNA fragments and performs DNA purification on silica columns (28). Method Y (29), as modified by (25, 30), utilizes similar purification, although after DNA has been concentrated on nitrocellulose membranes. In order to reduce the impact of post-mortem cytosine deamination on downstream analyses, half of the volume of the DNA extracts was treated with the USER enzyme mix (New England BioLabs Inc.), as described in (25).

We followed the procedure from (31), as modified in (32) and described in (25), to build blunt-end Illumina libraries from both USER and non-USER treated DNA extracts for the majority of the samples (Data File S1). Libraries were indexed by amplification with AmpliTaq Gold(R) (Thermo Fischer Scientific), or Accuprime Taq DNA polymerase (Thermo Fischer Scientific), as described in (25). Indexed DNA libraries were pooled and sequenced on an Illumina HiSeq 2500 instrument, at the Danish National High-Throughput DNA Sequencing Center. Details on sequencing runs and output are provided in Data File S2. Shotgun sequence data are available for download at the European Nucleotide Archive (Project PRJEB19970).

Sequencing reads were processed with the PALEOMIX pipeline (33). Briefly, adapters were trimmed with AdapterRemoval v2.1.1 (34), following (27). Alignments were performed with BWA v0.5.9 (35), separately against the horse nuclear (EquCab2.0) (36) and mitochondrial (NC_001640.1) (37) genomes, disabling seeding, filtering out duplicates and retaining reads with a minimum length of 25 bp and a mapping quality above 25 (see Data File S1 for a breakdown per sample and library of the number of high-quality read alignments).

Error rates were quantified for each genome, before and after filtering. We followed the procedure detailed in (27), using *Equus africanus somaliensis* (38) and the reference EquCab2.0 (36) as the outgroup and the “perfect” genomes, respectively. Calculations prior to filtering revealed BER11_L as the sample with the highest raw error rates (0.5096% per base vs. 0.1678-0.3667% per base for the remaining ancient genomes; Table S1).

To increase the quality of the alignments to the nuclear genome, the base qualities of non-USER treated libraries were rescaled using mapDamage2.0 (39). We used default parameters of post-mortem DNA damage, except for BER11_L. In the latter case, the sequence length to be rescaled at each read end was increased to 15 bp (option --seq-length 15).

For each USER-treated library, reads were trimmed to remove residual post-mortem DNA damage. Using as baselines the mis-incorporation rates found at the center of the reads, the 5' and 3' ends were trimmed when exceeding 0.75% of C→T and G→A rates, respectively (Data File S1). The error rates estimated following mapDamage2.0 analysis and quality filtering are shown in Fig. S1, and summarized in Table S1. These are, as expected, minimal for transversion substitutions across all samples (0.03% errors per base). Transition substitutions generally show higher error rates, but are limited across all

samples, except for BER06_F, BER09_I, BER11_L and BER12_M (0.05-0.36% errors per base).

Microbial profiling

We profiled the shotgun datasets obtained from the 14 ancient horses of the Berel', Arzhan and Sintashta sites for their microbial content, applying the metagenomic approach described in (40), and implemented in the metaBIT pipeline (41). For comparison, we also reanalyzed shotgun datasets previously published and generated from three pre-domestic horse remains, encompassing from the Late Pleistocene to the Early Holocene (14, 15), as well as the CGG101397 Yakutian horse sample from the 19th century A.D. (15) (Table S2).

We aligned shotgun DNA reads to the MetaPhlAn2 database, including eukaryotes (42) and using the Bowtie2 v2.1.0 aligner (43), with the --end-to-end mode and a sensitive global alignment strategy. For accurate relative abundance estimation, we removed PCR duplicates using, for single-end reads, the MarkDuplicates function of Picard Tools version 1.119 (<https://broadinstitute.github.io/picard/>), and for paired-end collapsed reads, a modified version of the FilterUniqueBAM Python script from PALEOMIX (33). Microbial taxon relative abundances were calculated at all taxonomic levels from kingdom to strain with the program MetaPhlAn2. In order to reduce the impact of false-positives, we excluded from further statistical analyses taxa showing abundances inferior to 1% across all samples.

Microbial profiles were generated first for each DNA library individually to rule out possible library cross-contamination. Microbial profiles were then generated on each individual sample by merging in metaBIT all the libraries obtained for a given sample. MetaPhlAn2 microbial profiles were compared to those obtained for 689 human (44) and 15 soil samples (41, 45) by Principal Coordinate Analyses (PCoA) of Bray-Curtis distances with the R (46) function *pcoa* and the comparative database provided as part of metaBIT (41).

Pathogen screening

metaBIT analyses identified a significant fraction of the reads generated from samples BER01_A, BER04_D and BER02_B as potential members of the *Rickettsia* genus, which contains a series of important pathogens. We thus attempted to identify whether the reads identified could be assigned to any particular species of pathogenic *Rickettsia* species, by mapping reads against a comparative genome database including at least one complete *Rickettsia* genome sequence per species (47). Read mapping was carried out using PALEOMIX (33) and BWA v0.5.9 (35), disabling seeding and filtering reads with mapping qualities strictly inferior to 25.

Genotype probabilities vs. genotype calls

TreeMix population trees (48) and f_3 -outgroup statistics (49) were estimated using explicit genotype calls. Genotype calling, quality filtering and the subsequent generation of individual-based VCF files were performed following the procedures described in (14, 15, 27, 38, 50). Genotype calling of diploid genomes can, however, be challenging in the presence of low-to-moderate coverage (up to 7-fold), as is the case for 9 out of the 14 ancient horse genomes reported here (Table S3). We thus conducted most of our

remaining analyses (Fig. S2) in a genotype likelihood framework, integrating inferences over the three genotype probabilities (AA/Aa/aa), as implemented in the ANGSD package (51). More specifically, we supplied ANGSD with the following general flags:

- Minimum base quality of 20 (-minQ 20)
- Minimum mapping quality of 25 (-minMapQ 25)
- Remove no primary, duplicate or unmated reads (-remove_bads 1)
- Remove reads with multiple hits (-uniqueOnly 1)
- Downgrade quality scores around indels (-baq 1)
- Adjust mapping quality for regions with excessive mismatches (-C 50)
- Genotype likelihoods were calculated using a procedure equivalent to that implemented in GATK (52) (-GL 2)

Finally, ABBA-BABA admixture tests and analyses based on haploid markers (Y-linked contigs and mitochondrial DNA) were carried out using dedicated pipelines (see corresponding sections below).

Adjusting the comparative panel to achieve a minimum number of high-quality sites

We applied specific coverage thresholds through different analyses, either for consistency with previous studies, or for optimizing the trade-off between the number of samples included (n), and the number of covered positions without missing alleles (p). The expected number of sites without missing alleles, $E(p)$, was calculated *a priori* as:

$$E(p) \approx GenomeSize \times \prod_{i=1}^{i=n} (1 - cdfPoisson(MinCov - 1, AvgCov_i))$$

where *cdfPoisson* is the cumulative Poisson distribution, with the variable $X = MinCov$ (the minimum site depth required to confidently conduct the analysis), and parameter $\lambda = AvgCov_i$ (the mean sequencing coverage for genome i). As *GenomeSize*, we used the length of the autosomal horse genome, 2,147,483,647 bp (EquCab 2.0).

Suppose that we aim at conducting an analysis that requires confident genotype calls, from positions with a minimum depth-of-coverage $X = 8$ -fold. Let also assume that we sequenced three horse genomes at a mean coverage of $\lambda = 4$ -fold. With such genome panel, the expected proportion of positions with a sequencing depth equal or higher than 8-fold would be, approximately, $1 - Poisson(7,4) = 0.0511$ for each genome, and $0.0511^3 = 0.00013$ for the panel of three genomes. In total, we would retain $2,147,483,647 \times 0.00013 = 287,111$ analyzable positions. Using this expectation, we have been able to adjust the individual composition in the comparative panel, so that the total number of positions considered for each individual analysis matches the requirements of the specific method.

Mitochondrial DNA (mtDNA)

The final sequence of the mitochondrial genome was generated for each ancient specimen using the procedure described in (27), which applies a strict majority rule based on a minimum depth of coverage of 3-fold, and mapping and base qualities equal or above 25.

Ancient mtDNA variability was assessed with both maximum likelihood (ML) and Bayesian methods, comparing the 14 horses sequenced here to a panel of 106 modern and

34 ancient horses previously published (Table S4). The mtDNA sequences were partitioned into six categories, namely the control region, rRNAs, tRNAs, and the three codon positions. Substitution models were estimated with ModelGenerator 0.85 (53) for each of the six individual partitions, and for the global alignment, representing the merger of all six partitions (Table S5).

We calculated the ML phylogenetic tree on the global alignment with PhyML 3.0 (54), and drawn it as an unrooted dendrogram with FigTree 1.4.2 (55), reporting node robustness based on 100 bootstrap pseudo-replicates (Fig. S3). BEAST 1.8.2 (56, 57) was similarly used to perform the Bayesian phylogenetic inference on the partitioned dataset, with tip calibration based on the age of the ancient samples, as reported in Table S4. We inferred a Bayesian Skyline (58) demographic model with a log-uncorrelated molecular clock, previously identified as best fit for the ancient and modern data tested here (15, 27, 50). The Bayesian phylogenetic inference was run for 500 million generations, sampling the chain every 1,000. Tracer 1.6.0 (59) was used to check convergence after excluding 10% burn-in, as well as for inferring the Bayesian Skyline profile (following down-sampling of the 500 million generations chain every 25,000 and a 10% burn-in), which was then plotted using ggplot2 (69) (Fig. S4).

The maximum credibility tree topology was obtained with TreeAnnotator 1.8.2 (56) and drawn with FigTree 1.4.2 (55) (Fig. S5). Estimates of mutation rates for each partition based on posterior distributions obtained with BEAST are reported in Table S6.

Y-chromosome

In absence of reference sequence for the horse Y-chromosome, we mapped the sequencing data underlying each ancient individual against 19 Y-chromosome contigs (22, 60), which cover 193,857 bp of the Y-chromosome, excluding one redundant contig. To achieve this, we followed the two-step procedure described in (14). Specifically, sequencing reads were firstly mapped against the Y-chromosome contigs. The alignable reads were then re-mapped against the whole nuclear genome, including the Y-chromosome contigs. Only those reads exclusively mapping against the Y-chromosome were further processed. We followed the same procedure for mapping against two regions that are absent in the EquCab2.0 reference sequence, including one linked to the wild coat color (Dun gene (61)), and another associated with racing performance (*MSTN* gene (62)).

The consensus sequence for the Y-chromosome contigs of ancient and modern stallions (Table S3 and Data File S3) was estimated with ANGSD, using the above-described filters, and restricting the analysis to those positions covered at least 3-fold. The resulting multi-fasta alignment was transformed to the nexus format with PGDSpider (63) and used to build a Minimum Spanning Network (64) using PopArt (<http://popart.otago.ac.nz/>).

Genotyping loci underlying important equine phenotypes

We investigated all 14 ancient horses for 41 Mendelian traits previously associated with coat color, racing performance, body size and congenital diseases (65–94). The shotgun sequence data allowed us to estimate the genotypes at many of the 41 phenotypic traits. Due to low coverage, however, the genotypes for some ancient specimens were not

confidently determined by solely using shotgun data. In the following, we describe additional procedures used to further validate the inferred genotypes.

The first procedure is based on in-solution target-enrichment, and relies on the methodology presented by Cruz-Dávalos and colleagues (95), with slight modifications. Here, we used 100-384 ngrams of DNA for enrichment, and 2-times dilutions of MYcroarray RNA probes. Library-probes annealing was performed for 40 hours at 55°C. DNA libraries were constructed following the procedure described in section '*DNA extraction and genome sequencing*', except that (i) new ancient DNA extracts were prepared from 295-989 mgrams of fossil material, aiming at testing the reproducibility of the results obtained, and; (ii) the end-repair and ligation steps were performed in 50µL final volumes. DNA extraction was performed following the same methodology as described in '*DNA extraction and genome sequencing*', using for all the samples but one the bone pellets left undigested after a first pre-digestion for one hour at 42°C. For the remaining sample (BER09_I, for which the largest amount of bone powder was used for extraction), a large amount of bone pellets was left undigested after an overnight digestion at 42°C. This fraction was centrifuged for two minutes at 3,000 rpm in order to replace the digestion buffer with a fresh digestion buffer and perform another digestion round for 24 hours at 42°C. The total number of PCR cycles required to obtain sufficient DNA library material for enrichment (7-19 cycles), and post-enrichment for sequencing (6-16 cycles), was determined using Real-time PCR. A total of 12µL of DNA library constructs were used during amplification prior to enrichment. One microliter of the full fraction eluted from the beads post-enrichment was used for Real-time PCR while the remaining 30µL were used for the final amplification post-enrichment. The final volumes of the latter reactions were 100µL, and were purified using MinElute (QIAGEN) into 15µL EB+Tween 0.05%, before final quantification and size profiling on the TapeStation 2200 instrument. Up to 16 uniquely-indexed DNA libraries were pooled together at equimolar concentrations and sequenced on the Illumina HiSeq 2500 instrument in a Single End mode (SE81). A total of 795,888-32,921,457 reads were obtained per DNA library, and processed as described in '*DNA extraction and genome sequencing*', including adapter trimming, mapping, damage-based recalibration of quality scores and read end-trimming. Four DNA libraries were target-enriched for each sample, targeting the ~5,000 loci presented by Cruz-Dávalos and colleagues (95), which include the 41 Mendelian traits investigated here. Targeted-capture sequence data are available for download at the European Nucleotide Archive (Project PRJEB20000).

As shotgun and target-enrichment sequencing data were found to be in strong agreement, we merged them into a single BAM file for each sample. These merged BAM files were exclusively used to investigate the 41 loci. The number of high-quality reads supporting each nucleotide variant is provided in Data File S4. These counts were converted into genotype likelihoods with ANGSD (51), except that heterozygous sites weakly supported by a single alternative read were conservatively downscaled as non-causative homozygous genotypes. Heatmaps based on these genotype likelihoods are presented in Fig. 1B and Fig. S6.

A limited number of such loci, all involved in driving the coat coloration phenotype, were also confirmed using the AMPLEx-Torrent methodology, at the Jacques Monod Institute, France. We used a set of seven primers pairs in a single multiplex PCR (Table S7) and directly sequenced using the Ion torrent technology (96), as described in (13, 97).

Each multiplex PCR reaction was carried out with 3 µL of aDNA extract in a 30 µL reaction volume with a final composition of 50 mM Tris/HCl, 10 mM KCl, 5 mM (NH₄)₂SO₄, 4 mM MgCl₂, pH 8.3 (25°C), 1 mg/mL BSA (bovine serum albumin), 0.15 µM of each primer, 0.25 mM dA/ dC/ dGTPs and 0.50 mM dUTP, 2 U of Fast Start DNA polymerase (Roche), 0.5 U of uracil-N-Glycosylase (UNG). Solutions were decontaminated following (98) and three independent amplifications per sample were performed, after UNG digestion at 37°C for 15 min (99), at 95°C for 10 min, 45 cycles at 95°C for 20 sec, 57°C for 30 sec and 72° for 30 sec, and a final extension step at 72°C for 4 min. Barcoded libraries were built from 10 µL of the multiplex PCR and pooled after the ligation of barcoded adapters and purification using a silica column (Macherey-Nagel, Nucleo-spin Gel and PCR clean-up), following (100). The pooled barcoded PCR products were then size-selected using Caliper Labchip XT (Perkin Elmer). Size-selected products were subjected to nick repair and amplification in a 40 µL final reaction volume using NEB OneTaq Hot Start and containing Ion Torrent primers A and P1 (0.5 µM). The size distribution and concentration of the library was assessed on the Agilent 2100 Bioanalyzer. Emulsion PCR and Ion Sphere Particle enrichment were conducted with the Ion OneTouch System (Life Technologies) using the Ion OneTouch 200 Template kit v2 DL according to the manufacturer's protocol. The DNA libraries were sequenced on the Ion Torrent Personal Genome Machine (PM) Sequencer using the Ion PGM 200 Sequencing Kit and Ion 316 semiconductor sequencing chips (Life Technologies). An average of ~20,000 reads were sequenced per multiplex PCR. Amplicon sequence reads were aligned with the Ion Torrent Mapping Alignment Program (TMAP), using concatenated sequences of the seven amplicons as a reference sequence. Output files in BAM format were further processed for typing using Geneious 6.1.8 (101).

Using the shotgun sequencing data and the same mapping strategy as for uncovering the Y-chromosome sequences, we also investigated all 14 ancient samples for two regions that are absent in the EquCab2.0 reference sequence: (i) the ~1.6kb insertion at the *TBX3* locus (61) associated with the Dun phenotype, and; (ii) the 227bp SINE insertion within the *MSTN* promoter (102) associated with short-distance racing performance in Standardbred horses. The Dun (61) phenotype is characterized by a pigmentary dilution in coat color, mostly associated with a dark dorsal stripe in wild horses (Przewalski's horses) and other wild non-caballine equids (donkeys, zebras, etc). Rather than its linked mutations, the SINE insertion at the 5' UTR of *MSTN* has been postulated to be the ultimate responsible for increased short-distance racing performance in Standardbred horses (102). Since the horse reference genome EquCab2.0 is devoid of both insertions, their sequences were added at the corresponding chromosome coordinates to generate an alternate reference genome, which was used for read (trimmed for adapters) mapping. Results on coat color variants, including shotgun and aMPlex-Torrent genotyping, are summarized in Tables S8 and S9. The SINE insertion was left undetected in all the 14 ancient samples investigated (Fig. S7).

Inferring the maximum population allele frequency under which alleles can remain undetected in our sample

As some alleles segregating in the ancient population may remain undetected in our Berel' sample set due to a limited sample size, we estimated the maximal frequency at which the allele could segregate in the Berel' population, while remaining undetected in

our sample. Assuming that Berel' horses represent a random subset of their contemporaneous population, the probability that an allele is found in the population at frequency p , given that the allele remained undetected in our sample ($k = 0$), can be calculated through a Binomial distribution, as:

$$P(p = X | k = 0) = \text{Binomial}(n, p, k = 0)$$

Of particular interest is a mutation in the *DMRT3* gene, an allele underlying alternative ambling gaits in horses, that was found to have a relatively recent evolutionary origin (103, 104), and is absent in our sample. We numerically searched for the maximum *DMRT3* population frequency compatible (at a 0.05 level of significance) with a n -sized sample where this allele is absent. We found that a population allele frequency as high as 0.1221 could remain undetected when assuming that each chromosome was sequenced at least once.

Outgroup f_3 statistics and TreeMix population tree

The VCF files encoding individual-based genotype calls were merged with bcftools v1.2 (105), and filtered for positions without missing genotypes. Only individuals with an average depth-of-coverage of at least 7-fold were considered, representing five Berel' and all modern horse genomes (Table S3 and Data File S3). The analysis was also restricted to biallelic sites passing all quality filters, which resulted in a panel of 78,467 single nucleotide polymorphisms (SNPs) across all individuals compared.

Using this SNP panel, we calculated the f_3 -statistics in the form of $f_3(\text{Berel}', \text{modern domestic; outgroup})$ with ADMIXTOOLS (49). This f_3 configuration is known as the outgroup f_3 statistics (49), and is devised to quantify the amount of genetic drift shared between Berel' and modern domestic horses. As outgroup genomes, we considered either that of a Przewalski's (Prz_M_0157_LO_SB293) horse without evidence of admixture with modern domestic horses (50), or a pre-domestic horse (Yak_M_0223_LO_Batagai) that belonged to a now extinct population (15). Results are shown in Fig. S8.

The SNP panel was also converted into TreeMix format with the plink2treemix.py utility (48). We ran TreeMix (48) with the following parameters:

- block resampling procedure (option -k), with groups of $k = 1,500$ variants, which approximately corresponds to ~42 megabases (Mb), far overpassing the blocks of linkage disequilibrium found in horses (106). Equivalent results were obtained using $k = 500$ (~14 Mb) (Table S10).
- a round of global rearrangements (option -global) to refine maximum-likelihood inferences.
- based on previous results (14, 15), the trees were explicitly rooted (option -root) in the branch separating the prehistoric (Yak_M_0225_LO_CGG10023, Yak_M_0223_LO_Batagai and Yak_F_0224_LO_CGG10022) from the rest of horses.
- an increasing number of migration edges (option -m), ranging from 0 to 15.

The variance explained by each one of these migration models was measured using the 'get_f()' R function supplied in the TreeMix suite. The variance explained

by different migration edges asymptotically increases until $m = 9$ (Table S10; Fig. S9). None of the nine migration edges was inferred to involve gene flow from/to Berel' horses.

Admixture test with D-statistics

We further tested for genetic introgression between wild horses and ancient domesticated horses. The wild group is represented by Przewalski's horses, or by the genome of three horses that lived prior to domestication and were previously characterized (14, 15). The latter belonged to a now-extinct lineage of wild horses, which diverged some ~150 kyr. ago from the lineage leading to present-day horses. The group of ancient domesticated horses consisted of the 14 ancient genomes reported in this study (Berel', Arzhan or Sintashta). We calculated the D-statistics (107, 108) as $D(((\text{ancient domesticates}, \text{modern domesticates}), \text{wild}), \text{outgroup})$, with the Somali wild ass (*E. africanus somaliensis*) representing the outgroup (38).

The D-statistic was computed following the approach originally developed by (108, 109), and subsequently adopted in (14, 15, 27, 38, 50). This entails a series of quality filtering steps that are especially suitable for genomes sequenced at low-coverage and/or with substantial error rates. As in the other analyses, reads with a mapping quality below 25 were excluded. At each position, we then grouped reads as A, C, G and T, including the strand information (eight categories). For each category, we removed the 50% lower-quality bases, before randomly sampling one of the remaining bases as representative of the genomic position. Finally, only nucleotide transversions were subsequently considered.

To assess for statistical significance, we expressed the D-statistics as a Z-score, by applying a jackknife procedure in 10 Mb-windows, following (27). Absolute Z-score values higher than three are usually considered significant. However, in order to control for multiple testing, we transformed Z-scores into p -values, and corrected them with the Benjamini-Hochberg procedure (110), as implemented in the 'p.adjust' function of the R programming language (Data File S5).

We also estimated the fraction of the genome (f) in present-day and ancient domestic horses that descend from the now-extinct lineage of prehistoric horses, using the approach defined in (107, 108, 111). In particular, we used the ABBA-BABA counts (numerator of the D-statistics, or S) in the following configuration:

$$f = \frac{S(((\text{Przewalski's}, H1), \text{CGG10022}), \text{outgroup})}{S(((\text{Przewalski's}, \text{Batagai}), \text{CGG10022}), \text{outgroup})}$$

where CGG10022 represents a member of the now-extinct lineage of pre-domestic horses, and the Przewalski's horse is Prz_F_0150_LO_KB3879, one of the less admixed with modern domesticates. Samples sequenced at low coverage, however, tend to exhibit reduced ABBA-BABA counts, regardless of the fraction of their genome descending from the pre-domestic horse population; *i.e.* the lower the coverage, the lower the counts, including ABBA-BABA events. To avoid biases in the f estimates, we thus restricted the analysis to Sintashta and Berel' horse genomes sequenced at a minimum coverage of 4-fold, since this appears to provide a stable number of ABBA-BABA counts across samples. We repeated the analysis replacing the genomes of the Berel' or Sintashta

horses by those belonging to present-day domestic horses. We found that f is significantly lower for the latter (median(Berel' and Sintashta) = 24.57% vs. median(modern-day domestic) = 17.96%, Wilcoxon test p -value < 0.00451). The significance consistently persists using other combinations of Przewalski's or pre-domestic horses (Table S11).

Genetic affinities

We estimated the genetic affinities between Berel', Arzhan and Sintashta horses, relative to a panel of 38 present-day and ancient samples (Table S12). We used two complementary approaches, including Principal Component Analysis (PCA) with ngsCovar (112), and phylogenetic reconstruction with ngsDist (113). For both approaches, we implemented a two-step Bayes Empirical Bayes (BEB) approach (114). The first step consisted in estimating a genome-wide site frequency spectrum (SFS) from biallelic sites with a minimum depth-of-coverage of 2-fold per sample (-SNP_pval 1e-06 -setMinDepthInd 2 -minInd 52). The second step used this SFS as a prior to estimate the posterior probabilities of the three possible genotypes. This analysis was restricted to biallelic sites without missing information (-SNP_pval 1e-06 -minInd 52 -doPost 3).

The two-step BEB procedure resulted in two SNP matrices. The first one contained the genotype probabilities for 100,855 nucleotide transversions, and included three Przewalski's and three pre-domestic wild horses as distantly-related populations. This matrix was used both in ngsCovar and ngsDist analyses. The pairwise genetic distances estimated from ngsDist were used to build a Neighbour-Joining tree with the FastME program (115), rooted following (15) and (14). We assessed the support of the clades recovered by ngsDist through 500 bootstrap pseudo-replicates.

Since distant populations compromise the PCA visualization, we also created a second matrix, excluding the three Przewalski's and the three pre-domestic horses. This second matrix contained a total of 86,485 nucleotide transversions. Results for the PCAs are in (Fig. S10)

Inbreeding and kinship calculation

We estimated the inbreeding coefficient (F) for each Berel' horse, as well as their pairwise kinship coefficients (θ), to characterize the horse breeding and mating strategies implemented by Scythian nomads. To achieve this, we used ANGSD to calculate the genotype likelihoods at biallelic positions (-SNP_pval 1e-6) sequenced in at least 90% of the samples (-minInd set according to the group size), and with a minimum count for the alternative allele superior to 1 (-minMaf set to exclude singletons). The latter makes sure that at least doubletons are considered, as testing whether alternative alleles are in homozygosis more often than expected under random mating requires that the alternative allele is observed at least twice. Such SNP panels were used as input for ngsF (17) and ngsRelate (18), which were ran with default parameters.

The inbreeding and kinship coefficients were estimated in different groups of horses. These groups were defined *a priori*, based on the monophyletic clades recovered by TreeMix (Fig. S9) and ngsDist (Fig. 3). These consisted of Berel' horses and: (i) the group formed by Mongolian and Yakutian horses (hereafter, referred to as *MonYak*), (ii) the group comprising Shetland, Icelandic and Fjord horses (*Nordic*), and; (iii) the remaining 20 modern domestic horses that have experienced intensive breeding practices (*Intensive*).

Of note, ngsRelate requires external population frequencies to estimate the kinship coefficient θ between pairs of sampled individuals. As such frequencies are not available for ancient specimens, we assumed that the horses from the Berel' sample provide an unbiased representation of the population allele frequencies. In practice, this assumption implies that ngsRelate can solely identify pairs of samples that are more related to each other than with the remaining set of sampled horses. Another possible drawback is that ngsF is sensitive to within-deme substructure. To monitor the bias possibly introduced by rampant within-deme structure, we calculated the inbreeding coefficient F in two genetically homogeneous breeds, for which a considerably larger number of individuals has been sequenced, namely Yakutian (15) and Franches-Montagnes horses (50) (Fig. S11). The set of Franches-Montagnes horses serves as control to contextualize inbreeding and kinship estimates, as only outbred Franches-Montagnes horses were carefully singled out prior to sequencing (50).

Nucleotide diversity (π)

We measured nucleotide diversity (π) at the population level (116), for loci with different modes of inheritance, including autosomes, the X- and Y-chromosomes, and mtDNA. For autosomal chromosomes, we estimated the SFS with ANGSD (-doSaf 2), while for haploid markers the SFS was directly calculated from the multiple sequence alignments generated in the 'Mitochondrial DNA (mtDNA)' and 'Y-chromosome' sections. We then used custom Perl scripts to fold the SFSs, and calculate π as:

$$\pi = \frac{2 \sum_{i=1}^{nb} (i(n-i) p_i)}{n(n-1)}$$

where n is the sample size (in number of chromosomes), nb the upper boundary of the SFS ($n/2$ for an even number of chromosomes, $(n-1)/2$ otherwise), and p_i the proportion of sites with mutations segregating at frequency i . We estimated the variance of the nucleotide diversity from 10,000 bootstrap SFS pseudo-replicates. We compared the Berel' population to other three groups defined a priori (*Nordic*, *Intensive* and *MonYak*), as well as to the merger of all three groups, and to the Yakutian and the Franches-Montagnes horse breeds (Fig. S12).

Genetic deleterious load

Evolutionary constraint appears as one of the best predictors for the fitness consequences of mutations (117). As the phyloP program shows a remarkable accuracy in measuring DNA sequence constraint (21), we downloaded phyloP scores pre-calculated for a 46-way placental mammal whole-genome alignment (<http://hgdownload.cse.ucsc.edu/goldenPath/hg19/phyloP46way/placentalMammals/>). These scores are, however, provided according to the genomic coordinates of the human reference sequence (hg19). We thus converted these coordinates into their respective orthologous positions in the horse genome (EquCab2.0) (36), using the command-line version of the liftOver tool, and the equCab2ToHg19 chain file available from the UCSC tables (<https://genome.ucsc.edu/cgi-bin/hgTables>).

For each tested horse genome, we calculated the genetic load based on positions showing a minimum phyloP score of 1.5 and a minimum coverage of 3-fold. The

minimal phyloP cutoff was selected because it discriminates zero-fold and four-fold degenerate positions, which approximate constrained and neutral evolution, respectively. The minimum coverage of 3-fold was selected in order to include a reasonable number of positions for genomes sequenced at low coverage, while limiting the corresponding genotype uncertainty. We also restricted our calculation to homozygous positions as (i) the number heterozygous sites can be overestimated under a genotype likelihood framework without prior knowledge on the allele frequencies, and (ii) estimating the genetic load at heterozygous positions would require additional assumptions about the dominance coefficient of mutations (e.g. overdominance, intermediate dominance, or recessive effect).

The genotype likelihoods for the set of analyzed positions, as well as the corresponding phyloP scores, were processed as input for an in-house developed Perl script, available upon request. This script multiplies the phyloP ($phyloP_i$) score at each given position (i), by the probability of observing an alternative (not conserved) allele (P_i). The total genetic load was then calculated as the average product observed over all the homozygous positions analyzed for a given individual.

$$Genetic\ Load = \frac{\sum_{i=1}^{hom} P_i(phyloP_i)}{hom}$$

Finally, we compared the mean genetic load observed in each protein-coding gene amongst Berel', Sintashta and Arzhan horses, relative to the rest of modern domestic breeds, by means of a non-parametric Wilcoxon test.

Copy Number Variants (CNVs)

We generated individual genome-wide copy number (CN) profiles for the 14 ancient horse genomes, as well as for a set of 14 genomes representing modern horse diversity, and that were previously sequenced (Table S13). Our procedure exploits read depth (RD) variation along genomes, as described previously (118), with algorithmic improvements, in five successive analytical steps:

1) Masking of repetitive regions in the horse reference genome. We first masked all regions in the EquCab2.0 reference genome (36) that are annotated as '*Repeat Masker*' and '*Simple repeats*' in the UCSC tracks. To account for unidentified short repeats, we estimated the frequency of 36-bp kmers along the entire nuclear genome, using a step window size of five bp. These kmers were mapped to the EquCab2.0 using mrsFast (119) masked whenever showing more than 20 placements along the genome.

2) Mapping 36-bp reads against the EquCab2.0 assembly. In order to exclude lower-quality ends, Illumina reads were trimmed using in-house developed scripts. We then split reads into non-overlapping 36-bp fragments, and mapped them against the version of EquCab2.0 that was previously masked for repetitive regions (step 1).

3) Estimating raw copy numbers. mrCaNaVaR (118) was used to estimate the copy number along the genome from the mapping RD. To achieve this, mean RD per base was calculated in one kb non-overlapping windows of unmasked sequence (that is, the effective window size, which may be larger than one kb due to the presence of repeats

and/or gaps). As reads will not map against masked positions, RD tends to be lower at the edges of these regions, which could potentially affect copy number calculation in the subsequent step. The 36 bp flanking any masked region or gap was, thus, further masked. A genome-wide RD distribution was calculated by iteratively excluding windows with the most extreme RD values, as identified by mrCaNaVaR, and retaining the remaining windows as control regions. Finally, we determined the copy number in the one kb windows of non-repetitive sequence as the GC-corrected RD of each window divided by the mean RD in the control region window set. The distribution of copy number values in control regions was centered then to the value of two, in line with the ploidy of the individual genomes.

4) HMM training and predictions. We further used a Hidden Markov Model (HMM) based method to transform raw copy number estimates into absolute copy number calls. More specifically, we declared separate HMM states for low copy numbers ranging from 0 to 20 (*CN0:CN20*). Higher copy numbers, ranging intervals *CN21-100* and *CN101-1000* respectively, were considered with only two extra HMM states. The HMM was trained processing the sequence data underlying the Mongolian horse genome KB7754 (50), which was sequenced to relatively high average depth-of-coverage, 19.5-fold. We predicted probabilities of each of the declared HMM states at each point by forward-reverse algorithm using the Pomegranate Python package. We additionally defined the probabilities of each HMM state, averaged over the 28 individuals, as a prior to refine CN calls through Bayesian inference around the window of interest (± 2 windows). We ordered HMM states according to their probability after re-calling, and calculated confidence intervals by considering one CN increment at a time, until reaching a cumulative probability of $p = 0.99$. Therefore, the method can assess whether for a particular window, any two individuals have the same CN (if we confidently call them with the exact value), different (if we confidently call them to belong to non-overlapping CN intervals), or unresolved (if we call the individuals to belong to overlapping intervals).

5) Defining segmental duplications. We defined a ‘*high confidence set of duplicated regions*’, as those regions encompassing at least five consecutive windows called with at least 3 copy numbers ($CN \geq 3$), in at least one of the genomes. We also applied a less stringent criterion, defining ‘*putative segmental duplications*’ as all genomic regions showing a minimum of two consecutive windows with $CN \geq 3$ in at least one of the genomes, be ancient or modern. The latter, used to conservatively mask CNVs in selection scans (see ‘*Functional enrichment analyses*’), is summarized in Fig. S13.

Genome-wide selection scans

F_{ST}. The F_{ST} index was calculated to assess the genetic differentiation along the X chromosomes between Berel’ horses and modern domesticates, the latter comprised of *MonYak*, *Nordic* and *Intensive* horse groups, as defined in ‘*Inbreeding and kinship calculation*’. Complementarily, we also estimated the decay in diversity during the last 2,300 years along the X chromosome, as $\log(\pi_{\text{modern}}/\pi_{\text{Berel'}}$). The F_{ST}, as well as $\pi_{\text{Berel'}}$ and π_{modern} , were calculated in non-overlapping sliding windows of 50 kb with mstatspop

v.0.1beta, available for download at <https://bioinformatics.cragenomica.es/numgenomics/people/sebas/software/software.html> (Fig. S14). This analysis was restricted to stallions (*i.e.* 10 Berel' after excluding BER11_L, and 18 present-day horses), as the absence of heterozygous sites on their X chromosomes greatly simplifies genotyping.

The X:40,375,001-53,375,001 region was identified as outlier by the changepoint R package (120). The 104 genes within this region were analyzed for functional enrichment (see below).

Population Branch Statistics. PBS (121) was applied as implemented in ANGSD (51) to identify autosomal regions showing outstanding levels of genetic differentiation in the Berel' horse population. We considered present-day domesticates as sister group, and the less admixed Przewalski's horses as outgroup (Data File S3). We scanned the autosomal chromosomes using 10 kb windows, overlapping every 5 kb. The 121 genes overlapping with the top-1000 PBS genomic regions (Table S14) were further analyzed for functional enrichment (see below).

Levels of exclusively Shared Differences (LSD). We developed the LSD method (Levels of exclusively Shared Differences) as a new framework that builds on low-to-moderate genome sequence data, to identify positively selected regions along pre-defined branches of a population tree. In contrast to 3P-CLR (122) and PBS (121), our method accommodates on tree topologies with an unlimited number of populations, including some represented by ancient genome data. LSD is, thus, well suited for handling genome time-series, which provide the most powerful datasets for detecting adaptive signatures. The method is described in great detail in '*Genome-wide selection scans*', where we also use a simulation framework to gauge for its overall accuracy (see '*Assessing the performance of LSD_{norm} using forward simulations*'). Forward simulations were performed using SLiM v2.0 (123) and were applied to a population tree representing our current knowledge on horse evolutionary histories, including present-day and Berel' horses, as well as a now-extinct population of pre-domestic horses (14, 15).

Using LSD_{norm} , we conducted a cross-population genome-wide selection scan using the Somali wild ass (*Equus a. somaliensis*) as an outgroup species, and considering a series of six horse groups:

- three wild pre-domestic horses (Yak_M_0223_LO_Batagai, Yak_F_0224_LO_CGG10022 and Yak_M_0225_LO_CGG10023)
- three Przewalski's horses (Prz_F_0150_LO_KB3879, Prz_M_0157_LO_SB293 and Prz_M_0160_LO_SB533),
- the four demes defined in the '*Inbreeding and kinship calculation*' section, namely *Berel'*, *MonYak*, *Intensive*, and *Nordic* horses. This comprised 46 individuals with a minimum average depth-of-coverage of 5-fold.

This panel includes seven contemporaneous Berel' horses, which are expected to provide reasonable allele frequency estimates of the ancient Scythian horse population (Table S15).

Gene models annotated in EquCab2.0 are extremely variable in size, ranging from a few hundred of base pairs to over one megabase. LSD_{norm} estimates the impact of positive selection along the full region represented in the local tree. If the local tree corresponds to a large gene interspersed by long introns, and the episode of positive selection is confined to a particular exon, the molecular signature of adaptation could be averaged out, and consequently missed. In order to avoid this, we refrained from a gene-centered approach and opted for examining sliding windows of fixed size, which provide an even power to detect the impact of positive selection across the whole genome.

We first ran ANGSD to generate a beagle output file, containing the posterior probabilities of sites with an overall missingness inferior to 10%. We then split the beagle file in sliding windows of 10 kb, with an overlapping step-size of 5 kb. For each window, we used FastME to build the local gene trees, based on pairwise genetic distances inferred with ngsDist, and rooted by using the Somali wild ass. Finally, we estimated the LSD_{norm} values in the ancestral population prior to the split of Berel' and modern domestic horses, a time range encompassing the period of horse domestication (1).

The top-1000 LSD_{norm} regions, representing ~1-2% of the horse genome, were considered candidates for positive selection, and further investigated for the presence of protein-coding exons (Table S16).

Functional enrichment analyses

Four different gene lists were independently analyzed for functional overrepresentation, including the F_{ST} , PBS and LSD candidates, as well as the protein-coding genes with more deleterious load in present-day than in the 14 ancient domesticates. Functional enrichment tests were conducted with the WebGestalt server (124). Additionally, we explored whether candidate genes are preferentially transcribed in certain developmental stages or parts of the horse anatomy, with the TopAnat tool of the Bgee database (125).

Since the function of horse genes stills poorly annotated, we relied on the 1:1 single-copy human:horse and mouse:horse orthologs, as defined by Ensembl. Autosomal, protein-coding and single-copy genes were considered as species-specific background set for the PBS, LSD and genetic load tests of overrepresentation. For genes within the X:40,375,001-53,375,001 region, the genes in the X chromosome were also included. Enrichment tests are summarized in Tables S17 to S26.

Supplementary Text

Sample information

The horse in the Scythian culture. Scythian nomads, also called Saka in the literature, inhabited the Eurasian steppes, from Hungary to the Great Wall of China, essentially during the 1st millennium BC (126). Scythian societies were characterized by nomadic pastoralism, improvements in horse harnessing and mounted warfare, representing one early militaristic society from the Iron Age (127).

Scythians were amongst the earliest-known mounted warriors, rode with saddles but no stirrups (128), and developed the aristocratic class of “Royal Scyths”, which is archaeologically documented by elaborated graves, including sacrificed horses with ornaments and harnesses (129). Their elaborated funerary rituals, where horses were carefully harnessed and decorated, then sacrificed and immediately positioned at the burial site before the *rigor mortis* set in, have been described in great detail by Herodotus, the father of history (129). This reflects the central role of the horse in Scythian societies, as a companion in both life and beyond. Scythians also used horses for their milk, meat and clothing, as well as for transportation both through riding and chariotry (127). The archaeological record is mainly based on their burial mounds, tumuli tombs known as *kurgan*, which include horses, horse ornaments, weapons and artifacts, decorated according to the Scythian iconography (130).

Berel’. Together with Pazyryk burials (131), Berel’ is one of the most iconic examples of the Scythian burial mounds. It is dated to around 2,300 years BP (9) and is located in a valley of the Altaï mountains, Kazakhstan at 1,200 m of altitude. Here, we investigated horse remains excavated within the *kurgan* number eleven (*kurgan* 11), which is itself located within an alignment of 30 *kurgans* oriented northwest-southeast. *Kurgan* 11 is a 33-meter-wide mound covering a sepulchral pit of 5x3x5 meters, completely embedded in permafrost. The burial contained two human mummies, a 40-50 years old adult man and a 60-70 years old woman. The woman was buried 5-40 years later than the man, and was not closely related to him, as revealed by a genetic study based on mtDNA and STR markers (132). *Kurgan* 11 also included a total number of 13 sacrificed horses, which were displayed in two layers separated by birch bark, outside the main burial chamber (133). The frozen conditions at the site helped to preserve the two mummies as well as the horse skin, fur and muscles in exceptional conditions. Despite plunders of the tomb, a large number of elements characteristic of Scythian funerary rituals could be retrieved, including clothing and decorative elements of the mummies, as well as the harnessing material, the saddles and the elaborated ornaments of the horses. All horses were harnessed, saddled and with limbs folded. Some of them wore horns, masks and golden elements of the Scythian animal iconography, including felines, deers, sheeps and the griffin, the mythological eagle-lion (19, 133).

Clear osteological signs, in at least six of the horse skeletons, point to a hit of an axe on the forehead as mean of death (129). Together with strangling, this was a common practice within the Scythian horse sacrifice. According to modern standards, the horses from Berel’ would be classified as medium size. Morphometric analyses of the length and width of eleven humerus, twelve radius, nine metacarpals, nine femurs, seven tibias and seven metatarsals revealed that they were relatively homogeneous in size, showing only 7% difference between the smallest and the largest, but were larger than the average of

the same period (9). All horses found in *kurgan 11* were males, possibly castrated (129), in line with stallions being predominantly used for sacrifices (although mares have also been found in other burials (10)). Dental eruption and degree of wear indicated that the stallions were 8 to 19 years old (10). No bone pathologies could be detected, except for extra bone growth at two thoracic vertebrae for individual BER05_E, possibly due to congenital disease, an advanced age (134) or a consequence of saddling (given that pad saddles directly resting on the thoracic vertebrae (10)). Additionally, nematode eggs (*Oxyuris equi* and unclassified *Strongylidae* members) could be isolated, but were not linked to serious diseases (135). This, and the relatively limited pathologic conditions observed in the horses found in *kurgan 11* likely contradicts theories suggesting specific selection of elder and diseased horses for sacrifice (136). The great care in harnessing and decoration, their positioning in the mound as well as their relatively wide age distribution and apparently good health suggest instead a highly symbolic sacrifice gesture (10). All 13 horse samples were screened for the present study, but only eleven showed DNA preservation levels compatible with economical genome-scale investigations (BER01_A, BER02_B, BER04_D, BER05_E, BER06_F, BER07_G, BER08_H, BER09_I, BER10_K, BER11_L and BER12_M; Table S3).

Arzhan I. The burial mound of Arzhan I is located ~600 km northeast of Berel', in the Tuva Republic, Russia, and dates back to the boundary between the 8th and 9th century BC (2,760–2,695 yBP (137)). This mound, excavated in the seventies (8), is considered the earliest Scythian or pre-Scythian monument in all Eurasia. It is particularly emblematic because of the massive size of its *kurgan*, which included 16 human remains and 160 sacrificed horses organized in different chambers. All sacrificed horses were harnessed and males, with an age above 12-15 years (8). This *kurgan* was also looted, and the few objects retrieved reflect its extraordinary wealth (131). The two samples included in this study come from chambers 1 and 2, close to the center of the mound, and are labeled as I-K2_Arz1 and I-K3_Arz2 (Table S3).

Sintashta. We included in this study a sacrificed horse from a different cultural context, pre-dating the Scythian culture by almost two millennia. This specimen was located in Sintashta site, Chelyabinsk Oblast, Russia (138), only 400 km west from Botai, the region where horses were probably domesticated (1). The archaeological site dates back to the Bronze Age (4,750–3,550 yBP) and includes a fortified settlement and five *kurgan* cemeteries, including the “Tsar” *kurgan* with 40 graves, seven of which with chariots (127). Sintashta was a metallurgical center and represents the first evidence of chariots in the archaeological record. The sample 838-64_NB46 analyzed here (Table S3) has been radiocarbon dated to 4,071-4,055 yBP, and was a mare buried together with three other horses and a chariot in the grave N 19 (138).

Comparative data set. We compared the 14 horse genomes sequenced in this study (Table S3) to a reference panel comprising a total of 54 horse genomes, 53 of which previously published (Data File S3). The latter includes 46 modern domestic horses from 15 different breeds (four American Quarter horses, two Arabian, one Dülmener, twelve Franches-Montagnes, two Hanoverian, three Icelandic, one Marwari, three Mongolian, one Morgan, one Norwegian Fjord, two Shetland ponies, four Standardbred, one Thoroughbred, and nine Yakutian), three Przewalski's horses showing no evidence of domestic introgression (50), one historical sample from the 19th century (CGG10397),

and three pre-domestic horses that belonged to a now-extinct population encompassing from the Late Pleistocene to the Early Holocene, namely CGG10022 (~42,000 yBP), CGG10023 (~16,000 yBP) and Batagai (~5,200 yBP) (14, 15, 27, 36, 50, 104, 139–143). The genome sequence of one modern Akhal-Teke stallion was obtained for the first time, at average depth-of-coverage of 22.8-fold, and following the methodology used by Der Sarkissian and colleagues to characterize the genome sequence of twelve Franches-Montagnes individuals (50). The sequence data underlying the Akhal-Teke genome are available to download at the European Nucleotide Archive, under the accession number SAMEA4075241 (ERS1246351).

Sequence authenticity

Post-mortem DNA damage. Laboratory contamination was strictly monitored with extraction, library and amplification blanks, all of which showed no sign of contamination. In the case of sample 838-64_NB46, all extraction blanks were over-amplified to provide sufficient library amounts for sequencing. These revealed a laboratory contamination range of 0.013% to 1.242%, depending on the extraction method considered (25). Deep sequencing was performed from successful DNA extractions conducted with method D (28) and method Y (29), as modified by (25, 30), both associated with low contamination bias (<0.06%), if any (25).

Sequence length distributions (Fig. S15), as well as fragmentation (Fig. S16) and deamination patterns (Figs. S17-S18) were examined with mapDamage2.0 (39), revealing typical patterns of post-mortem DNA degradation. The size distribution of endogenous horse reads was found to be on average larger in permafrost (Berel' site, 74.0±4.5 bp (average±sd)), than in non-permafrost sites (Sintashta and Arzhan I, 63.8±4.2 bp) (Fig. S15 and Data File S2). The ca. 10bp periodicity pattern proposed to reflect nucleosomal protection of the DNA post-mortem (144) was found across all samples (although admittedly less pronounced in 838-64_NB46; Fig. S15).

All non-USER treated libraries showed the typical ancient DNA (aDNA) deamination pattern, with increasing C→T and G→A mis-incorporation rates towards 5' and 3' ends, respectively (Fig. S17). Reduced mis-incorporation patterns were still detected in the sequencing data underlying USER-treated libraries (Fig. S17), as previously observed for ancient specimens (25, 145–147). This most likely reflects a reduced efficiency for the USER at template termini. Similarly, posterior distributions of deamination patterns at overhangs (δ_s) and double strands (δ_D), as well as the length of the overhangs (calculated as $(1/\lambda)-1$, following (27)) showed higher values in non-USER than in USER-treated libraries, as expected (Fig. S18).

Authenticity of results was further assessed for all the samples from Berel' by comparing their hypervariable mtDNA region to previous analyses carried out independently (11), revealing identical haplotypes. Independent replication was further provided by the analysis of markers underlying coat-coloration patterns (see '*Genotyping loci underlying important equine phenotypes*').

Sample-specific microbial profiles. We performed microbial profiling for each single library constructed from Berel', Arzhan and Sintashta horse samples. We used the metaBIT implementation of the *pvcust* R package (148), which enables to use Bray-Curtis distances, to hierarchically group microbial profiles by means of an average linkage clustering method and 10,000 bootstrap iterations. Libraries were found to

consistently cluster by sample, indicating a limited (in any) impact of contamination across samples (Fig. S19).

Microbial and pathogen screening

Ancient horse microbial profiles. The microbial profiles of the horses from Berel', Arzhan and Sintashta fell together with the microbial profiles of four other ancient horses (CGG10022, CGG10023, Batagai, CGG101397), outside the diversity of human-associated microbiomes, but close to the soil microbial profiles (Fig. S20). This advocates for a very limited (if any) human-driven contamination, *e.g.*, by saliva or skin cells, during excavation, storage or analyses of the samples. The similarity to soil microbial profiles can be explained by the presence of classes typically abundant and ubiquitous in soils (45): *Actinobacteria* (4.74-74.97%), *Alpha-proteobacteria* (1.74-35.67%), *Beta-proteobacteria* (2.40-51.08%), and *Gamma-proteobacteria* (2.12-58.88%) (Fig. S21), which most likely derive from a microbial colonization from the depositional soil, as shown in (40).

The structure within the microbial diversity present in the DNA extracts of all ancient horses was investigated through PCoA and hierarchical clustering at the genus level. The first PCoA dimension separated the microbial profiles of the Berel' horses from all others (Fig. S22). This structure was confirmed by hierarchical clustering, in which microbial profiles from Berel' form a strongly supported cluster (approximately unbiased *p*-value = 96%; bootstrap probability = 43%) (Fig. S23). This discrimination could not be indirectly explained by differences in numbers of uniquely mapped reads to MetaPhlAn markers (mean(Berel')=284,708 vs. mean(others)=183,318, Wilcoxon signed-rank test *p*-value = 0.07; Table S2) or in Shannon diversity indices calculated with the function *diversity* of the package *vegan* (<http://cran.r-project.org/package=vegan>) (mean (Berel')=2.71 vs. mean(others)=2.69, Wilcoxon signed-rank test *p*-value = 0.79; Table S2). We thus identified microbial taxa responsible for the differentiation of the microbial profiles from Berel' by the Linear Discriminant Analysis (LDA) implemented in the LEfSe program (LDA Effect Size) (149). This was done considering a significant alpha-value of 0.05 for the non-parametric Kruskal-Wallis sum-rank test and the unpaired Wilcoxon test, and discriminative features were identified if their logarithmic LDA score was found to be superior to 2. This analysis detected 67 features distinctive of the microbial profiles of the Berel' horses (Fig. S24). In particular, Berel' were differentiated from the rest of ancient horses by their relative abundances in environmental bacterial genera, characteristic of their depositional soil: *Glaciibacter* (3.26-38.07%), *Rhodococcus* (<22.87%), *Rhodanobacter* (<22.87%), *Streptomyces* (<17.85%), *Albidiferax* (<16.61%), *Cellulomonas* (<8.61%), *Thiobacillus* (<5.79%). Similarly, the remaining horses showed higher relative abundances than Berel' horses for the following bacterial groups: *Pseudomonas* (3.65-69.46%), *Clostridium* (<6.49%), *Leucobacter* (<3.84%), *Caulobacter* (<2.95%), *Variovorax* (<1.78%) (Fig. S24). The differential microbial environmental signature at Berel' might reflect water penetration into the burial following its plunder and later freezing as permafrost soil (see 'Sample information'). The innovative excavation technique used at Berel', which kept frozen blocks from the site until their analysis in the laboratory, could possibly explain some of the differences observed in the microbial profiles (19).

Pathogen detection. The microbial profiling in metaBIT (41) revealed the potential presence of pathogens belonging to the genera *Rickettsia* in BER01_A (15.4%), BER04_D (4.7%) and BER02_B (5.2%). Some *Rickettsia* species are obligate intracellular pathogens, and can cause skin rash, fever, headache, arthralgia and myalgia (150). Not all *Rickettsia* species are, however, represented in the MetaPhlAn2 database, which limited the resolution provided by metaBIT to the genus level. We tentatively identified potential *Rickettsia* pathogens by mapping all sample reads against a comparative database that includes at least one *Rickettsia* genome per species. Maximal genome coverage was obtained for all three horse remains against *R. felis*, which consisted of average depth-of-coverage of ~21-fold for BER01_A, ~2-fold for BER02_B and ~4-fold for BER04_D (Fig. S25). This represented a 2- to 128-fold enrichment compared to other *Rickettsia* species for these three horses, and an 8- to 3,958-fold enrichment compared to the coverage obtained for *R. felis* in the other ancient horses (after normalizing for the sequencing effort, i.e., the number of retained reads after adapter trimming/quality filtering).

For BER01_A, BER02_B and BER04_D, the depth-of-coverage was homogeneous along the genome sequence (Fig. S26), with the exception of the multicopy 23S, 5S and 16S ribosomal RNAs (positions 341,681-344,465, 344,780-344,822 and 456,883-457,890, respectively), which showed a 19 to 81-fold increase in coverage. No reads were found to map against regions 321,921-332,717 and 336,202-337,666, known to contain three of the 82 copies of transposase genes annotated in the *R. felis* genome sequence (151), and thus limit the region mappability (Fig. S26).

The even depth-of-coverage along the genome suggests that the *Rickettsia* sequences recovered from BER01_A, BER02_B and BER04_D most likely belong to *R. felis*. To verify this further, we mapped the sequence reads generated from ancient remains to contigs available for three additional *R. felis* strains. When comparing depths-of-coverage in *R. felis* genomes/contigs versus other *Rickettsia* genomes/contigs, we only considered high-quality reads, excluding those reads mapping to more than one *Rickettsia* species/strain reference sequence. This reduced the noise created by genomic regions conserved across several *Rickettsia* species/strains (Fig. S27), and revealed a significant enrichment of *R. felis* strains LSU (Accession number NZ_JSEL01000001.1) and LSU-Lb (NZ_JSEL01000001.1) (152).

Humans, cats, dogs, rodents, cattle, sheep and goats are common hosts of *R. sp.* (153) and *R. felis* was also previously detected in the serum of horses showing no fever symptoms (154). *R. felis* was initially found to be transmitted by the cat flea *Ctenocephalides felis* (155), however, it has now been isolated from a diversity of invertebrates (156). The strain LSU, for example, also in association with the cat flea, was originally detected from the booklouse *Liposcelis bostrychophila*. Unlike other *Rickettsia* species, *R. felis* is not an obligate intracellular pathogen, and its presence could arise from a modern source, e.g., soil microbial communities. To check if *R. felis* DNA shows damage patterns typical of ancient molecular, we used mapDamage 2.0 (39), and found no evidence of DNA damage mis-incorporation typical of ancient DNA (Fig. S28) using the sequencing data from the only library constructed from non-USER treated DNA yielding sufficient data for undertaking such analyses (BER01_A_E9.Dab2nd_NOuser_CTATCA). Moreover, fragment size distributions for *R. felis* DNA from BER01_A, BER02_B and BER04_D were skewed towards long

single-end DNA reads (i.e., ~100bp). We therefore conclude that the presence of *R. felis* in the sequence data generated from ancient Berel' horse remains probably consists of recent, soil-derived contamination and is uninformative about the health status of these horses.

Comparative genome analyses

Inbreeding and kinship coefficients. We found that Berel' horses tend to be less inbred than the remaining horse groups investigated (Fig. S11). Interestingly, BER10_K and BER11_L share a high proportion of identity-by-descent tracts or kinship coefficient (Fig. S11), as consistently shown by their close genetic affinity (Fig. 3), which almost quadruplicates those inferred for any other pair of Berel' horses. Altogether with their identical mtDNA sequences (Figs. S3 and S5), and their distant Y-chromosomes (Fig. 2A), this kinship coefficient suggests that BER10_K and BER11_L are likely to be maternally related. Given its close relatedness to BER10_K, the lower-quality genome of BER11_L was excluded from all subsequent analyses.

Genetic load. In order to investigate whether the differences in coverage could explain the reduced genetic load inferred for the 14 ancient domesticates, we downsampled four high-quality genomes, representing a combination of wild, domestic, ancient and modern horses. These include CGG10022 (wild from the Late Pleistocene), Batagai (wild from the mid-Holocene), as well as Mon_M_0065_LR_FM1798 (modern domestic) and Prz_M_0156_LO_SB281 (wild Przewalski's horse). We consistently found that, for reduced depths-of-coverage, the genetic load is overestimated (Fig. S29). This indicates that differences in deleterious loads between ancient and modern domesticates could be even more pronounced than actually observed (Fig. 2C).

Of note, we restricted this analysis to positions with a minimum coverage of 3-fold, which provides a reasonable trade-off between the number of analyzed positions and the certainty of their genotypes.

The genes showing significant differences in their genetic loads in Berel', Sintashta and Arzhan relative to modern domestic horses were submitted to a functional enrichment analysis, as detailed in '*Functional enrichment analyses*'. After correcting for multiple testing (110), we found that genes with an excess of genetic load in modern domestic horses were enriched in three functional categories, including 'dementia' (p -value = 0.0197), 'seizures' and 'abnormal seizure response to electrical stimulation' (p -value = 0.0390), as well as 'Other open wound of head and face' (p -value = 0.0402). This suggests that the increase in the deleterious load observed in the genomes of modern domestic breeds was not only driven by demographic collapses, but enriched in particular gene categories, likely reflecting an inadvertent hitchhiking of deleterious mutations through selective breeding.

Genome-wide selection scans. Serial sequence datasets catch evolution red-handed, allowing direct tracking of the genetic changes that accompanied the makeup of ancient and modern lineages. Several models have been implemented to explicitly exploit this temporal dimension (157). They are, however, often restricted to a single population (157), despite the increased power provided by the comparison of multiple populations in detecting adaptive signatures. With the exceptions of 3P-CLR (122) and PBS (121), which are merely limited to three populations, current cross-population methods (e.g. pairwise F_{ST} scans) are unable to polarize the direction of genetic differentiation, and consequently to identify the population targeted by positive selection. Currently available methods are, thus, poorly suited to our sampling scheme, which includes a mixture of modern and ancient genomes from multiple population backgrounds. To overcome these limitations, we developed a new framework that builds on low-to-moderate genome sequences, to identify positively selected regions along pre-defined branches of a population tree. This method, named LSD (Levels of exclusively Shared Differences), is presented here for the first time and can accommodate multiple (>2) populations and/or species. Of note, although suited to datasets including ancient genomes, it can also be applied to datasets consisting of present-day genomes.

Positive selection accelerates the fixation of genetic differences in the population it acts upon, yielding substitution rate accelerations in specific genomic regions (Fig. S30A). The number of genetic differences *exclusively shared* by the individuals of a population is expected, thus, to provide an excellent proxy for the selection coefficient (s). The number of population-specific genetic differences can be calculated by reconciling the population tree, which reflects population splits, with the local gene trees, which are built from particular genomic regions and summarize local evolutionary forces, potentially including positive selection.

Let suppose that we aim at identifying those genomic regions that have been selected in a given group of horses, labeled A. The first LSD step consists in locating on a local gene tree, the MRCA of group A and its sister clade, labeled B. Then, we calculate LSD as

$$LSD(A) = \frac{(D(MRCA, A) - \frac{d(A)}{2}) + (k(A, B) - \frac{d(A)}{2} - \frac{d(B)}{2}) - (D(MRCA, B) - \frac{d(B)}{2})}{2}$$

$$= \frac{D(MRCA, A) - d(A) + k(A, B) - D(MRCA, B)}{2}$$

where $D(MRCA, A)$ is the average distance (in branch length units) from the MRCA of A and B to individuals from group A, $d(A)$ is the average pairwise distance between individuals from A, and $k(A, B)$ the average pairwise distance between A and B individuals.

Fig. S30B provides a visual representation of the mathematical terms. $D(MRCA, A) - d(A)$, represented by the red arrow, accounts for the amount of differences *shared* among individuals from population A. Genomic regions where the red branch is long could potentially reflect the action of positive selection acting specifically on population A (Fig. S30A).

Due to incomplete lineage sorting (ILS) and/or admixture, however, some of these differences are not *exclusive* among individuals from A, but shared with B. Since

variants shared between populations are unlikely to underlie local adaptations, LSD removes them through the $k(A,B)$ term. The effect of $k(A,B)$ on LSD is shown in Fig. S30C, where $k(A,B) = 4$ in absence of ILS, but decreases to 1 when population A is not fully sorted. In the example with ILS, one individual from population A, labeled as A3, clusters within population B (Fig. S30C). LSD is maximized in situations of complete lineage sorting, and for local gene trees where the branch leading to population A has accumulated a large amount of fixed genetic differences.

According to its definition, changes in population B also contribute to $k(A,B)$. Nevertheless, population B is just considered to better identify exclusive allele changes in population A. To remove the impact of population B on $k(A,B)$, and keep $LSD(A)$ independent from population B, the last LSD operation involves subtracting $D(MRCA,B)$. By interpreting $LSD(A)$ as operations in the paths of the population tree (Fig. S30B), it can be intuitively shown that the lineage leading to population A has been counted twice, and thus requires to be finally divided by two.

LSD can be similarly applied to any other external population. For population B

$$LSD(B) = \frac{D(MRCA,B) - d(B) + k(A,B) - D(MRCA,A)}{2}$$

The method can be also extended to investigate the impact of positive selection in populations parental to the sampled ones, such as for instance, that leading to all A and B individuals (referred to as AB, and depicted in green in Fig. S30D). Note that, in this case, applying the previous formula

$$LSD(AB) = \frac{D(MRCA,AB) - d(AB) + k(AB,C) - D(MRCA,C)}{2}$$

would produce biased LSD estimates. This is so because $d(AB)$ includes $d(A)$ and $d(B)$, which do not reflect evolutionary forces operating on the parental population AB, but within A and within B. The above formula would create a spurious LSD correlation between parental and descendant populations, where episodes of positive selection occurring in A and/or B groups would be partially attributed to AB. To make parental and descendant lineages fully independent, the LSD estimation in the internal branches of a population tree requires a simple modification, where $d(AB)$ is replaced by $k(A,B)$, as illustrated in Fig. S30D. The calculations relative to AB must also be averaged over the A and B subclades, to correct for possible unbalanced sample sizes:

$$LSD(AB) = \frac{\frac{D(MRCA,A) + D(MRCA,B)}{2} - k(A,B) + \frac{k(A,C) + k(B,C)}{2} - D(MRCA,C)}{2}$$

LSD corresponds to the allele frequency changes exclusively shared by a group of individuals, while accounting for the hierarchical structure of the metapopulations (Fig. S30D). As F_{ST} , LSD can be also normalized:

$$LSD_{norm}(AB) = \frac{\frac{D(MRCA,A) + D(MRCA,B)}{2} - k(A,B) + \frac{k(A,C) + k(B,C)}{2} - D(MRCA,C)}{\frac{D(MRCA,A) + D(MRCA,B)}{2} + \frac{k(A,C) + k(B,C)}{2} - D(MRCA,C)}$$

to provide an unbiased quantification of the LSD *fraction* shared amongst all AB individuals. The LSD_{norm} statistics is conceptually similar to the PBS developed by Yi and colleagues (121), excepting that it is not merely restricted to three populations, and can thus investigate episodes of positive selection in the internal branches of a population tree. Positive values indicate genetic differentiation, while regions with negative LSD scores are likely to represent ancestral substructure or tracts introgressed from an outgroup population.

Assessing the performance of LSD_{norm} using forward simulations. A desirable property of any summary statistics aimed at detecting positive selection is to scale with the selection coefficient (s). In order to test this, we used SLiM v2.0 (123) to simulate eight different population histories, including various selection scenarios under a wide range of s values. Values of fitness $(1 + s) = 1$ correspond to $s = 0$, and thus to neutrality. Values lower and higher than one imply negative and positive selection, respectively. The following parameters were fixed in our simulations:

- 500 replicates per condition
- regions of 10 kb
- recombination rate = 1×10^{-8} crossovers/(generation x bp)
- mutation rate = 1×10^{-8} mutations/(generation x bp)
- population tree as provided in Fig. S31.

The population tree reflects current knowledge on the equine evolutionary history, and assumes a 4-4.5 myr divergence between horses and donkeys (38) and a ~150 kyr divergence between a now-extinct lineage of wild pre-domestic horses (14, 15). The last 300 generations of divergence between populations A and B approximate the time difference between Berel' and modern domestic horses (~2,300 years assuming a generation time of 8 years). We simulated the following scenarios (Fig. S32):

Scenario 1: It includes 300 generations of positive selection (PS) in population A (Berel'), since the split from population B (present-day domestic horses) to the sampling time, as depicted by the blue area in Fig. S32. The remaining part of the tree topology is assumed to evolve under neutrality.

Scenario 2: It assumes 300 generations of PS in the branch directly ancestral to populations A and B, just prior to their split, as depicted by the by the red area in Fig. S32. The remaining part of the tree topology is assumed to evolve under neutrality.

The multiple sequence alignments resulting from simulations were used to estimate the local gene trees with the FastME program (115), following the same procedure as that used for the empirical data set (see '*Genome-wide selection scans*'). The resulting LSD_{norm} estimates are shown in Fig. S33 in blue for population A, and red for the lineage directly ancestral to populations A and B.

Fitness values lower than one equate to negative selection, implying that derived alleles are eliminated, or alternatively that ancestral alleles are selected. Consequently, the signature of negative selection (NS) resembles that of positive selection, whenever

acting on a genetic background of neutrally evolving alleles. This leads to a “V-shaped” LSD_{norm} profile for population A, with a valley around neutrality (Fig. S33). The results show that in both, scenarios 1 and 2, LSD_{norm} scores scale asymptotically with the fitness of mutations. In contrast, the non-selected lineage tends to be flat, indicating that the method can accurately encapsulate the episode of PS in a branch-specific manner.

The lineage parental to populations A and B in scenario 1 deserves further attention as it shows LSD_{norm} values departing from zero, despite evolving under neutrality. This is so because even in the absence of positive selection, genetic drift can result in genetic differentiation, which is captured by LSD_{norm} as higher-than-zero values. Absolute LSD_{norm} values, thus, depend on the full demographic history of a given population. This implies that, as a proxy for selection coefficients, LSD_{norm} can be compared among loci within the same population, but not between populations with different evolutionary histories, as is also the case for selection scans based on F_{ST} values.

In scenario 1, the red line also shows an unexpected peak around neutrality (Fig. S33). As demonstrated by scenario 8 (see below), this likely reflects situations in which two descending branches experience similar changes (at this particular fitness = 1 condition, both descending lineages are evolving under neutrality). Part of this parallel evolution is erroneously attributed to their ancestral branch, creating a spurious signal of acceleration.

Although illustrative, scenarios 1 and 2 conveniently assume neutral evolution, except when population A (or its parental) lineage undergoes positive selection. As this might be unrealistic, we explored six additional scenarios. These consider that novel mutations appearing along the entire population history can have slightly or mildly deleterious fitness effects, except for a very particular time period in which they are favored by means of positive selection, or purged out by means of negative selection. For these six scenarios (scenarios 3-8, Fig. S32), we assumed that new mutations have a coefficient of selection drawn from an exponential distribution with parameter -0.1.

Scenarios 3 and 4 are identical to scenarios 1 and 2, respectively, except for periods when mutations are assumed to be slightly deleterious instead of neutral throughout most of the evolutionary history.

Scenarios 5 and 6 are identical to scenarios 3 and 4, but we assumed that PS only lasted for 100 generations (instead of 300). After this period, NS acted against the derived allele during 200 generations prior to sampling (scenario 5), or during 800 generations until the split of populations A and B (scenario 6) (Fig. S32). As reflected by the lower LSD_{norm} values in scenario 5 (and 6) than in scenario 3 (and 4), this selective regime created a weaker impact on the DNA variation. Nonetheless, LSD_{norm} appropriately scales with the range of selection coefficients.

Scenario 7: PS jointly acts in population A and its ancestral branch, following the same selective regimes as those simulated for scenarios 5 and 6. For fitness values larger than two, LSD_{norm} scores calculated for population A become lower than in scenario 5, Fig. S33). This suggests that adding an ancestral episode of positive selection can slightly interfere in those inferred for population A.

Scenario 8: To further evaluate the consequences of multiple PS episodes on LSD, we simulated the most confounding scenario, where exactly the same allele is promoted in population A and population B. This scenario of converging evolution affects LSD_{norm}

by attributing the selective episode to the parental branch (red line, Fig. S33). This is mostly due to inaccurate gene tree reconstructions.

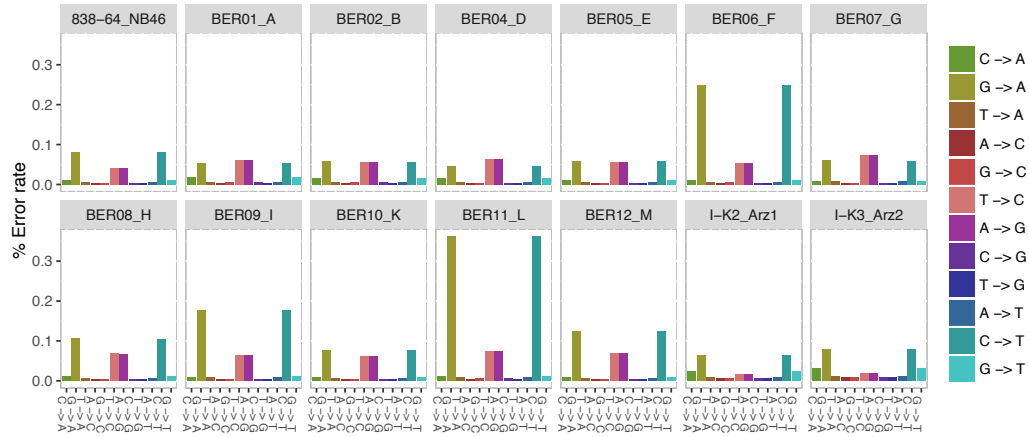


Fig. S1.

Error rates for the 14 ancient horse genomes, after mapDamage2.0 trimming/base quality rescaling (see mapDamage2.0 parameters in ‘*DNA extraction and genome sequencing*’) and filtering (see ANGSD filters in ‘*Genotype probabilities vs. genotype calls*’). Error rates were inferred separately for each type of nucleotide substitution, revealing increased error rates for transitions, known to accumulate most in ancient sequences as a result of post-mortem DNA damage. BER11_L shows high error rates, in line with a larger contribution of DNA sequence reads generated from non-USER treated libraries for this sample (19.5%) compared to the other samples (0.0-11.2%) (Table S1).

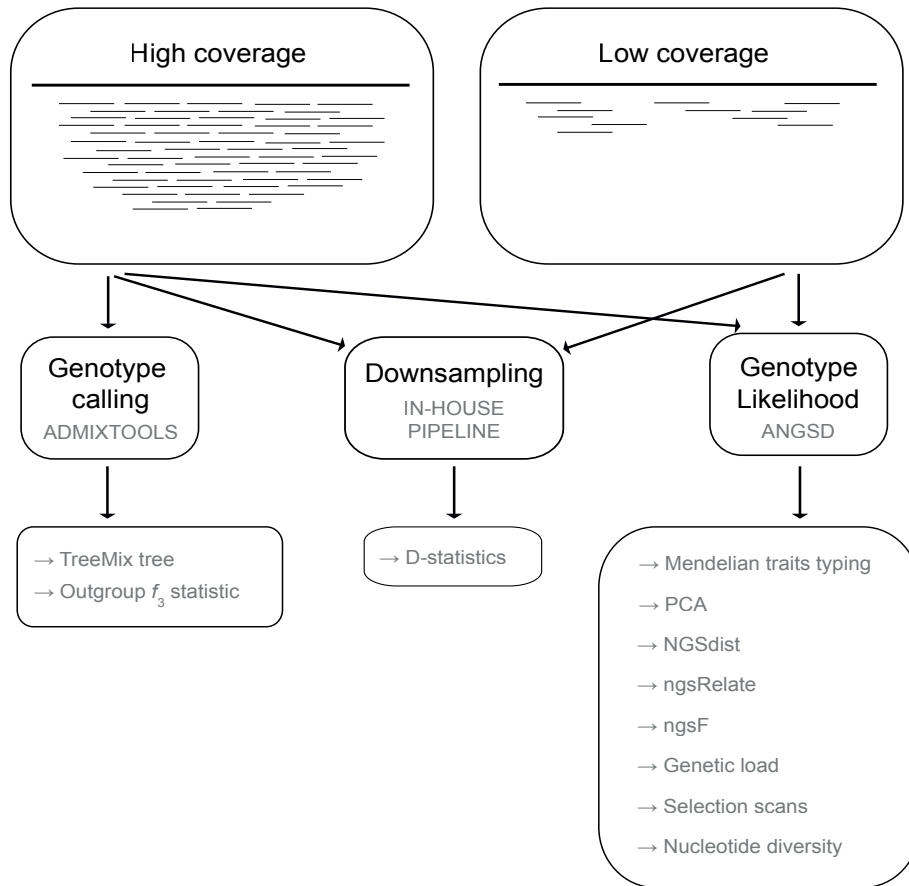


Fig. S2.

Methodological framework applied for the comparative genomics analyses. Analyses based on genotype probabilities integrate inferences over the uncertainty of the three possible genotypes (AA/Aa/aa). The TreeMix population tree and the output- f_3 statistics were inferred after explicit genotype calling, but restricting to a subset of high-quality biallelic sites.

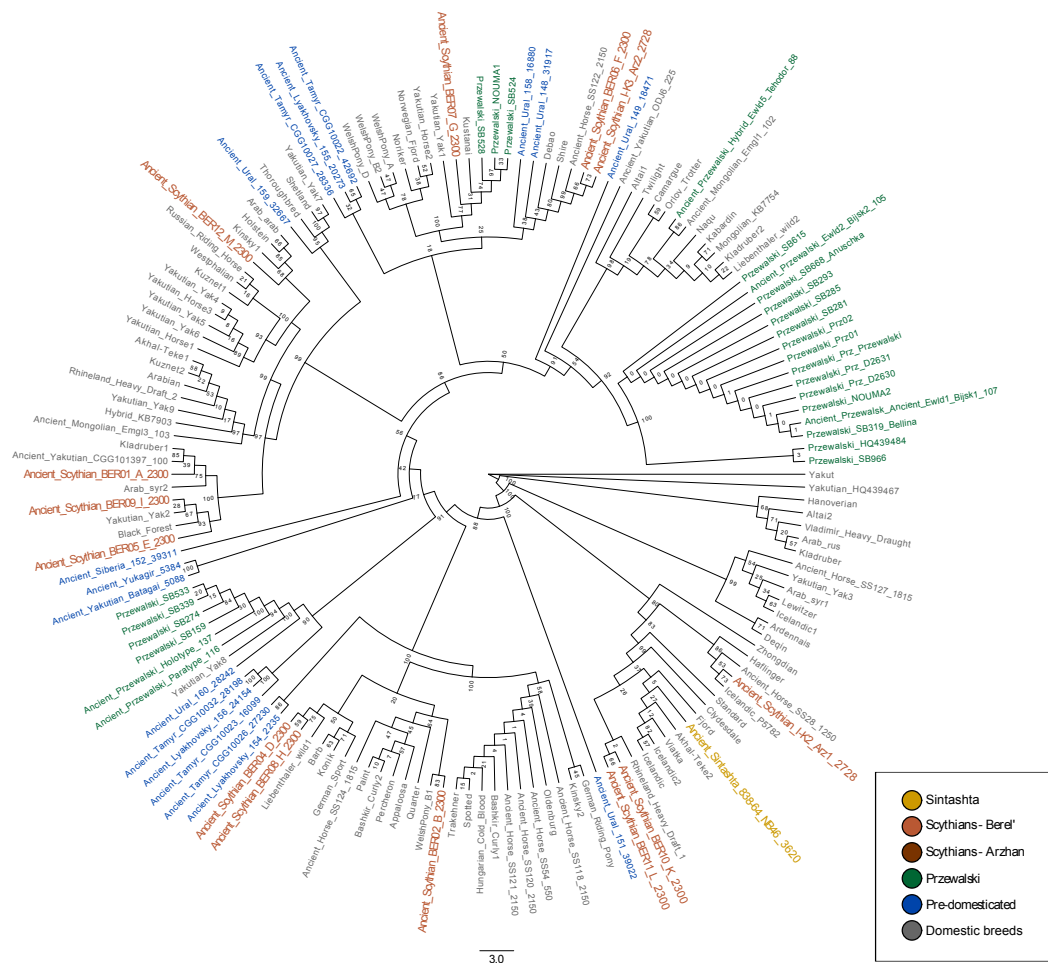


Fig. S3. Maximum likelihood phylogeny of the mitochondrial genomes, as estimated by PhyML 3.0 (54). Node labels show bootstrap support values. Horse labels are color-coded according to their respective breed/population.

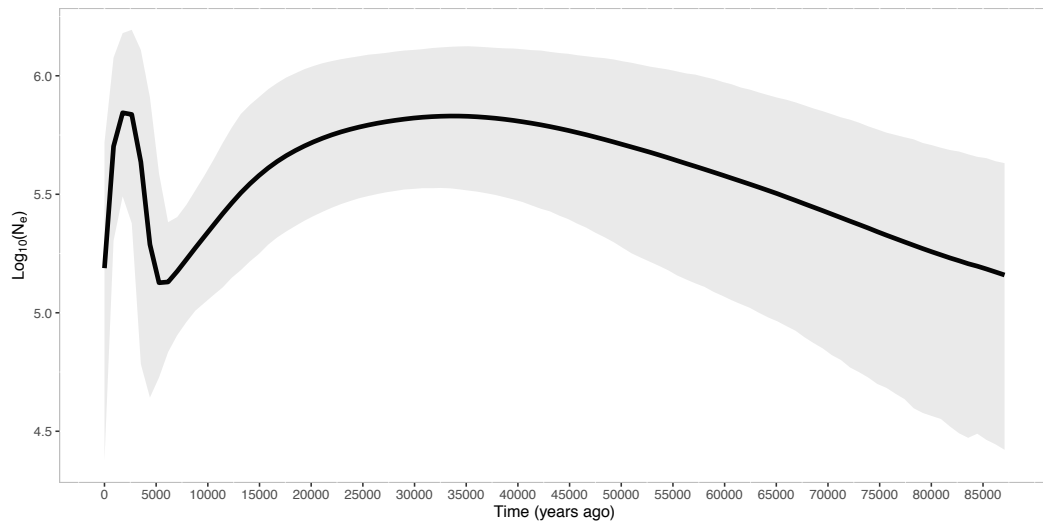


Fig. S4.

Bayesian Skyline Plot (BSP) showing changes in the effective population size of horses over the last 85,000 years, as estimated by BEAST 1.8.2 (57) from mitochondrial data. The shaded area depicts the confidence interval around the mean posterior probability. The BSP was inferred from 106 modern and 48 ancient mitochondrial genomes (Tables S3 and S4).

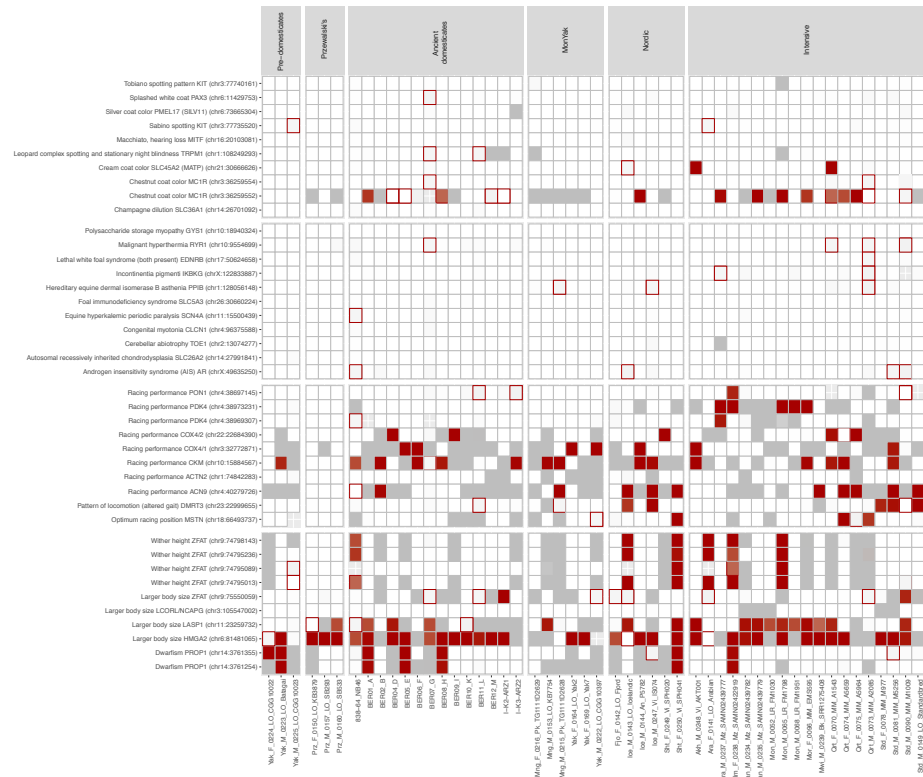


Fig. S6.

Heatmap where the gradient color represents the probability that a horse (x-axis) carries the allele causative of one of the 41 phenotypic traits (y-axis). Absence of the allele is represented in white, and its presence as homozygous is shown in red. Intermediate grey colors symbolize heterozygous sites, except for cells surrounded by a red frame (coverage < 3), where genotypes are uncertainly determined. Grey crosses indicate non-genotyped positions.

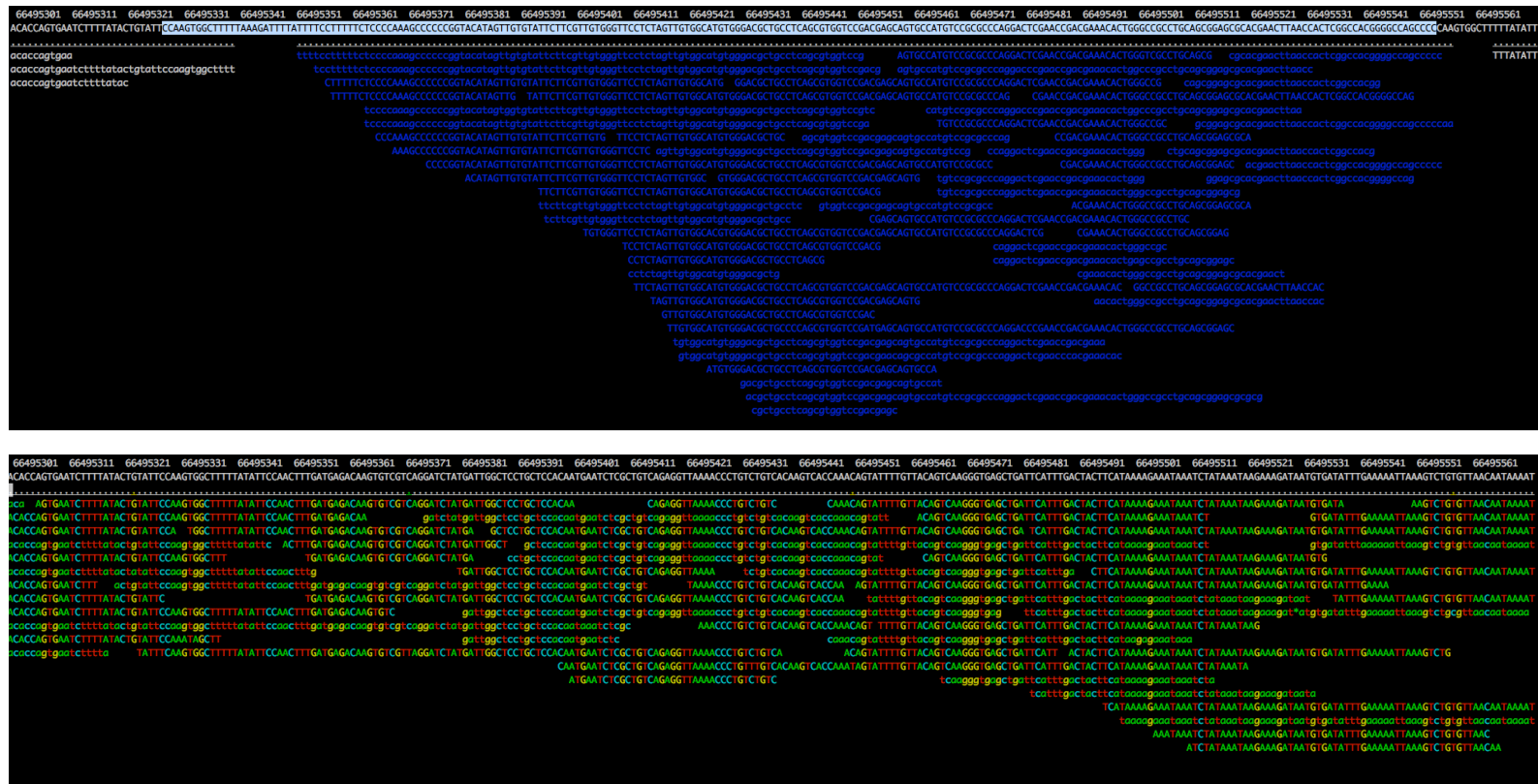


Fig. S7.

Samtools *tview* profiles showing reads mapping against the BER06_F genome, at the *Myostatin* (*MSTN*) promoter. In the top panel, reads mapping against the reference sequence including the *in silico* SINE insert. The highlighted consensus sequence corresponds to the 227bp SINE insertion. Reads mapping against the insert were not filtered out just to show their mapping quality below nine, as coded by their blue color, and as expected for reads mapping against a repetitive element. No reads span, and thus validate, the region centered at the insert site breakpoints. In the bottom, reads mapping against the *MSTN* promoter, without the *in silico* SINE insertion. Similar profiles were observed in the sequencing data underlying all remaining 13 ancient horse genomes.

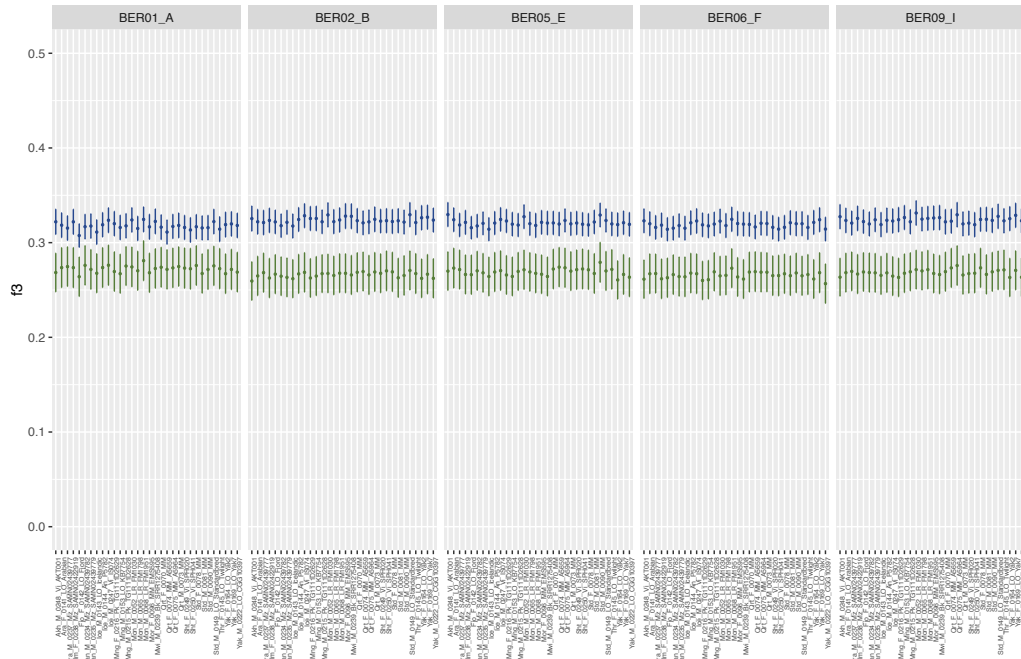


Fig. S8.

Genetic drift shared between Berel' and modern-day domestic horses, measured via the outgroup- f_3 statistic, in the form $f_3(\text{Berel}', \text{modern-day domestic}; \text{outgroup})$. Only high-quality Berel' genomes (minimum depth-of-coverage of 7-fold) were compared to modern-day domestic horses. Two outgroup genome sequences were considered, namely a horse from the mid-Holocene, Batagai (in blue), and the Przewalski's horse SB293 (green). Lower red f_3 values show that Przewalski's horses are phylogenetically closer to Berel' and modern domestic animals than pre-domestic horses.

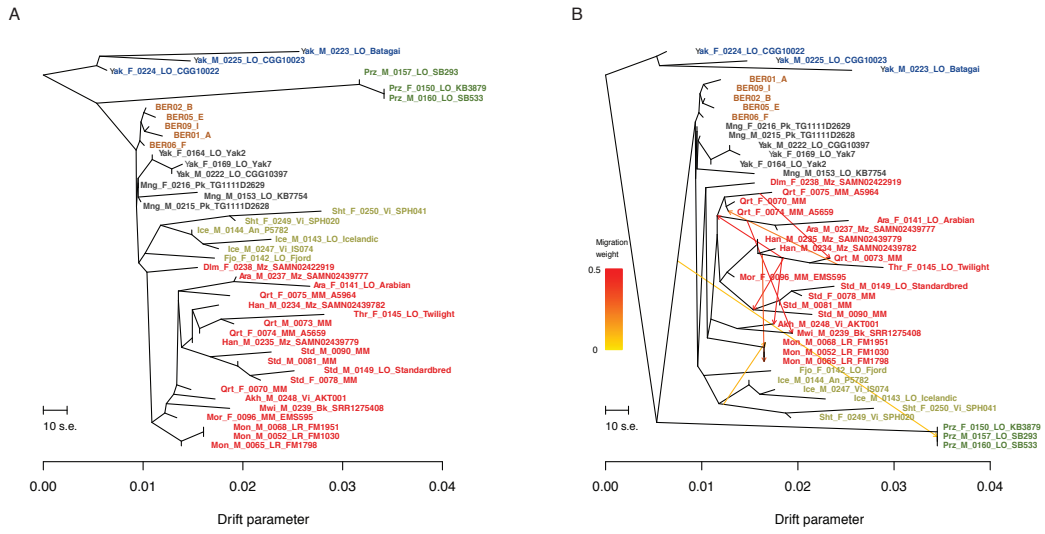


Fig. S9. TreeMix tree with zero (A) and nine (B) migration edges (Table S10). The genomes of the three pre-domestic horses (Yak_F_0224_LO_CGG10022, Yak_M_0225_LO_CGG10023 and Yak_M_0223_LO_Batagai) were used as outgroups to root the tree. Branch lengths are measured in drift units, and arrows in (B) indicate potential admixture events.

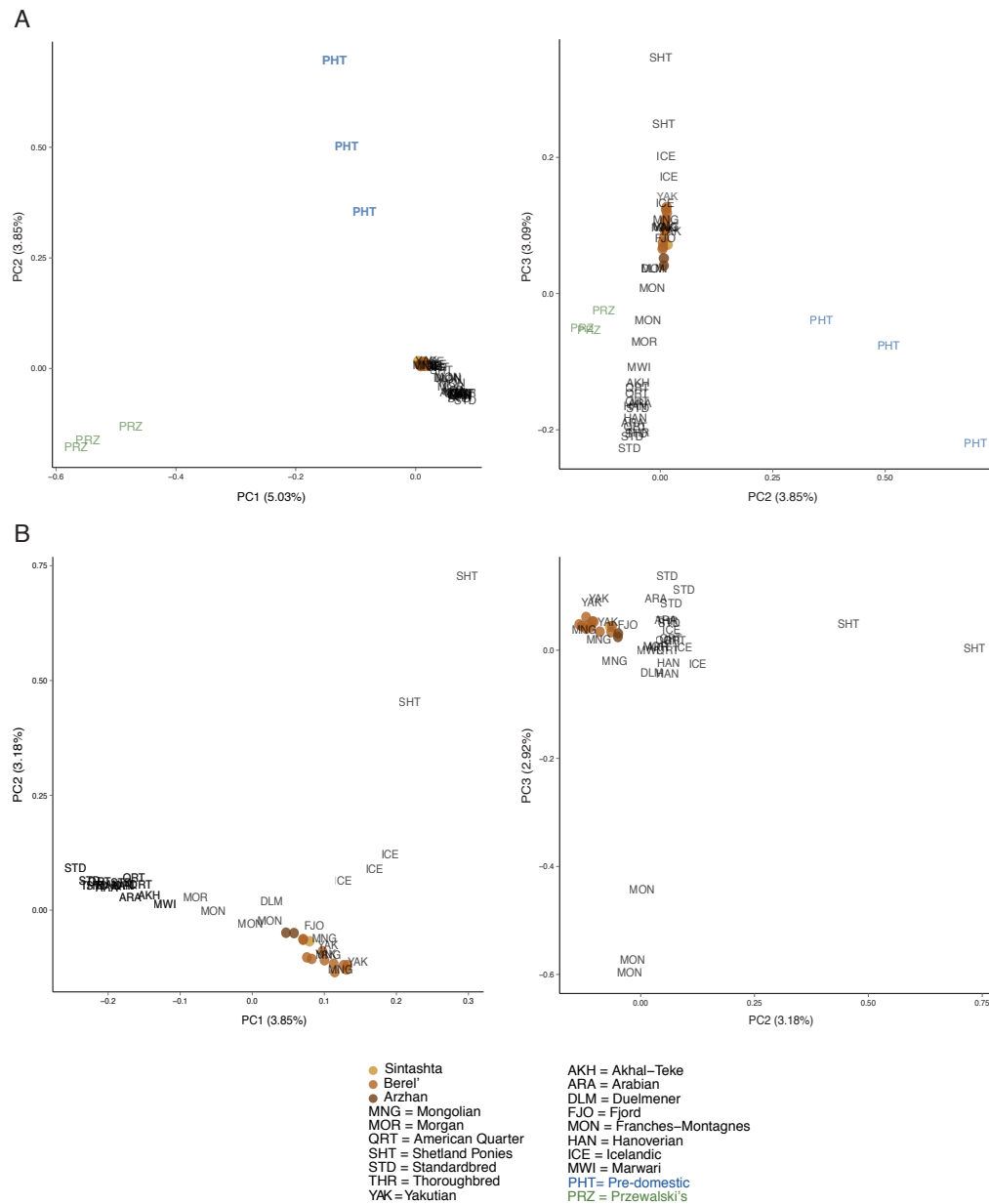


Fig. S10.

Principal Component Analysis (PCA) based on genotype likelihoods, with (A) and without outgroup individuals (B). Sintashsta, Berel' and Arzhan horses are illustrated as color-coded dots. The comparative genome panel is labeled following a three-letter code.

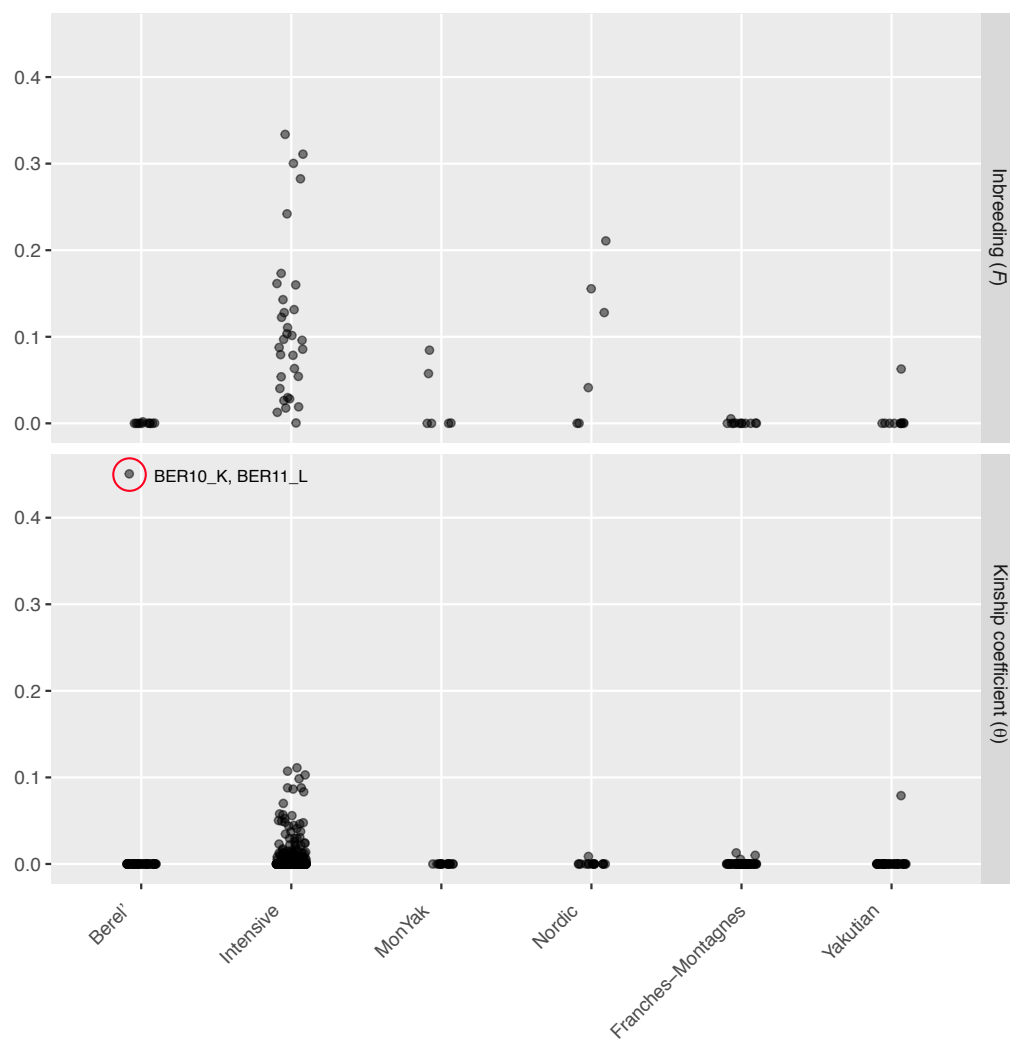


Fig. S11.

Inbreeding estimates (F) and kinship coefficients (θ) estimates calculated with ngsF (17) and ngsRelate (18), respectively. Each point in the top panel represents a single individual, while a pairwise comparison in the bottom panel. We highlighted the BER10_K-BER11_L pairwise comparison showing that these two horses share a high proportion of identity-by-descent tracts.

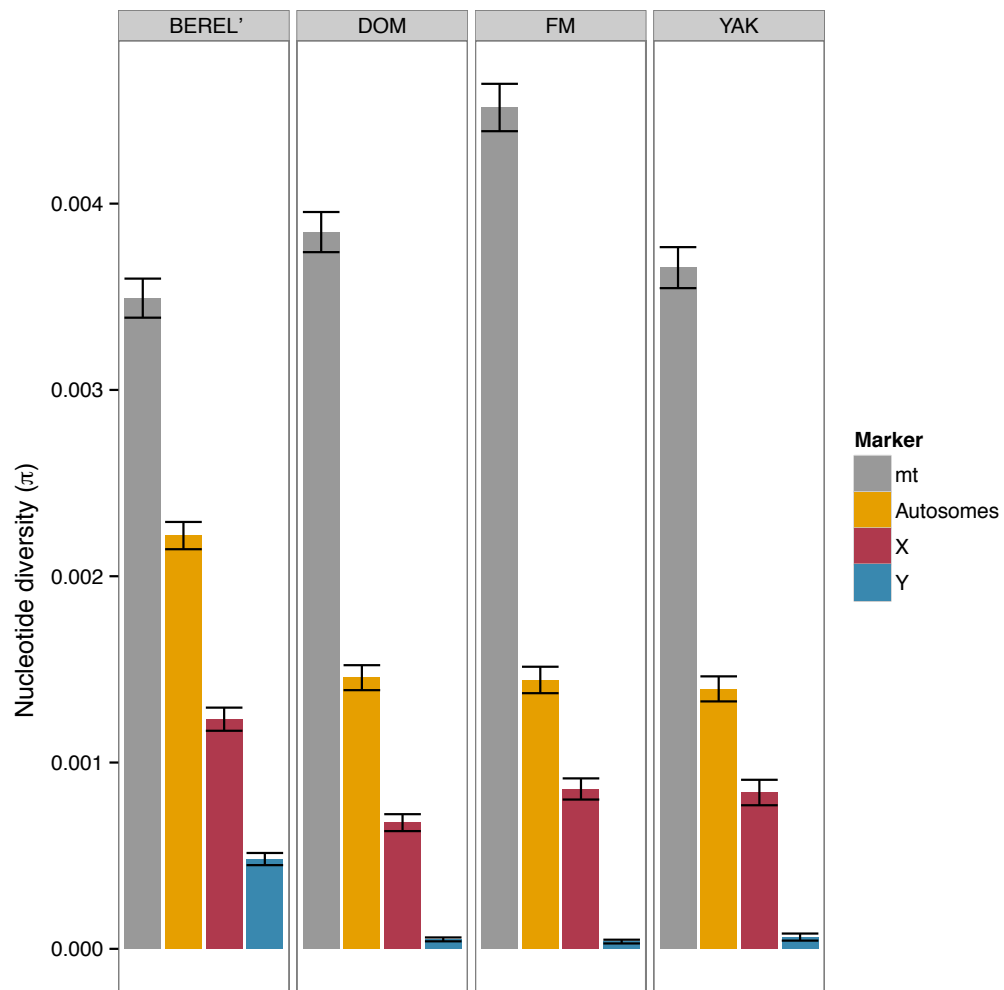


Fig. S12.

Nucleotide diversity across markers with different modes of inheritance. Berel' nucleotide diversity is compared to those estimated for Franches-Montagnes (FM) and Yakutian (YAK) modern breeds, as well as for a merger of *MonYak*, *Intensive* and *Nordic* of domestic horses (DOM).

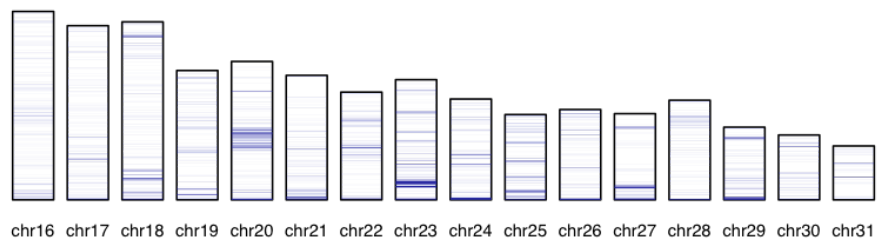
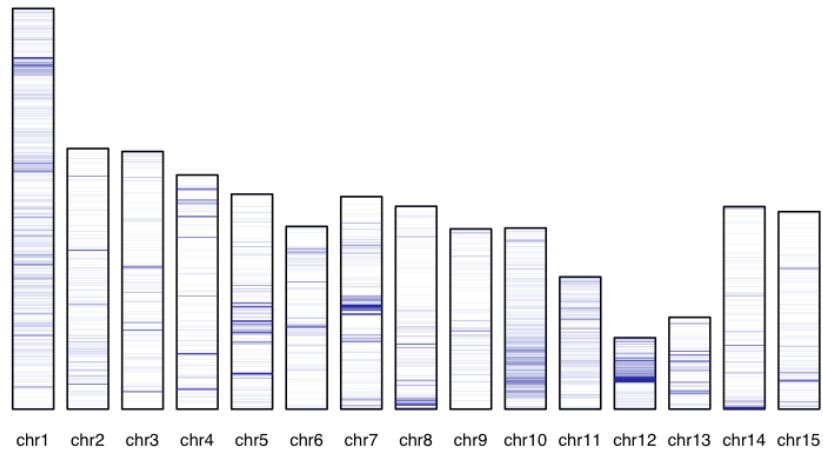


Fig. S13.

Chromosomal map of Copy Number Variants (CNVs) that are present in at least one of the 28 analyzed horses (Table S13). All blue regions, representing CNVs, were excluded from functional enrichment tests (see '*Functional enrichment analyses*').

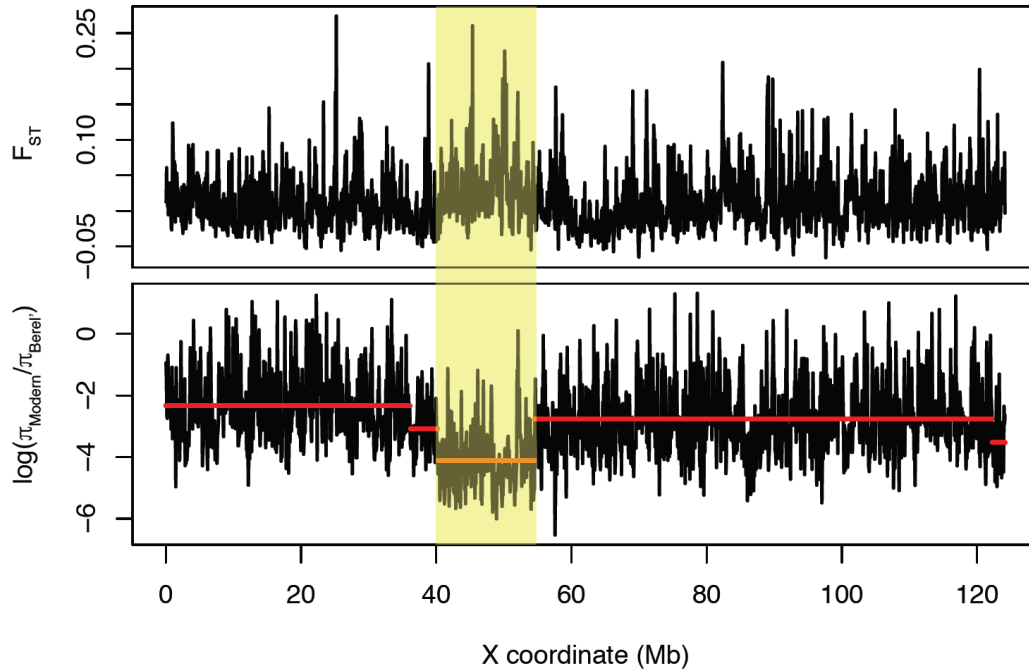


Fig. S14.

F_{ST} and relative levels of nucleotide diversity ($\log(\pi_{\text{Modern}}/\pi_{\text{Berel}'})$), calculated from Berel' and present-day stallions. The average $\log(\pi_{\text{Modern}}/\pi_{\text{Berel}'})$ value, consistently lower than zero, indicates an overall decay of X-linked diversity during the last 2,300 years. Discontinuities in the red line (bottom panel) depict changes in the $\log(\pi_{\text{Modern}}/\pi_{\text{Berel}'})$ mean, as partitioned according to the changepoint package of the R programming language. The yellow area highlights the exacerbated $\log(\pi_{\text{Modern}}/\pi_{\text{Berel}'})$ decay in the X:40,375,001-53,375,001 region.

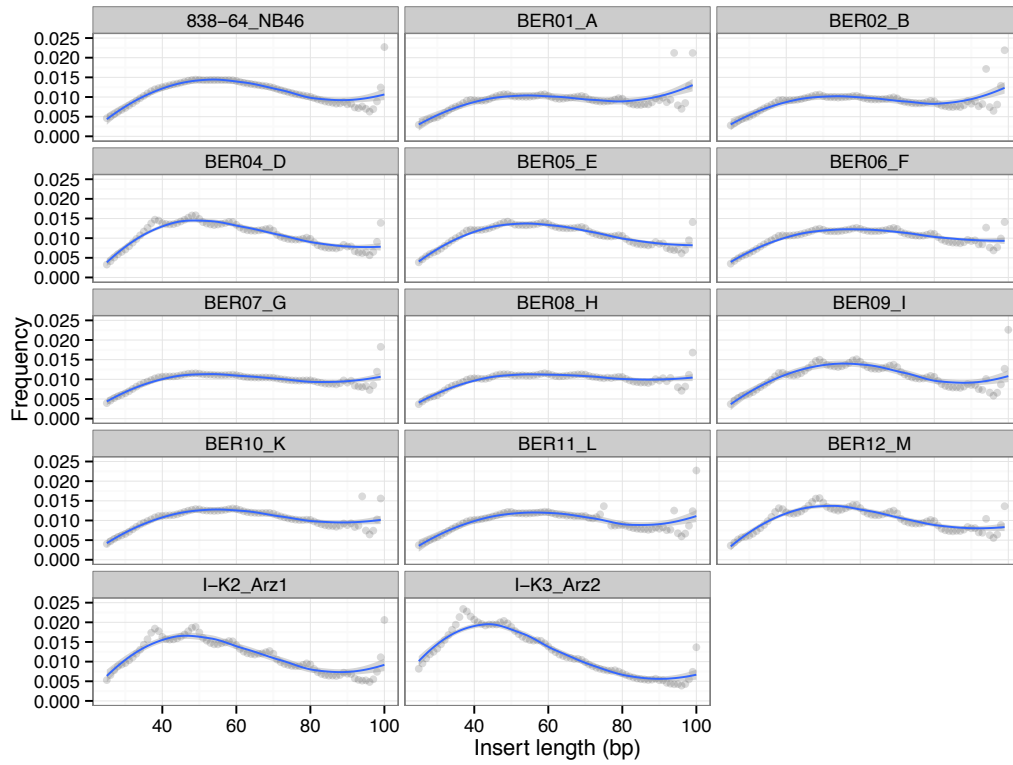


Fig. S15.

Raw (grey) and smoothed (blue) read length distributions in ancient samples. The average values much lower than the Illumina 2500 standard result from post-mortem DNA fragmentation. The 10-bp periodicity delineated by the grey line likely results from differential DNA fragmentation due to nucleosome protection.

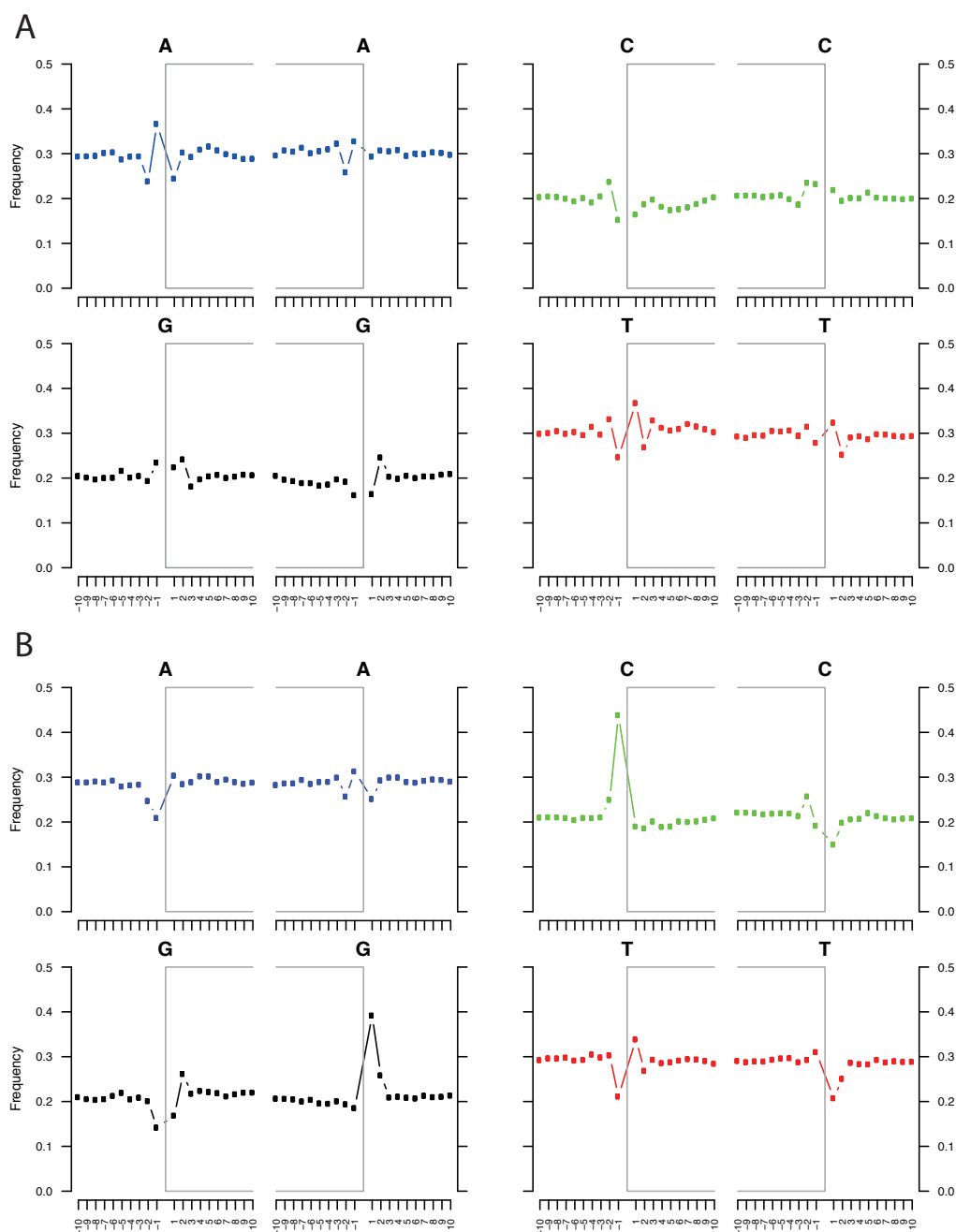


Fig. S16.

DNA fragmentation patterns for a non-USER (top panel, BER06_F_E10.Dab2nd_NOuser_CAGCTA) and a USER treated library (bottom panel, BER06_F_SCY2.1_user_TGAACA). The 10 bp preceding read starts (positions -1 to -10 on the left panels) and the 10 bp following read ends (positions 1 to 10 on the right panels) are shown.

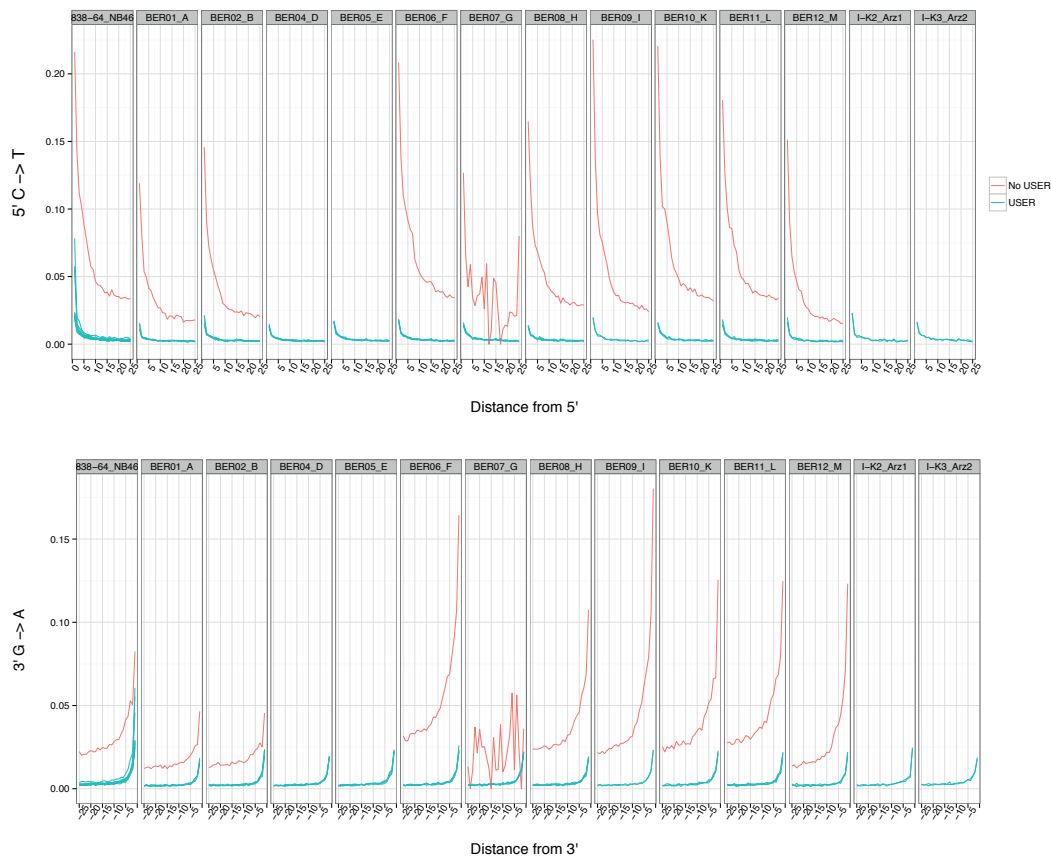


Fig. S17.

C→T and G→A mis-incorporation patterns at the 5'- and 3'-read ends, along the first and last 25 read positions of USER (turquoise) and non-USER (red) treated libraries. The erratic trend in the non-USER treated library of BER07_G is due to the low amount of non-treated reads (1,435) (Data File S2), which precludes accurate inference of DNA damage parameters.



Fig. S18. Posterior distributions for deamination patterns in single- (δ_S) and double- (δ_D) stranded DNA, as well as length of the overhangs (as given by the proxy $1/\lambda-1$). USER and non-USER libraries are shown in turquoise and red, respectively.

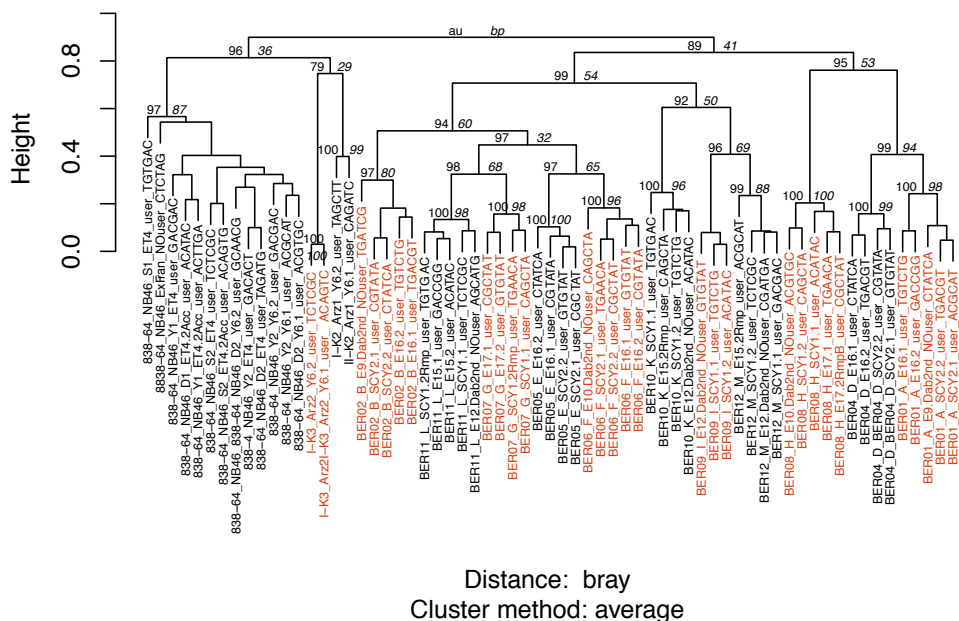


Fig. S19.

Hierarchical clustering of microbial DNA profiles obtained at the genus level for the libraries generated from the Sintashta, Arzhan and Berel' horses. The clustering is based on Bray-Curtis distances, and its support evaluated through 10,000 bootstrap replicates. "au": approximately unbiased p-value and "bp": bootstrap probability, both in %.

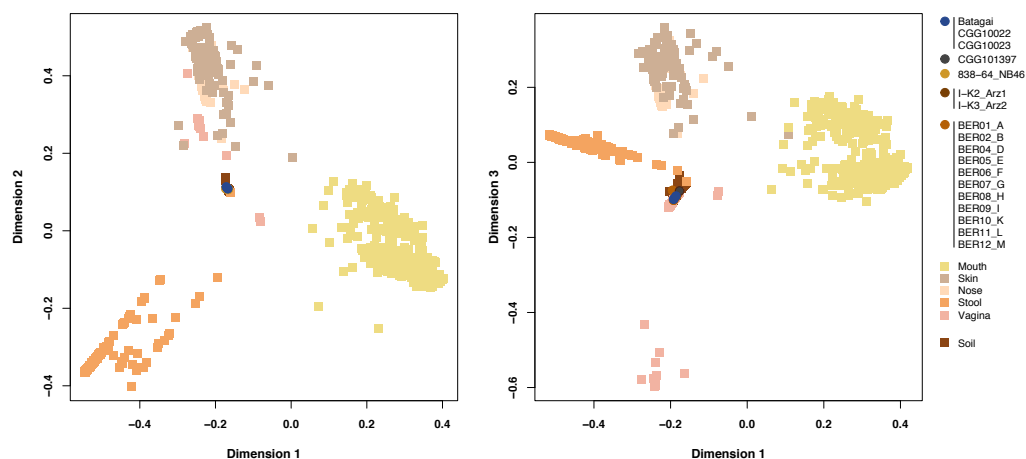


Fig. S20.

Principal Coordinate Analysis (PCoA) of microbial DNA profiles at the genus level in various human associated microbiomes, soils, and horse DNA extracts, including the horses from Sintashta, Arzhan and Berel' sites (this study), as well as pre-domestic horses (CGG10022, CGG10023 and Batagai) and from the 19th century (CGG101397). The PCoA is based on Bray-Curtis distances.

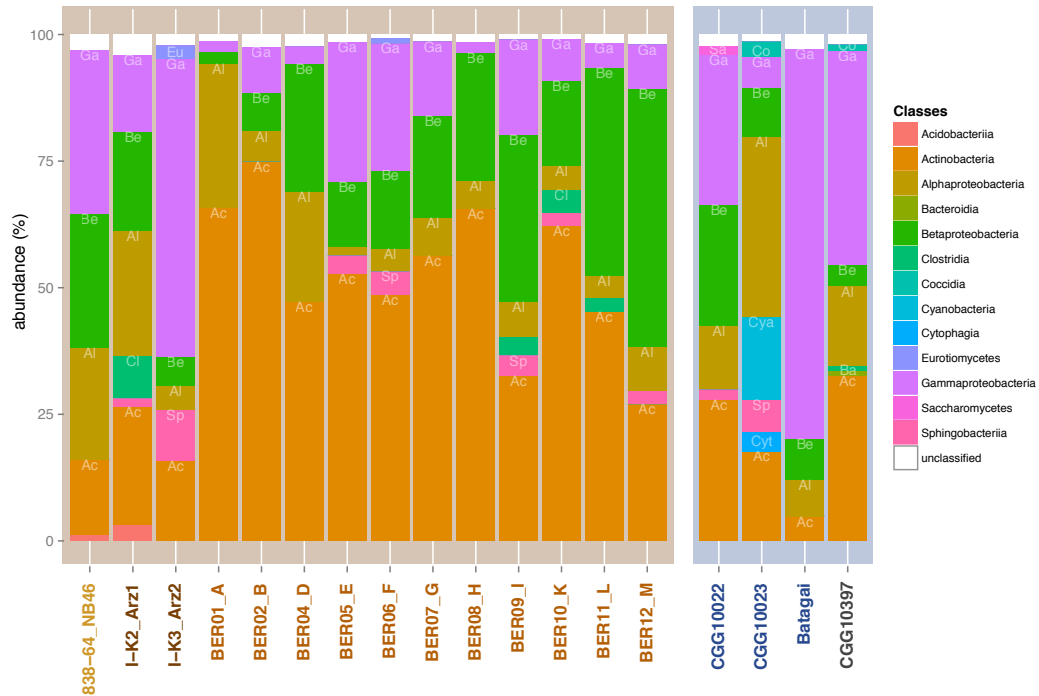


Fig. S21.

Stacked barplot summarizing the relative abundances of microbial classes in ancient horse DNA extracts. Horses from the Sintashta, Arzhan and Berel' sites, from the Late Pleistocene (CGG10022 and CGG10023), the early Holocene (Batagai) and from the 19th century (CGG101397). See Tables S3 and S4 for sample dates.

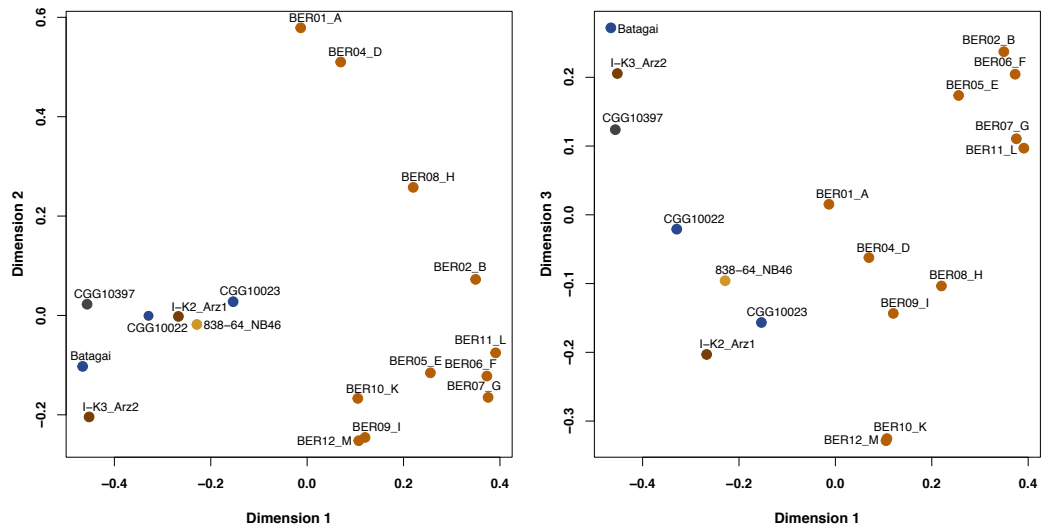


Fig. S22.

Principal Coordinate Analysis (PCoA) of Bray-Curtis distances between microbial DNA profiles at the genus level in horses from the Sintashta, Arzhan and Berel' sites, from the Late Pleistocene (CGG10022 and CGG10023) and the early Holocene (Batagai) and from the 19th century (CGG10397).

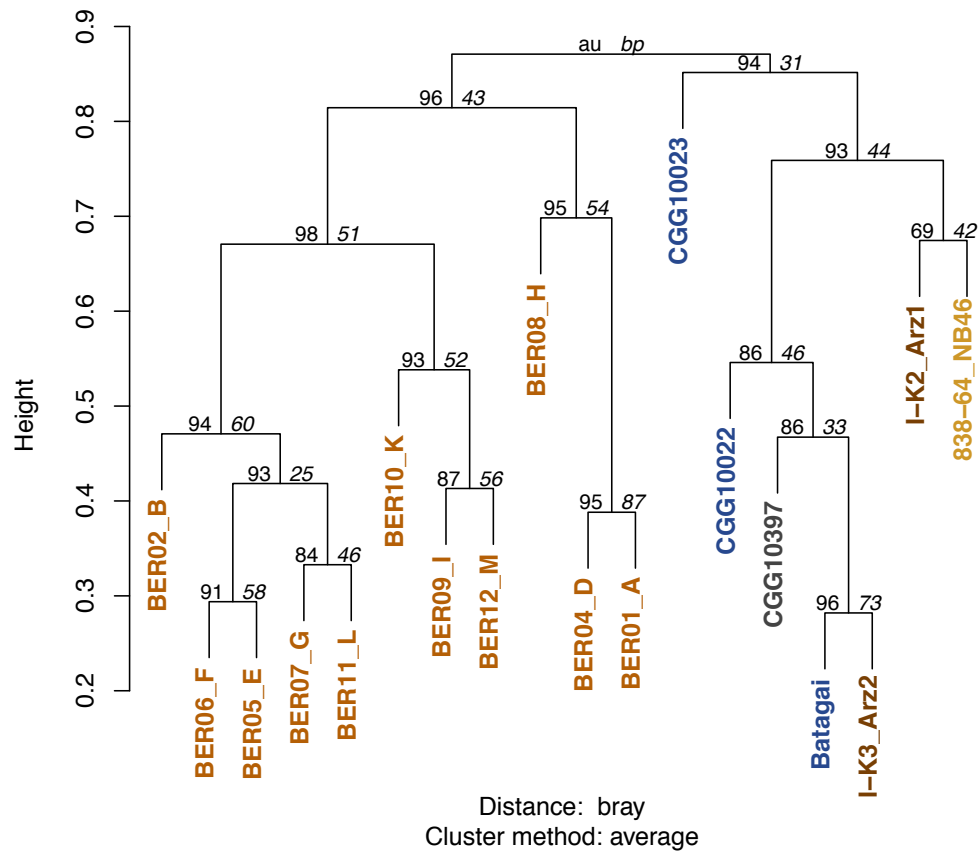


Fig. S23.

Hierarchical clustering of Bray-Curtis distances between microbial DNA profiles. Clade support is evaluated through 10,000 bootstrap replicates. “au”: approximately unbiased p-value and “bp”: bootstrap probability, both in %. Pre-domestic horses CGG10022, CGG10023 and Batagai, as well as the one from the 19th century, CGG101397, are included for comparison.

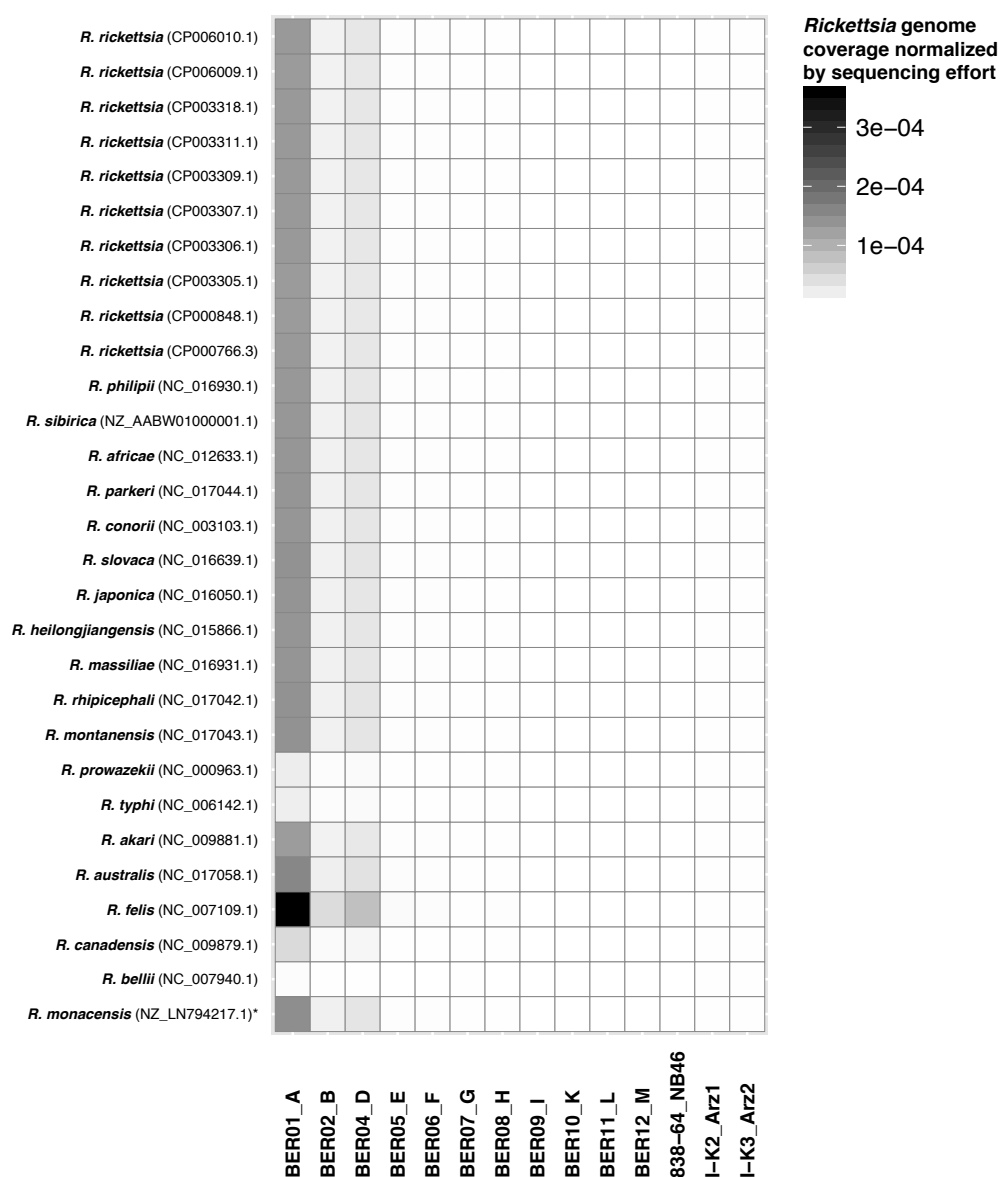


Fig. S25.

Reads mapping to representative *Rickettsia* *sp.* complete genomes. Genbank accession numbers for the *Rickettsia* reference genome sequences are given in parentheses. Genomes are ordered according to the *Rickettsia* core genome phylogenetic tree in (47) except for the additional genome sequences marked by “*”. The raw coverage was normalized by dividing by the sequencing effort (number of reads sequenced for each horse sample).

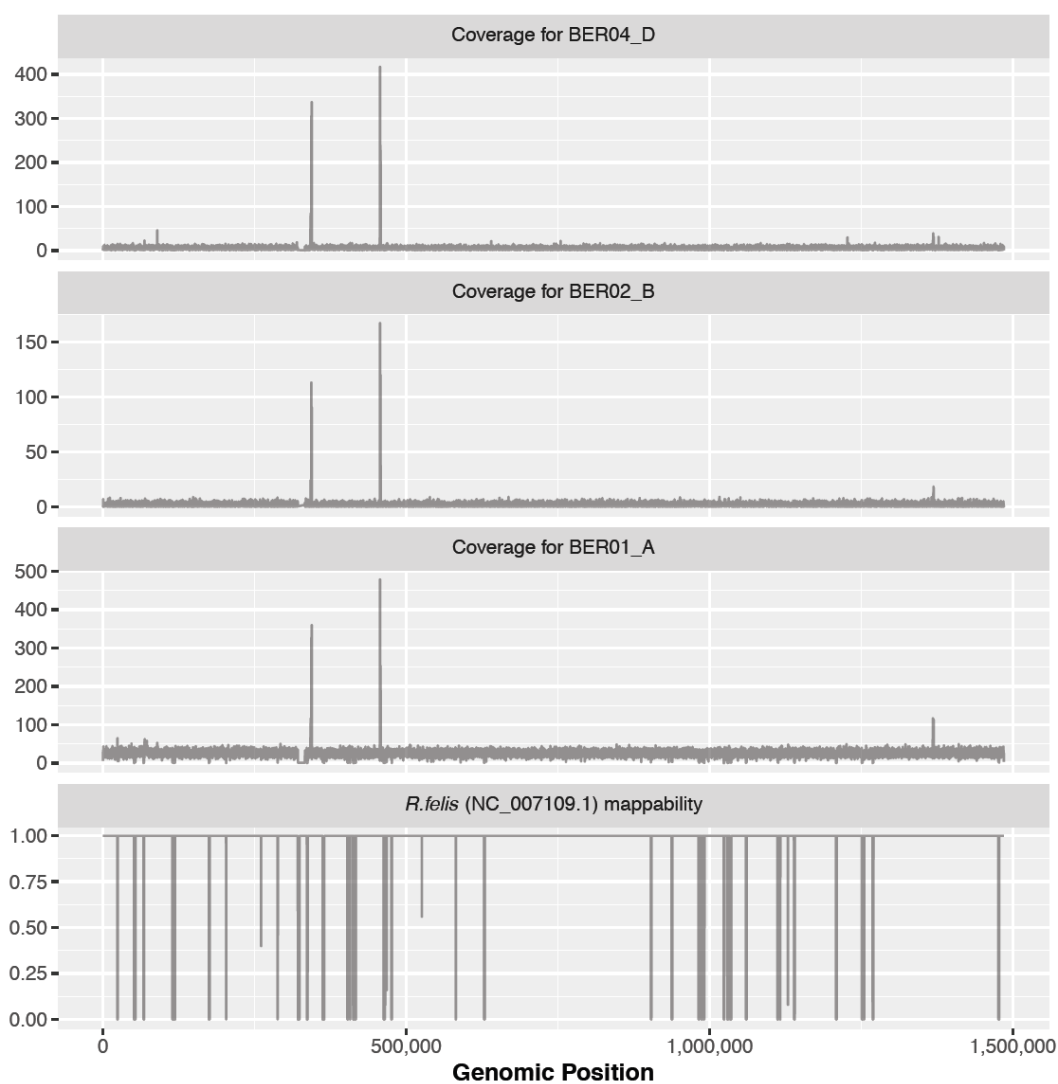


Fig. S26.

Depth-of-coverage along the *R. felis* complete genome (NC_007109.1), as recovered from BER01_A, BER02_B and BER04_D DNA extracts, and averaged in 25 bp windows. Mappability was calculated in 50 bp windows with step size of 25 bp, following the method described in (38).

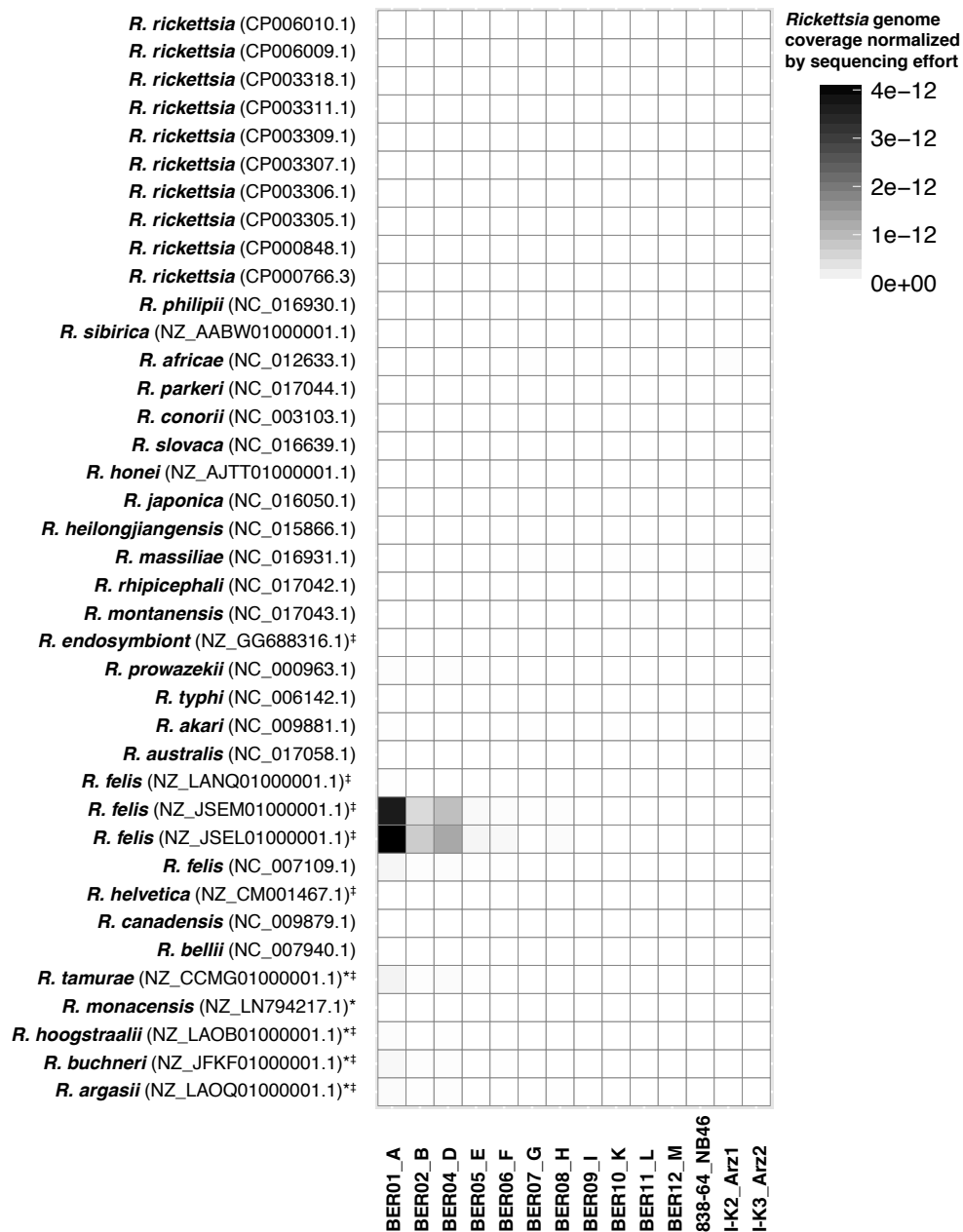


Fig. S27.

Reads mapping to representative *Rickettsia* sp. complete genomes and contigs. Genbank accession numbers for the *Rickettsia* reference sequences are given in parentheses. “†” indicates that the reference sequences are contigs and not complete genomes. The “corrected coverage” accounts for reads uniquely and perfectly mapping to only one single *Rickettsia* genome sequence, divided by the sequencing effort (number of reads sequenced for each horse sample).

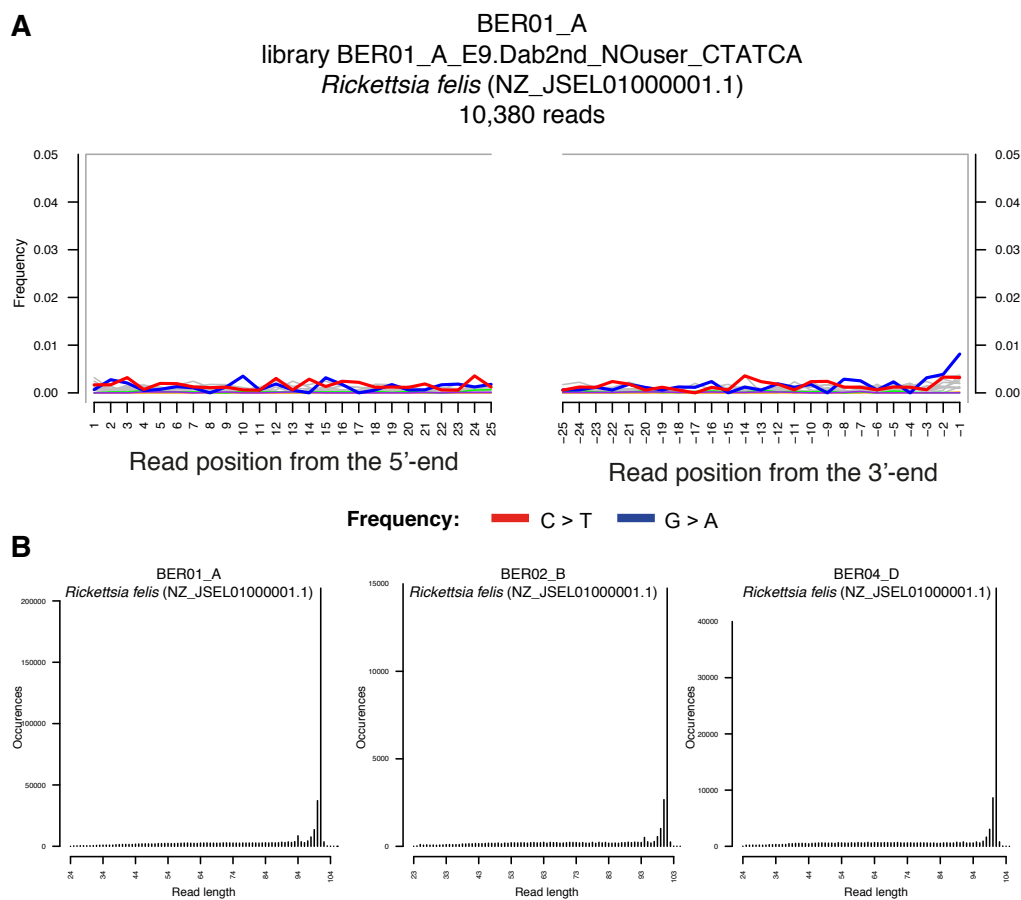


Fig. S28.
C→T and G→A mis-incorporation patterns at the 5' and 3' read ends, respectively, and fragment size distributions for *R. felis* DNA (NZ_JSEL01000001.1).

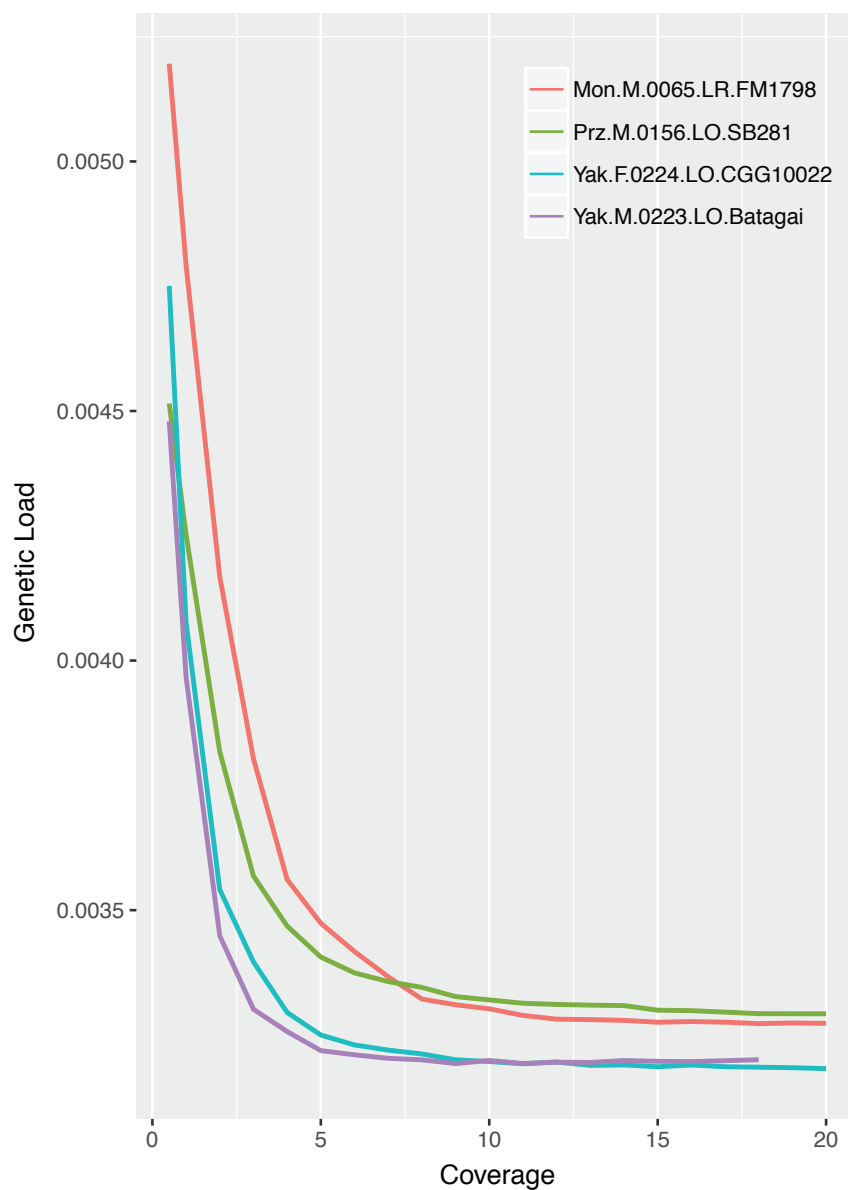


Fig. S29.

Estimates of genetic load as a function of sequencing coverage. The resulting relation shows how genetic load is overestimated for low sequencing coverages. This pattern is consistent across the four genomes that were computationally downsampled, namely Batagai, CGG10022, SB281 and FM1798, which are representative of ancient pre-domestic, wild and modern domesticates.

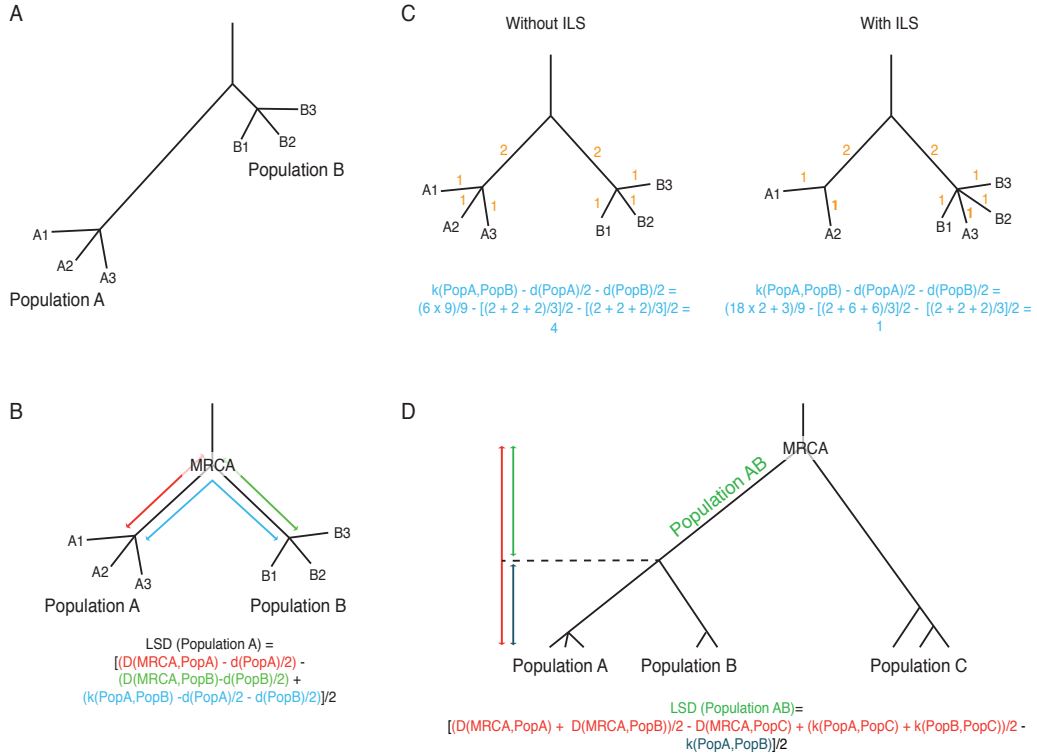


Fig. S30.

Illustration of the LSD mathematical terms. (A) Local gene genealogy showing a disproportionate number of differences that are exclusively shared by individuals from population A, a molecular signature compatible with the action of positive selection. (B) Assuming a clade with two completely sorted populations, the genetic differences exclusively shared by Population A or Population B (the length of their ancestral branches) are colored in red and green, respectively. The average pairwise distance (k) between Population A and Population B is colored in blue. The colors also map to their corresponding mathematical terms in the bottom equation. (C) Exemplification of the impact of incomplete lineage sorting (ILS) on the LSD statistic. Assuming the genetic distances depicted in orange [# differences / 100 positions], the third term of LSD equals to 4 without ILS, but just to 1 if Population A carries an alternative allele. (D) Visualization of the LSD adapted for internal branches, which represent populations that were parental to the ones sampled.

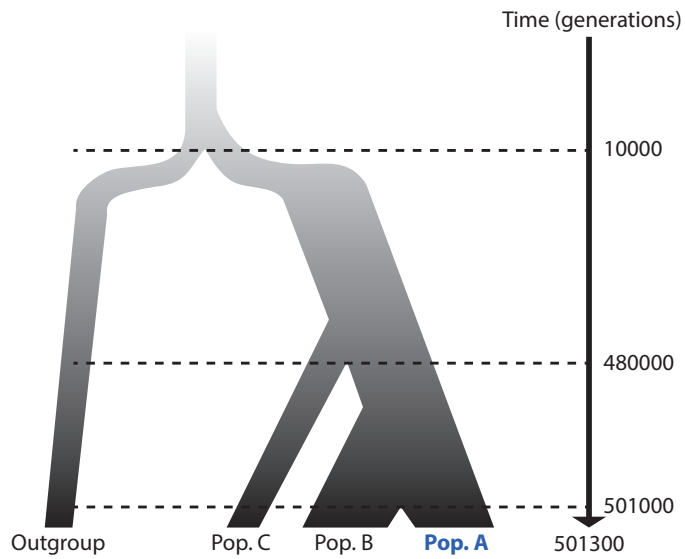


Fig. S31.

Population tree fixed for forward simulations. The model approximately resembles our knowledge about horse evolution, with the outgroup and population C (Pop. C) representing the Somali wild ass and the Przewalski's horses, respectively. Pop. A and B represent Berel' and present-day domestic horses. For illustrative purposes, the time (measured in generations forward in time) is not linearly scaled.

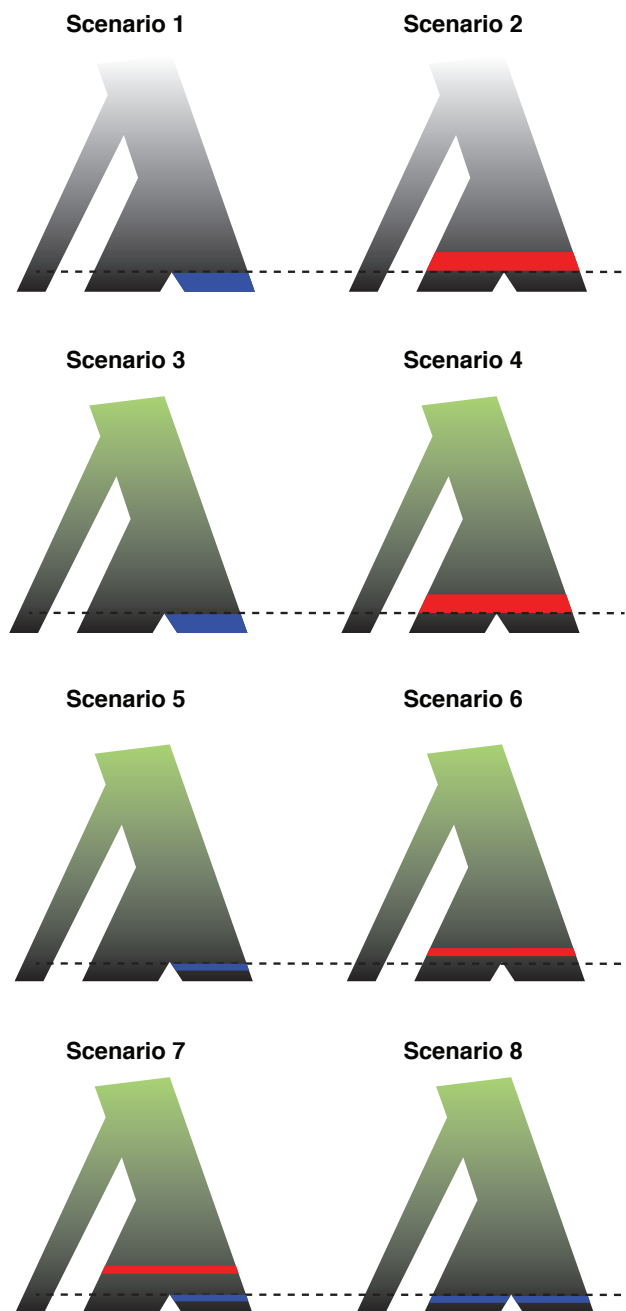


Fig. S32.

Shaded areas on the demographic model depict the time intervals encompassing episodes of positive selection in extant populations (blue) and/or its parental lineage (red). The grey and green colors, filling the demographic model, indicate that the background mode of evolution was neutral or negative selection, respectively.

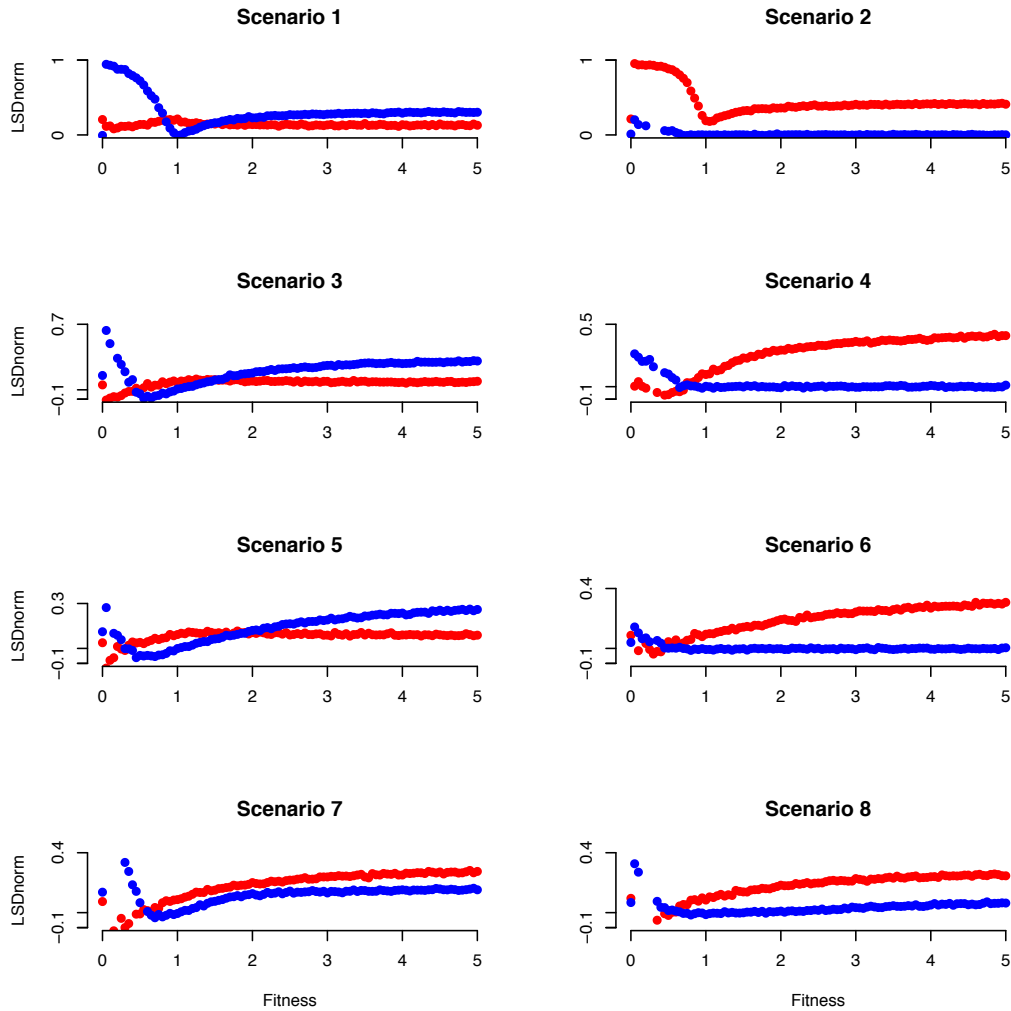


Fig. S33.

LSD_{norm} estimates in the eight simulated scenarios. The x-axis comprises the full range of fitness values fixed for the eight simulated scenarios (see text for their description). The thick Fitness = 1 delineates neutrality. The y-axis recapitulates the LSD_{norm} estimates for population A (in blue) or its ancestral lineage (in red). The maximum LSD_{norm} value is always 1, but the y-axis has been rescaled to facilitate visualization. Each point in each plot is the average over 500 replicates.

Table S1

Error rates estimated for the 14 ancient horse genomes. Error rates are expressed in %. For example, 0.0709 corresponds to seven errors every 10000 nucleotides.

ID	Error rates pre-filtering	Error rates post-filtering
838-64_NB46	0.1761	0.0709
BER01_A	0.1855	0.0741
BER02_B	0.1965	0.0725
BER04_D	0.1824	0.0728
BER05_E	0.178	0.0706
BER06_F	0.3667	0.1483
BER07_G	0.1829	0.0801
BER08_H	0.2363	0.0957
BER09_I	0.3281	0.1244
BER10_K	0.1967	0.0779
BER11_L	0.5096	0.2066
BER12_M	0.2654	0.1057
I-K2_Arz1	0.1678	0.0573
I-K3_Arz2	0.2082	0.0716

Table S2

Microbial profiling: mapping statistics (against the MetaPhlAn2 database) and alpha-diversity (SI: Genus-level Shannon Index)

Sample/library	Total reads	Mapped	Uniquely mapped	SI
SCYTHIAN & SINTASHTA HORSES				
838-64_NB46	1,700,313,073	640,799	232,585	3.48
838-64_NB46_D2_Y6.2_user_GCAACG	9,972,745	3,637	3,394	2.41
838-64_NB46_S1_ET4_user_TGTGAC	6,299,118	3,202	2,302	1.95
838-64_NB46_D1_ET4.2Acc_user_ACATAC	27,115,327	11,601	9,936	2.77
838-64_NB46_Y1_ET4_user_GACGAC	11,823,936	6,661	6,106	2.62
838-64_NB46_S2_ET4_user_TCTCGC	12,584,957	6,028	5,258	2.67
838-64_NB46_S2_ET4.2Acc_user_ACAGTG	21,508,984	9,322	5,532	2.58
838-64_NB46_ExFran_NOuser_CTCTAG	8,087,270	3,124	2,902	1.50
838-64_NB46_Y2_Y6.1_user_ACGCAT	5,725,432	2,363	2,267	2.10
838-64_NB46_Y2_ET4_user_GACACT	723,053,143	282,104	104,867	3.44
838-64_NB46_D2_Y6.1_user_ACGTGC	6,706,641	2,448	2,336	2.07
838-64_NB46_Y2_Y6.2_user_GACGAC	6,078,444	2,816	2,519	2.52
838-64_NB46_Y1_ET4.2Acc_user_ACTTGA	26,451,753	14,071	12,141	2.84
838-64_NB46_D2_ET4_user_TAGATG	834,905,323	293,422	73,025	3.28
I-K2_Arz1	86,233,241	23,413	15,390	3.23
I-K2_Arz1_Y6.2_user_TAGCTT	19,576,499	5,957	4,495	2.76
I-K2_Arz1_Y6.1_user_CAGATC	66,656,742	17,456	10,895	3.12
I-K3_Arz2	209,400,853	122,869	70,763	2.10
I-K3_Arz2_I-K3_Arz2_Y6.2_user_TCTCGC	102,543,033	60,276	34,750	2.02
I-K3_Arz2_I-K3_Arz2_Y6.1_user_ACAGTC	106,857,820	62,593	36,013	1.96
BER01	1,037,359,106	942,901	534,762	2.66
BER01_A_SCY2.2_user_TGACGT	402,131,113	361,902	188,049	2.60
BER01_A_E16.1_user_TGTCTG	13,780,607	10,808	10,086	2.29
BER01_A_E9.Dab2nd_NOuser_CTATCA	17,080,289	16,646	14,064	2.25
BER01_A_E16.2_user_GACCGG	60,636,334	50,563	40,052	2.45
BER01_A_SCY2.1_user_ACGCAT	543,730,763	502,982	282,511	2.65
BER02	614,105,379	178,733	90,781	2.41
BER02_B_E16.2_user_TGTCTG	23,397,270	10,837	7,520	1.96
BER02_B_SCY2.1_user_CGTATA	330,098,860	88,774	42,250	2.37
BER02_B_E16.1_user_TGACGT	47,668,680	18,661	11,793	2.01
BER02_B_E9.Dab2nd_NOuser_TGATCG	8,291,774	2,341	1,982	1.35
BER02_B_SCY2.2_user_CTATCA	204,648,795	58,120	27,236	2.25
BER04	996,026,825	713,574	328,248	3.29
BER04_D_E16.1_user_CTATCA	624,810,317	430,241	186,788	3.32
BER04_D_SCY2.2_user_CGTATA	15,850,323	17,787	16,025	2.89
BER04_D_SCY2.1_user_GTGTAT	20,865,590	23,249	21,037	2.95
BER04_D_E16.2_user_TGACGT	334,500,595	242,297	104,398	3.23

BER05	1,137,469,664	1,039,810	435,127	2.48
BER05_E_E16.2_user_CTATCA	15,535,468	21,763	15,956	2.08
BER05_E_SCY2.2_user_GTGTAT	677,201,799	613,485	219,045	2.49
BER05_E_SCY2.1_user_CGCTAT	430,395,629	385,888	186,050	2.49
BER05_E_E16.1_user_CGTATA	14,336,768	18,674	14,076	2.23
BER06	920,685,963	631,031	347,549	2.41
BER06_F_SCY2.1_user_TGAACA	607,972,637	399,799	197,010	2.41
BER06_F_SCY2.2_user_CGCTAT	181,271,316	120,665	79,119	2.29
BER06_F_E10.Dab2nd_NOuser_CAGCTA	102,944,375	75,639	41,547	2.26
BER06_F_E16.1_user_GTGTAT	12,788,528	15,430	13,639	2.47
BER06_F_E16.2_user_CGTATA	15,709,107	19,498	16,234	2.37
BER07	316,599,378	311,521	187,778	2.48
BER07_G_E17.1_user_CGCTAT	30,744,544	35,900	29,395	2.47
BER07_G_SCY1.2Rmp_user_TGAACA	13,961,067	13,719	11,808	2.14
BER07_G_E17.2_user_GTGTAT	17,588,870	21,411	17,308	2.40
BER07_G_SCY1.1_user_CAGCTA	254,303,462	240,489	129,265	2.41
BER07_G_E10.Dab2nd_NOuser_TGAGCC	1,435	2	2	0.00
BER08	436,239,951	276,487	170,084	2.42
BER08_H_E10.Dab2nd_NOuser_ACGTGC	9,723,554	6,834	6,313	1.33
BER08_H_E17.1_user_TGAACA	25,729,947	18,206	16,070	2.14
BER08_H_SCY1.2_user_CAGCTA	10,148,484	6,616	6,219	1.61
BER08_H_SCY1.1_user_ACATAC	384,926,977	240,613	137,474	2.47
BER08_H_E17.2RmpB_user_CGCTAT	5,710,989	4,218	4,008	1.82
BER09	1,100,612,099	663,992	364,891	3.13
BER09_I_E12.Dab2nd_NOuser_GTGTAT	115,622,248	91,912	50,481	2.89
BER09_I_SCY1.1_user_TGTCTG	565,023,631	328,317	194,707	3.15
BER09_I_SCY1.2_user_ACATAC	419,966,220	243,763	119,703	3.06
BER10_K	384,489,475	216,200	124,097	2.97
BER10_K_SCY1.1_user_TGTGAC	342,008,987	188,654	103,231	3.03
BER10_K_SCY1.2_user_TGTCTG	15,583,604	9,156	8,055	2.32
BER10_K_E12.Dab2nd_NOuser_ACATAC	6,040,530	3,039	2,862	1.93
BER10_K_E15.2Rmp_user_CAGCTA	20,856,354	15,351	9,949	2.39
BER11	637,138,773	882,000	388,800	2.58
BER11_L_SCY1.2Rmp_user_TGTGAC	12,192,255	16,902	14,978	2.21
BER11_L_SCY1.1_user_TCTCGC	467,001,930	635,131	234,173	2.65
BER11_L_E12.Dab2nd_NOuser_AGCATG	124,003,272	176,223	93,212	2.52
BER11_L_E15.1_user_GACCGG	21,805,021	34,746	29,802	2.28
BER11_L_E15.2_user_ACATAC	12,136,295	18,998	16,635	2.29
BER12	1,410,239,476	718,861	159,671	2.98
BER12_M_E15.2Rmp_user_ACGCAT	20,728,024	15,755	5,197	2.40
BER12_M_E12.Dab2nd_NOuser_CGATGA	66,994,757	41,094	21,022	2.68
BER12_M_SCY1.1_user_GACGAC	469,882,378	233,847	91,038	2.96
BER12_M_SCY1.2_user_TCTCGC	852,634,317	428,165	42,414	2.90

ANCIENT HORSES					
Batagai	1,240,821,408	158,989	70,645	1.61	
CGG10022	1,068,302,433	162,655	54,703	2.76	
CGG10023	5,397,659,810	1,256,022	674,348	2.82	
CGG101397	1,678,398,841	214,349	164,796	2.84	

Table S3

Temporal, cultural and geographical information for the 14 ancient horses sequenced in this study, altogether with sex and sequencing statistics

Period	Age (BP)	Site	Culture	Location	Sample	Sex	Number of Sequencing Reads	Endogenous content	mtDNA coverage	nuDNA coverage after trimming/rescaling
Bronze Age	4,071–4,055	Sintashta	Sintashta	Chelyabinsk Oblast, Southeastern Russia	838-64_NB46	F	1,700,313,073	11.51%	336.23	4.936
Iron Age	2,300	Berel' Kurgan 11	Scythian kurgan	Berel, Eastern Kazakhstan	BER01_A	M	1,037,359,106	29.36%	396.82	9.210
					BER02_B	M	614,105,379	57.41%	318.50	10.704
					BER04_D	M	996,026,825	25.47%	300.24	6.808
					BER05_E	M	1,137,469,664	27.85%	424.60	8.540
					BER06_F	M	920,685,963	40.91%	523.37	10.654
					BER07_G	M	316,599,378	21.22%	229.54	1.888
					BER08_H	M	443,478,247	17.44%	211.46	2.221
					BER09_I	M	1,100,612,099	36.14%	370.66	10.957
					BER10_K	M	384,489,475	21.92%	182.72	2.296
					BER11_L	M	637,138,773	9.01%	253.57	1.665
					BER12_M	M	1,410,239,476	13.80%	163.08	5.496
Iron Age	2,760–2,695	Arzhan I	Scythian kurgan	Arzhan, Southern Russia	I-K2_Arz1	M	86,233,241	56.10%	48.05	1.220
					I-K3_Arz2	M	209,400,853	27.89%	191.53	1.292

Table S4

Samples used for the mitochondrial DNA analyses, including 106 and 34 ancient horses. For the latter, dates are given in years BP.

Sample	Population	Tip Accession Date number	Reference
Akhal-Teke1	Modern	0 HQ439441	Lippold et al. 2011 (158)
Akhal-Teke2	Modern	0 HQ439442	Lippold et al. 2011 (158)
Altai1	Modern	0 HQ439443	Lippold et al. 2011 (158)
Altai2	Modern	0 HQ439444	Lippold et al. 2011 (158)
Appaloosa	Modern	0 HQ439446	Lippold et al. 2011 (158)
Arabian	Modern	0 n/a	Orlando et al. 2013 (27)
Arab/arab	Modern	0 HQ439488	Lippold et al. 2011 (158)
Arab/rus	Modern	0 HQ439447	Lippold et al. 2011 (158)
Arab/syr1	Modern	0 HQ439448	Lippold et al. 2011 (158)
Arab/syr2	Modern	0 HQ439449	Lippold et al. 2011 (158)
Ardennais	Modern	0 HQ439450	Lippold et al. 2011 (158)
Barb	Modern	0 HQ439453	Lippold et al. 2011 (158)
Bashkir Curly1	Modern	0 HQ439451	Lippold et al. 2011 (158)
Bashkir Curly2	Modern	0 HQ439452	Lippold et al. 2011 (158)
Black Forest	Modern	0 HQ439487	Lippold et al. 2011 (158)
Camargue	Modern	0 HQ439454	Lippold et al. 2011 (158)
Clydesdale	Modern	0 HQ439455	Lippold et al. 2011 (158)
Debao	Modern	0 EU939445	Jiang et al. 2010 (159)
Deqin	Modern	0 EF597514	Xu et al. 2007 (160)
Fjord	Modern	0 n/a	Orlando et al. 2013 (27)
German Riding Pony	Modern	0 HQ439461	Lippold et al. 2011 (158)
German Sport	Modern	0 HQ439456	Lippold et al. 2011 (158)
Haflinger	Modern	0 HQ439464	Lippold et al. 2011 (158)
Hanoverian	Modern	0 HQ439457	Lippold et al. 2011 (158)
Holstein	Modern	0 HQ439458	Lippold et al. 2011 (158)
Hungarian Cold Blood	Modern	0 HQ439494	Lippold et al. 2011 (158)
Hybrid/KB7903	Modern	0 KT368757	Der Sarkissian et al. 2015 (50)
Icelandic	Modern	0 n/a	Orlando et al. 2013 (27)
Icelandic/P5782	Modern	0 n/a	Orlando et al. 2013 (27)
Icelandic1	Modern	0 HQ439465	Lippold et al. 2011 (158)
Icelandic2	Modern	0 HQ439466	Lippold et al. 2011 (158)
Kabardin	Modern	0 HQ439468	Lippold et al. 2011 (158)
Kinsky1	Modern	0 HQ439469	Lippold et al. 2011 (158)
Kinsky2	Modern	0 HQ439470	Lippold et al. 2011 (158)
Kladruber	Modern	0 HQ439445	Lippold et al. 2011 (158)
Kladruber1	Modern	0 HQ439471	Lippold et al. 2011 (158)
Kladruber2	Modern	0 HQ439472	Lippold et al. 2011 (158)

Konik	Modern	0 HQ439473	Lippold et al. 2011 (158)
Kustanai	Modern	0 HQ439476	Lippold et al. 2011 (158)
Kuznet1	Modern	0 HQ439474	Lippold et al. 2011 (158)
Kuznet2	Modern	0 HQ439475	Lippold et al. 2011 (158)
Lewitzer	Modern	0 HQ439477	Lippold et al. 2011 (158)
Liebenthaler/wild1	Modern	0 HQ439478	Lippold et al. 2011 (158)
Liebenthaler/wild2	Modern	0 HQ439479	Lippold et al. 2011 (158)
Mongolian/KB7754	Modern	0 KT368741	Der Sarkissian et al. 2015 (50)
Naqu	Modern	0 EF597513	Xu et al. 2007 (160)
Noriker	Modern	0 HQ439480	Lippold et al. 2011 (158)
Norwegian Fjord	Modern	0 HQ439463	Lippold et al. 2011 (158)
Oldenburg	Modern	0 HQ439459	Lippold et al. 2011 (158)
Orlov Trotter	Modern	0 HQ439481	Lippold et al. 2011 (158)
Paint	Modern	0 HQ439482	Lippold et al. 2011 (158)
Percheron	Modern	0 HQ439483	Lippold et al. 2011 (158)
Przewalski/HQ439484	Modern	0 HQ439484	Lippold et al. 2011 (158)
Przewalski/NOUMA1	Modern	0 NC_024030	n/a
Przewalski/NOUMA2	Modern	0 AP013095	n/a
Przewalski/Prz_D2630	Modern	0 n/a	Do et al. 2014 (143)
Przewalski/Prz_D2631	Modern	0 n/a	Do et al. 2014 (143)
Przewalski/Prz_Przewalski	Modern	0 n/a	Orlando et al. 2013 (27)
Przewalski/Prz01	Modern	0 JN398402	Achilli et al. 2012 (161)
Przewalski/Prz02	Modern	0 JN398403	Achilli et al. 2012 (161)
Przewalski/SB159	Modern	0 KT368751	Der Sarkissian et al. 2015 (50)
Przewalski/SB274	Modern	0 KT368752	Der Sarkissian et al. 2015 (50)
Przewalski/SB281	Modern	0 KT368746	Der Sarkissian et al. 2015 (50)
Przewalski/SB285	Modern	0 KT368744	Der Sarkissian et al. 2015 (50)
Przewalski/SB293	Modern	0 KT368748	Der Sarkissian et al. 2015 (50)
Przewalski/SB319 (Bellina)	Modern	0 AP012267	Goto et al. 2011 (162)
Przewalski/SB339	Modern	0 KT368749	Der Sarkissian et al. 2015 (50)
Przewalski/SB524	Modern	0 KT368742	Der Sarkissian et al. 2015 (50)
Przewalski/SB528	Modern	0 KT368743	Der Sarkissian et al. 2015 (50)
Przewalski/SB533	Modern	0 KT368750	Der Sarkissian et al. 2015 (50)
Przewalski/SB615	Modern	0 KT368747	Der Sarkissian et al. 2015 (50)
Przewalski/SB668 (Anuschka)	Modern	0 AP012268	Goto et al. 2011 (162)
Przewalski/SB966	Modern	0 KT368745	Der Sarkissian et al. 2015 (50)
Quarter	Modern	0 n/a	Orlando et al. 2013 (27)
Rhineland Heavy Draft	Modern	0 HQ439485	Lippold et al. 2011 (158)
Rhineland Heavy Draft	Modern	0 HQ439491	Lippold et al. 2011 (158)
Russian Riding Horse	Modern	0 HQ439486	Lippold et al. 2011 (158)
Shetland	Modern	0 HQ439489	Lippold et al. 2011 (158)
Shire	Modern	0 HQ439490	Lippold et al. 2011 (158)
Spotted	Modern	0 HQ439492	Lippold et al. 2011 (158)

Standard	Modern	0	n/a	Orlando et al. 2013 (27)
Thoroughbred	Modern	0	HQ439462	Lippold et al. 2011 (158)
Trakehner	Modern	0	HQ439493	Lippold et al. 2011 (158)
Twilight	Modern	0	n/a	Orlando et al. 2013 (27)
Viatka	Modern	0	HQ439495	Lippold et al. 2011 (158)
Vladimir Heavy Draught	Modern	0	HQ439500	Lippold et al. 2011 (158)
WelshPony/A	Modern	0	HQ439497	Lippold et al. 2011 (158)
WelshPony/B1	Modern	0	HQ439498	Lippold et al. 2011 (158)
WelshPony/B2	Modern	0	HQ439499	Lippold et al. 2011 (158)
WelshPony/D	Modern	0	HQ439496	Lippold et al. 2011 (158)
Westphalian	Modern	0	HQ439460	Lippold et al. 2011 (158)
Yakut	Modern	0	HQ439467	Lippold et al. 2011 (158)
Yakutian_Horse1	Modern	0	KT368727	Librado et al. 2015 (15)
Yakutian_Horse2	Modern	0	KT368728	Librado et al. 2015 (15)
Yakutian_Horse3	Modern	0	KT368729	Librado et al. 2015 (15)
Yakutian_HQ439467	Modern	0	HQ439467	Lippold et al. 2011 (158)
Yakutian_Yak1	Modern	0	KT368730	Librado et al. 2015 (15)
Yakutian_Yak2	Modern	0	KT368731	Librado et al. 2015 (15)
Yakutian_Yak3	Modern	0	KT368732	Librado et al. 2015 (15)
Yakutian_Yak4	Modern	0	KT368733	Librado et al. 2015 (15)
Yakutian_Yak5	Modern	0	KT368734	Librado et al. 2015 (15)
Yakutian_Yak6	Modern	0	KT368735	Librado et al. 2015 (15)
Yakutian_Yak7	Modern	0	KT368736	Librado et al. 2015 (15)
Yakutian_Yak8	Modern	0	KT368737	Librado et al. 2015 (15)
Yakutian_Yak9	Modern	0	KT368738	Librado et al. 2015 (15)
Zhongdian	Modern	0	EF597512	Xu et al. 2007 (160)
Ancient/148	Ancient	31,917	n/a	Orlando et al. 2013 (27)
Ancient/149	Ancient	18,471	n/a	Orlando et al. 2013 (27)
Ancient/151	Ancient	39,022	n/a	Orlando et al. 2013 (27)
Ancient/152	Ancient	39,311	n/a	Orlando et al. 2013 (27)
Ancient/154	Ancient	2,235	n/a	Orlando et al. 2013 (27)
Ancient/155	Ancient	20,273	n/a	Orlando et al. 2013 (27)
Ancient/156	Ancient	24,154	n/a	Orlando et al. 2013 (27)
Ancient/158	Ancient	16,880	n/a	Orlando et al. 2013 (27)
Ancient/159	Ancient	32,667	n/a	Orlando et al. 2013 (27)
Ancient/160	Ancient	28,242	n/a	Orlando et al. 2013 (27)
Ancient/Batagai	Ancient	5,088	KT368725	Librado et al. 2015 (15)
Ancient/CGG10022	Ancient	42,692	n/a	Schubert et al. 2014 (14)
Ancient/CGG10023	Ancient	16,099	n/a	Schubert et al. 2014 (14)
Ancient/CGG10026	Ancient	27,230	n/a	Orlando et al. 2013 (27)
Ancient/CGG10027	Ancient	28,336	n/a	Orlando et al. 2013 (27)
Ancient/CGG10032	Ancient	28,198	n/a	Orlando et al. 2013 (27)
Ancient/CGG101397	Ancient	100	KT368726	Librado et al. 2015 (15)

Ancient/ODJ6	Ancient	225	KT368724	Librado et al. 2015 (15)
Ancient/SS118	Ancient	2,150	n/a	Sawyer et al. 2012 (163)
Ancient/SS120	Ancient	2,150	n/a	Sawyer et al. 2012 (163)
Ancient/SS121	Ancient	2,150	n/a	Sawyer et al. 2012 (163)
Ancient/SS122	Ancient	2,150	n/a	Sawyer et al. 2012 (163)
Ancient/SS124	Ancient	1,815	n/a	Sawyer et al. 2012 (163)
Ancient/SS127	Ancient	1,815	n/a	Sawyer et al. 2012 (163)
Ancient/SS28	Ancient	1,250	n/a	Sawyer et al. 2012 (163)
Ancient/SS54	Ancient	550	n/a	Sawyer et al. 2012 (163)
Ancient/Yukagir	Ancient	5,384	KT368723	Librado et al. 2014 (15)
Hybrid/Ewld5_Theodor	Ancient	88	KT368758	Der Sarkissian et al. 2015 (50)
Mongolian/Emgl1	Ancient	102	KT368739	Der Sarkissian et al. 2015 (50)
Mongolian/Emgl3	Ancient	103	KT368740	Der Sarkissian et al. 2015 (50)
Przewalski/Ewld1_Bijsk1	Ancient	107	KT368753	Der Sarkissian et al. 2015 (50)
Przewalski/Ewld2_Bijsk2	Ancient	105	KT368754	Der Sarkissian et al. 2015 (50)
Przewalski/Holotype	Ancient	137	KT368755	Der Sarkissian et al. 2015 (50)
Przewalski/Paratype	Ancient	116	KT368756	Der Sarkissian et al. 2015 (50)

Table S5

Best substitution model for each partition and the global dataset, as assessed by ModelGenerator v0.85 and the Bayesian Information Criterion.

Partition	Best-fit model
All regions	TrN+I+G
Control Region	TrN+I+G
Genes - codon position 1	HKY+I+G
Genes - codon position 2	HKY+I
Genes - codon position 3	TIM+G
rRNA	HKY+I+G
tRNA	TrN+I+G

Table S6

Mutation rate (μ = [mutations/(generation x site)]) estimated for the different partitions of the mitochondrial genome, altogether with its standard deviation (SE), as inferred by BEAST 1.8.2.

Partition	Mean μ	SE
Control Region	3.68E-07	8.06E-08
Genes first position	1.81E-08	3.44E-09
Genes second position	8.92E-09	1.97E-09
Genes third position	7.21E-08	1.11E-08
rRNA genes	2.56E-08	4.99E-09
tRNA genes	5.61E-07	1.73E-07

Table S7

Primer pairs used to corroborate the genotype of seven loci with Mendelian inheritance, and underlying coat-color variations.

Gene	Primer	Target on EquCab2.0	Sequence 5' to 3'	Fragment length
KIT	Pk-KIT-I13 F	chr3:77,740,163	CGT CAT GAC TCA TTC GTG AGA A	63 bp
	Pk-KIT-I13 R		GCT CTG AAG GTA ACA AGC AAC TAA	
KIT	Pk-KIT-I16 F	chr3:77,735,520	TTT AAA TGG CTT TCT TTT CTC C	59 bp
	Pk-KIT-I16 R		TGC CAA GTC CCT ATG AAT ACA C	
MATP	P-MATP-E2F	chr21:30,666,626	GCC ATA ACC ATC ACC ATG ATA G	65bp
	Pk-MATP-E2 R		GGC CCA TCA ATG AAG TCA G	
MC1R	Pk-MC1R-E1 F	chr3: 36,259,552	GCA CTC ACC CAT GTA CTA CTT CAT	71 bp
	Pk-MC1R-E1 R		GCA CGT TGC TCA TGC TCA C	
SILV	Pk-SILV-E11 F	chr6:73,665,304	TCC TTC TTC TTC TCC CAA ATC A	52 bp
	P-SILV-E11 R		GAG CTG AGC CCT GCT TCA TAA	
ASIP	Pk-ASIP-E2 F	chr22:25,168,579-25,168,589del	CAA GAA ATC CAA AAA GAT CAG C	67/78 bp
	Pk-ASIP-E2 R		ATG AGA AGT CCA AGG CCT ACC T	
TRPM1	LP F	chr1:108,249,293	AGA TCG AGC TGC TGA ACT GG	62 bp
	LP R		TCT CCA TGA TCC CAA GCA AT	

Table S8

Coat-color variation inferred for 16 ancient horses, based on their genotypes in informative genetic markers. The coefficient of dominance for derived alleles is specified below the corresponding column. The '^' symbol indicates positions solely typed by PCR, while '*' solely by shotgun sequencing. The '**' symbol indicates inconsistencies between both methods, and where the shotgun sequencing has been assured to be the correct. Shaded cells highlight horses with a given phenotype, as revealed by their heterozygous (grey) or homozygous alternative (red) genotype.

Sample	Color	Spotting	Dilution	Dun	Black (ASIP)	Tobiano spotting (KIT)	Sabino Spotting (KIT)	Cream (SLC45A2 or MATP)	Chestnut (MC1R)	Silver (PMEL17 or SILV11)	Leopard complex (TRPM1)
838-64_NB46	Bay	No	No	d1/d1*	A/A^	KM0/KM0	sb1/sb1	C/C	E/E	z/z	lp/lp
I-K2_Arz1	Chestnut Leopard	Leopard	No	d1/d2?*	a/A^	KM0/KM0	sb1/sb1	C/C	e/e	z/z	lp/Lp
I-K3_Arz2	Black Silver	No	Silver	d1/d2?*	a/a^	KM0/KM0	sb1/sb1	C/C	e/E	z/Z	lp/lp
BER01_A	Chestnut	No	No	d1/d1*	a/A^	KM0/KM0	sb1/sb1	C/C	e/e	z/z	lp/lp
BER02_B	Black Dun?	No	No	D/d1*	a/a^	KM0/KM0	sb1/sb1	C/C	e/E	z/z	lp/lp
BER03_C	Chestnut	No	No	—	a/A^	KM0/KM0^	sb1/sb1^	C/C^	e/e^	z/z^	lp/lp^
BER04_D	Chestnut	No	No	d1/d1*	a/a^	KM0/KM0	sb1/sb1	C/C	e/e	z/z	lp/lp
BER05_E	Chestnut	No	No	d1/d1*	a/a^	KM0/KM0	sb1/sb1	C/C	e/e^	z/z	lp/lp
BER06_F	Bay (Incontinentia pigmenti?)	No	No	d1/d1*	a/A^	KM0/KM0	sb1/sb1	C/C	e/E	z/z	lp/lp
BER07_G	Bay Silver Dun?	No	Silver	D/d1*	A/A^	KM0/KM0^	sb1/sb1	C/C	e/E^	z/Z^	lp/lp^
BER08_H	Chestnut	No	No	d1/d1*	a/A^	KM0/KM0	sb1/sb1	C/C	e/e^	z/z^	lp/lp
BER09_I	Bay Cream	No	No	d1/d1*	—	KM0/KM0*	sb1/sb1*	c/C*	e/E**	z/z	lp/lp*
BER10_K	Bay	No	No	d1/d1*	a/A^	KM0/KM0	sb1/sb1	C/C	e/E	z/z	lp/lp
BER11_L	Chestnut	No	No	d1/d1*	a/A^	KM0/KM0	sb1/sb1	C/C	e/e^	z/z	lp/lp^
BER12_M	Bay Leopard	Leopard	No	d1/d1*	A/A^	KM0/KM0*	sb1/sb1*	C/C*	E/E*	z/z*	lp/Lp*
BER13_N	Bay	No	No	—	A/A^	KM0/KM0^	sb1/sb1^	C/C^	e/E^	z/z^	lp/lp^
				Co-dominant?	Recessive	Dominant	Dominant	Dominant	Recessive (Dominant over ASIP)	Dominant	Co-dominant

Table S9

Genotypes at the loci underlying the Dun phenotype for the samples reported in Imsland et al. (2015), and the 14 ancient horses sequenced here. The color code follows Imsland et al. 2015 (61): red denotes genotypes associated with the Dun phenotype, shaded red refers to heterozygotes Dun/non-Dun 1, blue genotypes associated with non-Dun 1, shaded blue heterozygotes non-Dun 1/non-Dun 2, grey genotypes associated to non-Dun 2 and shaded grey heterozygotes associated with non-Dun 2/other. Inferred phenotypes from ancient samples are reported with a question mark.

Sample	Predicted phenotype	Genotype	SNP2 (chr8:18,226,905)	SNP1 (chr8:18,227,267+1,066)	Deletion (D=deletion)	Reference
Domestic horse	non-Dun	d2/d2	AA	Deletion	DD	Imsland et al. 2015
Domestic horse	non-Dun	d1/d1	AA	TT	dd	Imsland et al. 2015
Domestic horse	Dun	D/D	GG	GG	dd	Imsland et al. 2015
Domestic horse (Estonian)	Dun	D/D	AA	GG	dd	Imsland et al. 2015
Przewalski's horse	Dun	D/D	GG	GG	dd	Imsland et al. 2015
Przewalski's horse	Dun	D/D	GG	GG	dd	Imsland et al. 2015
4,400-year-old wild horse (Batagai)	non-Dun?	d1/d1	AA	TT	dd	Imsland et al. 2015
42,700-year-old wild horse (CGG10022)	Dun?	D/d1	AG	GT	dd	Imsland et al. 2015
African wild ass	Dun	D/D	GG	GG	dd	Imsland et al. 2015
Domestic donkey	Dun	D/D	GG	GG	dd	Imsland et al. 2015
Kiang	Dun	D/D	GG	GG	dd	Imsland et al. 2015
Onager	Dun	D/D	GG	GG	dd	Imsland et al. 2015
Grevy's zebra	Zebra	D/D	GG	GG	dd	Imsland et al. 2015
Muntain zebra	Zebra	D/D	GG	GG	dd	Imsland et al. 2015
Plains zebra (Burchell's)	Zebra	D/D	GG	GG	dd	Imsland et al. 2015
Plains zebra (quagga)	Zebra/Dun	D/D	GG	GG	dd	Imsland et al. 2015
838-64_NB46	non-Dun?	d1/d1	5A	7T	dd	This study
BER01_A	non-Dun?	d1/d1	10A	15T	dd	This study
BER02_B	Dun?	D/d1	6A12G	10G3T	dd	This study
BER04_D	non-Dun?	d1/d1	4A	7T	dd	This study

BER05_E	non-Dun?	d1/d1	10A		11T	dd	This study
BER06_F	non-Dun?	d1/d1	14A		9T	dd	This study
BER07_G	Dun?	D/d1	4A		1G2T	dd	This study
BER08_H	non-Dun?	d1/d1	4A		2T	dd	This study
BER09_I	non-Dun?	d1/d1	14A		22T	dd	This study
BER10_K	non-Dun?	d1/d1	2A		1T	Dd	This study
BER11_L	non-Dun?	d1/d1	2A		1T	Dd	This study
BER12_M	non-Dun?	d1/d1	2A		7T	dd	This study
I-K2_Arz1	non-Dun?	d1/d2?	1A		NA	dd	This study
I-K3_Arz2	non-Dun?	d1/d2?	4A		NA	dd	This study

Table S10

Variance explained by TreeMix models with an increasing number of migration edges ($m = 0$ to $m = 15$). The shaded row, at $m = 9$, highlights the number of migrations for which the variance explained by the model stops monotonically increasing.

Number of migrations	Variance (K = 500)	Variance (K = 1500)
0	0.9821725	0.9817440
1	0.9828998	0.9829094
2	0.9835324	0.9833155
3	0.9839743	0.9847349
4	0.9856714	0.9855656
5	0.9861469	0.9860962
6	0.9875058	0.9875283
7	0.9882171	0.9878166
8	0.9888783	0.9879859
9	0.9890056	0.9901243
10	0.9889707	0.9892418
11	0.9896994	0.9898968
12	0.9900114	0.9895256
13	0.9903639	0.9907263
14	0.9907750	0.9909750
15	0.9908181	0.9905481

Table S11

Minimum fraction of the genome (f) in 'H2_numerator' descending from pre-domestic horses (H3). See the '*Admixture test with D-statistics*' section in the supplementary information for a description of the method.

H1	H2_numerator	H2_denominator	H3	f
KB3879	Sintashta and Berel'	Batagai	CGG10022	0.2457115666
KB3879	Modern-day domestic	Batagai	CGG10022	0.1796383557
KB3879	Sintashta and Berel'	CGG10022	Batagai	0.1782351117
KB3879	Modern-day domestic	CGG10022	Batagai	0.1433584124
SB533	Sintashta and Berel'	Batagai	CGG10022	0.2677888294
SB533	Modern-day domestic	Batagai	CGG10022	0.2060511274
SB533	Sintashta and Berel'	CGG10022	Batagai	0.1987727425
SB533	Modern-day domestic	CGG10022	Batagai	0.1659105742

Table S12

Horse genomes used to estimate genetic affinities. At the bottom, the 14 whole-genome sequenced in this study. At the top, the 38 included for comparative purposes. “yBP”, years Before Present.

ID	Breed/Population	Date	Coverage (fold)
Akh_M_0248_Vi_AKT001	Akhal-Teke	Modern	22.77
Qrt_F_0070_MM_A1543	American Quarter Horse	Modern	12.55
Qrt_F_0074_MM_A5659	American Quarter Horse	Modern	13.67
Qrt_F_0075_MM_A5964	American Quarter Horse	Modern	13.97
Qrt_M_0073_MM_A2085	American Quarter Horse	Modern	12.32
Yak_M_0223_LO_Batagai	Ancient Yakutian	5,200 ± 35 yBP	18.29
Yak_M_0222_LO_CGG10397	Historical Yakutian	100 yBP	20.25
Ara_M_0237_Mz_SAMN02439777	Arabian	Modern	12.32
Ara_F_0141_LO_Arabian	Arabian	Modern	10.82
Dlm_F_0238_Mz_SAMN02422919	Duelmener	Modern	12.93
Mon_M_0052_LR_FM1030	Franches-Montagnes	Modern	17.02
Mon_M_0065_LR_FM1798	Franches-Montagnes	Modern	22.62
Mon_M_0068_LR_FM1951	Franches-Montagnes	Modern	19.24
Han_M_0234_Mz_SAMN02439782	Hannoverian	Modern	10.09
Han_M_0235_Mz_SAMN02439779	Hannoverian	Modern	23.40
Ice_M_0247_Vi_IS074	Icelandic	Modern	9.19
Ice_M_0143_LO_Icelandic	Icelandic	Modern	8.51
Ice_M_0144_An_P5782	Icelandic	Modern	32.67
Yak_F_0224_LO_CGG10022	Late Pleistocene	42,692 ± 891 yBP	25.60
Yak_M_0225_LO_CGG10023	Late Pleistocene	16,099 ± 192 yBP	7.72
Mwi_M_0239_Bk_SRR1275408	Marwari	Modern	28.34
Mng_M_0153_LO_KB7754	Mongolian	Modern	19.46
Mng_F_0216_Pk_TG1111D2629	Mongolian	Modern	24.09
Mng_M_0215_Pk_TG1111D2628	Mongolian	Modern	23.54
Mor_F_0096_MM_EMS595	Morgan	Modern	14.02
Fjo_F_0142_LO_Fjord	Norwegian Fjord	Modern	7.73
Prz_F_0150_LO_KB3879	Przewalski	Modern	22.88
Prz_M_0157_LO_SB293	Przewalski	Modern	19.30
Prz_M_0160_LO_SB533	Przewalski	Modern	20.36
Sht_F_0249_Vi_SPH020	Shetland Pony	Modern	21.58
Sht_F_0250_Vi_SPH041	Shetland Pony	Modern	19.78
Std_F_0078_MM_M977	Standardbred	Modern	12.35
Std_M_0081_MM_M5256	Standardbred	Modern	11.44
Std_M_0090_MM_M1009	Standardbred	Modern	7.68
Std_M_0149_LO_Standardbred	Standardbred	Modern	12.31
Thr_F_0145_LO_Twilight	Thoroughbred	Modern	41.14
Yak_F_0164_LO_Yak2	Yakutian	Modern	18.38

Yak_F_0169_LO_Yak7	Yakutian	Modern	21.63
838-64_NB46	Sintanshta	4,055–4,071 yBP	4.936
BER01_A	Berel'	2,300 yBP	9.210
BER02_B	Berel'	2,300 yBP	10.704
BER04_D	Berel'	2,300 yBP	6.808
BER05_E	Berel'	2,300 yBP	8.540
BER06_F	Berel'	2,300 yBP	10.654
BER07_G	Berel'	2,300 yBP	1.888
BER08_H	Berel'	2,300 yBP	2.221
BER09_I	Berel'	2,300 yBP	10.957
BER10_K	Berel'	2,300 yBP	2.296
BER11_L	Berel'	2,300 yBP	1.665
BER12_M	Berel'	2,300 yBP	5.496
I-K2_Arz1	Arzhan	2,760–2,695 yBP	1.220
I-K3_Arz2	Arzhan	2,760–2,695 yBP	1.292

Table S13

The 14 ancient and 14 modern horse genomes scanned for the presence of copy number (CNs) variants. Their sequencing coverage varies with those on Tables S4 and S5 due to differences in raw data treatment required for CNV calling. The last column indicates the DNA spanned within segmental duplications, in Megabases (Mb).

	Sample	Coverage	Mb in Segmental Duplications
Ancient domesticates	I-K2_Arz1	0.8	111.8
	I-K3_Arz2	0.8	108.5
	838-64_NB46	3.5	138.0
	BER01_A	6.7	140.4
	BER02_B	7.8	139.3
	BER04_D	4.8	133.8
	BER05_E	6.1	131.7
	BER06_F	7.8	139.0
	BER07_G	1.4	123.2
	BER08_H	1.6	125.3
	BER09_I	7.9	134.0
	BER10_K	1.6	124.5
	BER11_L	1.2	120.4
	BER12_M	4.0	131.7
Modern domesticates	Akh_M_0248_Vi_AKT001	27.5	188.7
	Ara_M_0237_Mz_SAMN02439777	10.8	179.0
	Dlm_F_0238_Mz_SAMN02422919	11.1	182.2
	Fjo_F_0142_LO_Fjord	25.5	180.8
	Han_M_0235_Mz_SAMN02439779	20.1	182.1
	Ice_M_0144_An_P5782	27.1	181.4
	Mng_F_0216_Pk_TG1111D2629	29.0	177.0
	Mng_M_0215_Pk_TG1111D2628	19.3	173.7
	Mon_M_0065_LR_FM1798	17.8	180.9
	Mor_F_0096_MM_EMS595	11.9	178.8
	Mwi_M_0239_Bk_SRR1275408	28.4	179.9
	Sht_F_0249_Vi_SPH020	18.2	185.8
	Std_F_0078_MM_M977	10.8	181.2
	Yak_F_0169_LO_Yak7	27.0	187.0

Table S14

The 121 genes overlapping the top-1000 PBS windows, when estimated for the Berel' population.

Ensembl Gene ID	Chromosome Name	Gene Start (bp)	Gene End (bp)	Associated Gene Name
ENSECAG00000023813	1	5872903	5981766	
ENSECAG00000010116	1	26079852	26096939	<i>GSTO2</i>
ENSECAG00000007119	1	47004378	47008447	
ENSECAG00000014378	1	77932913	78063676	
ENSECAG00000000579	1	105238087	105572574	<i>ADAMTS17</i>
ENSECAG00000017780	1	139619109	139705216	<i>TRPM7</i>
ENSECAG00000000916	1	148061665	148068584	<i>BAHD1</i>
ENSECAG00000007693	1	156280102	156281019	
ENSECAG00000003013	5	3508131	3517432	<i>POGK</i>
ENSECAG00000009640	5	3523778	3534148	<i>TADA1</i>
ENSECAG00000009274	5	17091236	17111030	<i>NPHS2</i>
ENSECAG00000017037	5	17122146	17199566	<i>TDRD5</i>
ENSECAG00000003048	5	38642851	38643774	<i>OR6K6</i>
ENSECAG00000003379	5	38662244	38663158	
ENSECAG00000003416	5	38670629	38671601	
ENSECAG00000003607	5	38691627	38692592	<i>OR6K2</i>
ENSECAG00000008749	5	51931355	51952069	<i>VTGNI</i>
ENSECAG00000012474	5	51998441	52037789	<i>TTF2</i>
ENSECAG00000016838	5	65559918	65696607	<i>CDC14A</i>
ENSECAG00000011008	5	78791187	78838921	<i>MCOLN2</i>
ENSECAG00000011427	6	20655754	20732147	<i>TRPM8</i>
ENSECAG00000014269	6	22070624	22434439	<i>AGAP1</i>
ENSECAG00000020862	6	58720925	58796131	<i>ABCD2</i>
ENSECAG00000011563	6	60897553	60919966	<i>GXYLT1</i>
ENSECAG00000005506	7	12803636	12815411	<i>MMP13</i>
ENSECAG00000018059	7	28787971	28792258	<i>SC5D</i>
ENSECAG00000000557	7	34951880	35037258	<i>RPUSD4</i>
ENSECAG00000005590	7	35052024	35079140	<i>FAM118B</i>
ENSECAG00000008874	7	48739516	48741791	
ENSECAG00000017070	7	50507618	50508612	
ENSECAG00000017115	7	50588844	50591649	
ENSECAG00000001169	7	50617975	50618913	
ENSECAG00000019206	7	50828293	50847391	
ENSECAG00000002629	7	51431459	51432361	
ENSECAG00000021649	7	51473311	51477731	
ENSECAG00000002775	7	51511683	51512600	
ENSECAG00000002839	7	51520913	51521839	
ENSECAG00000003018	7	51564093	51565028	<i>OR7D2</i>

ENSECAG00000003131	7	51572976	51573623
ENSECAG00000003155	7	51583911	51584831
ENSECAG00000022001	7	51603285	51607794 <i>ZNF317</i>
ENSECAG00000003527	7	51623722	51624639
ENSECAG00000003652	7	51656774	51657712
ENSECAG00000002436	7	51662491	51667297
ENSECAG00000003811	7	51804657	51805595
ENSECAG00000003885	7	51819498	51820400
ENSECAG00000004014	7	51841312	51842229
ENSECAG00000004090	7	51856802	51857719
ENSECAG00000004254	7	51914336	51915268
ENSECAG00000004323	7	51942403	51943305
ENSECAG00000004396	7	51970629	51971276
ENSECAG00000004454	7	51987761	51988336
ENSECAG00000004487	7	52115358	52116296
ENSECAG00000012861	8	18209492	18218248
ENSECAG00000009611	8	26661332	26688281 <i>SLC15A4</i>
ENSECAG00000018454	8	38891761	38975728 <i>RBBP8</i>
ENSECAG00000018963	8	40895845	40937320 <i>COLEC12</i>
ENSECAG00000013468	8	43362962	43487652 <i>MYOM1</i>
ENSECAG00000005248	10	7355786	7356910 <i>ZNF146</i>
ENSECAG00000024840	10	21052877	21061281
ENSECAG00000007783	10	21481345	21482329
ENSECAG00000009253	10	33162887	33347973
ENSECAG00000000590	11	23696398	23711958
ENSECAG00000016107	11	23889368	23904015 <i>TBKBPI</i>
ENSECAG00000018160	11	23932876	23934853
ENSECAG00000010544	11	24247881	24428988
ENSECAG00000024417	11	47220897	47225142 <i>TAX1BP3</i>
ENSECAG00000024635	11	47225475	47226274 <i>EMC6</i>
ENSECAG00000024637	11	47234738	47240801
ENSECAG00000003158	11	57608554	57644242 <i>ZNF624</i>
ENSECAG00000003836	11	57776310	57820215 <i>LRRC75A</i>
ENSECAG00000010792	13	5603955	5639522 <i>INTS1</i>
ENSECAG00000014383	13	18476085	18501261 <i>ITGAD</i>
ENSECAG00000010439	13	18678936	18688121 <i>PRSS36</i>
ENSECAG00000013572	13	18690553	18694200 <i>PRSS8</i>
ENSECAG00000018567	13	18695047	18705063 <i>KAT8</i>
ENSECAG00000010442	13	30842320	30850452
ENSECAG00000017922	14	21691499	21767122 <i>ADAM19</i>
ENSECAG00000021907	14	34923787	34945363 <i>NDFIP1</i>
ENSECAG00000004274	14	88274367	88497811 <i>IQGAP2</i>
ENSECAG00000015844	14	93067977	93103018 <i>ZNF496</i>

ENSECAG00000014976	15	28768171	28852206 <i>SLC4A5</i>
ENSECAG00000012644	15	50648132	50723439
ENSECAG00000008483	16	17474066	17553670 <i>GXYLT2</i>
ENSECAG00000004471	16	18569283	18591172 <i>EIF4E3</i>
ENSECAG00000024577	16	39539791	39559497 <i>KLHL18</i>
ENSECAG00000001282	16	40489968	40491047 <i>CCR3</i>
ENSECAG00000025146	16	41406568	41459966 <i>CDCP1</i>
ENSECAG00000011371	16	59022492	59096414 <i>UBE2E1</i>
ENSECAG00000004932	16	66168348	66205124 <i>CAPN7</i>
ENSECAG00000022460	16	76866332	76912293 <i>U2SURP</i>
ENSECAG00000012777	17	577668	618759 <i>MPHOSPH8</i>
ENSECAG00000017286	17	650118	715609 <i>PSPC1</i>
ENSECAG00000021688	17	19133042	19150721 <i>NEK3</i>
ENSECAG00000018885	17	26424829	26507135
ENSECAG00000020440	17	26517958	26541745 <i>CCDC122</i>
ENSECAG00000011453	18	12990397	13091785
ENSECAG00000021980	18	54590477	54591870 <i>HOXD8</i>
ENSECAG00000012544	18	56684352	56950669
ENSECAG00000022850	18	59278576	59341367 <i>CERKL</i>
ENSECAG00000012181	18	59405959	59410120 <i>NEUROD1</i>
ENSECAG00000018546	19	27207319	27428546 <i>TP63</i>
ENSECAG00000010255	19	43927895	43971754 <i>CCDC191</i>
ENSECAG00000009518	20	15858379	15879996
ENSECAG00000016121	21	30719594	31030477 <i>ADAMTS12</i>
ENSECAG00000001093	22	22504563	22505387
ENSECAG00000015750	22	22510931	22515509
ENSECAG00000019217	22	22594366	22634766 <i>HM13</i>
ENSECAG00000015942	23	17800873	17921080 <i>TRPM6</i>
ENSECAG00000006599	23	40152950	40153537 <i>IFN-OMEGA2</i>
ENSECAG00000014712	24	1608419	1664797 <i>DDHD1</i>
ENSECAG00000006500	26	24397034	24439348 <i>ADAMTS5</i>
ENSECAG00000010747	26	35449812	35495111 <i>SH3BGR</i>
ENSECAG00000011449	28	23154909	24182143
ENSECAG00000016646	28	25969409	26028564 <i>PARBPB</i>
ENSECAG00000012629	29	9908789	9940852 <i>MASTL</i>
ENSECAG00000000006	29	10454293	10532795 <i>GAD2</i>
ENSECAG00000019039	30	5815577	5925775 <i>CNST</i>
ENSECAG00000011256	30	24462153	24540348
ENSECAG00000014636	31	9493976	9561808 <i>WDR27</i>
ENSECAG00000007095	31	16096654	16182421 <i>PLEKHG1</i>

Table S15

Horse genomes used for the LSD_{norm} analysis, all with a minimum coverage of 5-fold.

ID	Deme	Date	Coverage (fold)
Akh_M_0248_Vi_AKT001	Intensive	Modern	22.77
Qrt_F_0070_MM_A1543	Intensive	Modern	12.55
Qrt_F_0074_MM_A5659	Intensive	Modern	13.67
Qrt_F_0075_MM_A5964	Intensive	Modern	13.97
Qrt_M_0073_MM_A2085	Intensive	Modern	12.32
Ara_M_0237_Mz_SAMN02439777	Intensive	Modern	12.32
Ara_F_0141_LO_Arabian	Intensive	Modern	10.82
Dlm_F_0238_Mz_SAMN02422919	Intensive	Modern	12.93
Mon_M_0052_LR_FM1030	Intensive	Modern	17.02
Mon_M_0065_LR_FM1798	Intensive	Modern	22.62
Mon_M_0068_LR_FM1951	Intensive	Modern	19.24
Han_M_0234_Mz_SAMN02439782	Intensive	Modern	10.09
Han_M_0235_Mz_SAMN02439779	Intensive	Modern	23.40
Mwi_M_0239_Bk_SRR1275408	Intensive	Modern	28.34
Mor_F_0096_MM_EMS595	Intensive	Modern	14.02
Std_F_0078_MM_M977	Intensive	Modern	12.35
Std_M_0081_MM_M5256	Intensive	Modern	11.44
Std_M_0090_MM_M1009	Intensive	Modern	7.68
Std_M_0149_LO_Standardbred	Intensive	Modern	12.31
Thr_F_0145_LO_Twilight	Intensive	Modern	41.14
Yak_M_0222_LO_CGG10397	MonYak	100 yBP	20.25
Mng_M_0153_LO_KB7754	MonYak	Modern	19.46
Mng_F_0216_Pk_TG1111D2629	MonYak	Modern	24.09
Mng_M_0215_Pk_TG1111D2628	MonYak	Modern	23.54
Yak_F_0164_LO_Yak2	MonYak	Modern	18.38
Yak_F_0169_LO_Yak7	MonYak	Modern	21.63
Ice_M_0247_Vi_IS074	Nordic	Modern	9.19
Ice_M_0143_LO_Icelandic	Nordic	Modern	8.51
Ice_M_0144_An_P5782	Nordic	Modern	32.67
Fjo_F_0142_LO_Fjord	Nordic	Modern	7.73
Sht_F_0249_Vi_SPH020	Nordic	Modern	21.58
Sht_F_0250_Vi_SPH041	Nordic	Modern	19.78
Yak_M_0223_LO_Batagai	Pre-domestic	5,200 $\pm$ 35 yBP	18.29
Yak_F_0224_LO_CGG10022	Pre-domestic	42,692 $\pm$ 891 yBP	25.60
Yak_M_0225_LO_CGG10023	Pre-domestic	16,099 $\pm$ 192 yBP	7.72
Prz_F_0150_LO_KB3879	Przewalski	Modern	22.88
Prz_M_0157_LO_SB293	Przewalski	Modern	19.30
Prz_M_0160_LO_SB533	Przewalski	Modern	20.36
BER01_A	Berel'	2,300 yBP	9.210

BER02_B	Berel'	2,300 yBP	10.704
BER04_D	Berel'	2,300 yBP	6.808
BER05_E	Berel'	2,300 yBP	8.540
BER06_F	Berel'	2,300 yBP	10.654
BER09_I	Berel'	2,300 yBP	10.957
BER12_M	Berel'	2,300 yBP	5.496

Table S16

The 174 genes overlapping the top-1000 LSD_{norm} windows, in the lineage prior to the Berel' split ~2300 years ago (Fig. 3), which overlaps early domestication stages.

Ensembl Gene ID	Chromosome	Gene Start (bp)	Gene End (bp)	Associated Gene Name
ENSECAG00000010116	1	26079852	26096939	<i>GSTO2</i>
ENSECAG00000013814	1	37175278	37236746	<i>HECTD2</i>
ENSECAG00000013338	1	75859281	75907593	<i>B3GALNT2</i>
ENSECAG00000009930	1	77783531	77901353	
ENSECAG00000014378	1	77932913	78063676	
ENSECAG00000008162	1	78554968	78660425	<i>SIPA1L2</i>
ENSECAG00000022561	1	106759249	106770015	<i>TM2D3</i>
ENSECAG00000004287	1	146583181	146622250	<i>VPS39</i>
ENSECAG00000017216	1	146795854	146867842	<i>EHD4</i>
ENSECAG00000013267	1	153939326	154278140	<i>RYR3</i>
ENSECAG00000018235	1	154279548	154404419	
ENSECAG00000006709	1	155483884	155484822	
ENSECAG00000016649	2	25359004	25362976	<i>NKAIN1</i>
ENSECAG00000025069	2	37417515	37422963	<i>CTRC</i>
ENSECAG00000020537	2	53159199	53222398	<i>ADAM7</i>
ENSECAG00000018924	2	55991005	56050782	<i>PTK2B</i>
ENSECAG00000003068	2	56161552	56190605	
ENSECAG00000013893	2	56473864	56582249	<i>SCARA5</i>
ENSECAG00000009107	2	115349258	115422839	<i>ENPEP</i>
ENSECAG00000022031	3	44794591	45895007	<i>GRID2</i>
ENSECAG00000019173	3	48991132	49261132	<i>FAM13A</i>
ENSECAG00000015983	3	112675614	112802326	<i>STX18</i>
ENSECAG00000014825	3	116735515	116746079	<i>LRPAP1</i>
ENSECAG00000024955	3	117887707	117931757	<i>ZFYVE28</i>
ENSECAG00000020468	4	4129555	4518186	<i>RELN</i>
ENSECAG00000010375	4	5512782	5645301	<i>SRPK2</i>
ENSECAG00000000344	4	38674493	38697716	<i>PON1</i>
ENSECAG00000020648	5	22943039	22950657	
ENSECAG00000020916	5	37076491	37077460	
ENSECAG00000019807	5	37352960	37361248	<i>CASQ1</i>
ENSECAG00000021715	5	41249384	41254498	
ENSECAG00000020111	5	41693697	41732576	<i>IQGAP3</i>
ENSECAG00000020654	5	47915670	47935445	<i>LIX1L</i>
ENSECAG00000022581	5	48419700	48444979	<i>ACP6</i>
ENSECAG00000021854	5	53134872	53195934	<i>CASQ2</i>
ENSECAG00000003066	5	57666620	57668119	<i>KCNA2</i>
ENSECAG00000021537	5	59240709	59270584	<i>FNDCC7</i>
ENSECAG00000021205	5	63514498	63709742	<i>COL11A1</i>

ENSECAG00000010571	5	66063233	66131884
ENSECAG00000001074	5	66248402	66294287 <i>PALMD</i>
ENSECAG00000015661	5	72725415	72765268 <i>BRDT</i>
ENSECAG00000024997	5	78365481	78486705 <i>DDAH1</i>
ENSECAG00000014027	5	89631942	90446366 <i>NEGR1</i>
ENSECAG00000022014	5	95728977	95800866 <i>RAVER2</i>
ENSECAG00000019096	6	28057667	28285464 <i>CACNA1C</i>
ENSECAG00000000564	6	31520884	31559085
ENSECAG00000019111	6	34883509	34902028 <i>CD163</i>
ENSECAG00000003085	6	34990864	34991241
ENSECAG00000014867	6	42011009	42060794 <i>PLBD1</i>
ENSECAG00000018813	6	42105527	42180124 <i>GUCY2C</i>
ENSECAG00000020056	6	68281548	68427716 <i>SLC11A2</i>
ENSECAG00000013371	6	68390765	68404279 <i>HIGD1C</i>
ENSECAG00000024023	6	70604367	70618797 <i>CALCOCO1</i>
ENSECAG00000008246	6	81826749	82051443 <i>GRIP1</i>
ENSECAG00000009829	7	25405517	25501065
ENSECAG00000016371	7	45130884	45287133 <i>CACNA1A</i>
ENSECAG00000015051	7	45328509	45331513 <i>NACCI</i>
ENSECAG00000016149	7	45346016	45351774 <i>TRMT1</i>
ENSECAG00000019690	7	45363359	45422044 <i>NFIX</i>
ENSECAG00000012495	7	45509412	45515116 <i>GCDH</i>
ENSECAG00000018970	7	45524767	45526789 <i>DNASE2</i>
ENSECAG00000022473	7	45528500	45548290 <i>MAST1</i>
ENSECAG00000014343	7	45577330	45600350 <i>HOOK2</i>
ENSECAG00000007103	7	45610152	45615940 <i>ASNA1</i>
ENSECAG00000012154	7	45617437	45620601 <i>C19orf43</i>
ENSECAG00000022546	7	45660594	45674937 <i>MAN2B1</i>
ENSECAG00000014042	7	45732240	45734894
ENSECAG00000016771	7	45973358	45995362
ENSECAG00000000785	7	46603984	46606643
ENSECAG00000008702	7	46630271	46661483 <i>TBPL2</i>
ENSECAG00000010608	7	47096680	47149578
ENSECAG00000017424	7	47648542	47650560
ENSECAG00000018210	7	47907075	47911114
ENSECAG00000004383	7	47911299	47911895
ENSECAG00000020668	7	48319300	48366144
ENSECAG00000008222	7	48605491	48608571
ENSECAG00000020060	7	48941441	48943612 <i>TMEM205</i>
ENSECAG00000020419	7	48946651	48952423 <i>RAB3D</i>
ENSECAG00000022479	7	48964014	48972958 <i>TSPAN16</i>
ENSECAG00000022584	7	48996078	49036503 <i>DOCK6</i>
ENSECAG00000017469	7	49066334	49071794 <i>SPC24</i>

ENSECAG00000008334	7	49126926	49196223 <i>SMARCA4</i>
ENSECAG00000004638	7	49616090	49617052
ENSECAG00000020260	7	49624651	49640724 <i>TYK2</i>
ENSECAG00000024723	7	49644897	49650142 <i>ICAM3</i>
ENSECAG00000013883	7	49685895	49692789 <i>ICAM1</i>
ENSECAG00000025081	7	49840263	49844697
ENSECAG00000002539	7	51153076	51153996
ENSECAG00000002721	7	51190124	51190762
ENSECAG00000003264	7	74313703	74315388 <i>UBQLN3</i>
ENSECAG00000005874	7	77446582	77447496
ENSECAG00000006238	7	77490604	77491548
ENSECAG00000021124	7	88171886	88304987 <i>PRMT3</i>
ENSECAG00000010920	7	93233667	93279841 <i>SLC5A12</i>
ENSECAG00000019113	7	96116000	96117889
ENSECAG00000008712	8	16219679	16433412 <i>FBXW8</i>
ENSECAG00000017514	8	19931373	19935238
ENSECAG00000011774	8	35165671	35197430 <i>APCDD1</i>
ENSECAG00000018523	8	36548925	36572379 <i>RNMT</i>
ENSECAG00000007436	8	36593500	36603164 <i>FAM210A</i>
ENSECAG00000008041	8	36619934	36621874 <i>LDLRAD4</i>
ENSECAG00000017880	8	37262217	37284526 <i>SEH1L</i>
ENSECAG00000018454	8	38891761	38975728 <i>RBBP8</i>
ENSECAG00000013468	8	43362962	43487652 <i>MYOM1</i>
ENSECAG00000021760	8	70765480	71472757 <i>DCC</i>
ENSECAG00000000728	8	78517759	78626204 <i>PIGN</i>
ENSECAG00000020472	8	79629648	79666241 <i>KDSR</i>
ENSECAG00000005031	9	28473950	28474324
ENSECAG00000000235	9	63327236	63408454 <i>MTBP</i>
ENSECAG00000018382	9	78106853	78373475 <i>COL22A1</i>
ENSECAG00000023361	9	82006749	82013814 <i>RHPN1</i>
ENSECAG00000013121	10	6639494	6652250 <i>ATP4A</i>
ENSECAG00000008821	10	24105412	24113519 <i>FCAR</i>
ENSECAG00000009253	10	33162887	33347973
ENSECAG00000023776	10	34214578	34250082 <i>TTK</i>
ENSECAG00000009840	10	34312823	34533153 <i>BCKDHB</i>
ENSECAG00000008253	10	41689058	41729551 <i>GABRR2</i>
ENSECAG00000002203	10	51695277	51696101
ENSECAG00000016887	10	57961902	57993673 <i>OSTM1</i>
ENSECAG00000022222	10	59709982	59729250 <i>WASF1</i>
ENSECAG00000022780	11	7525457	7557309 <i>TTYH2</i>
ENSECAG00000014219	11	23478525	23543480 <i>SRCIN1</i>
ENSECAG00000013061	11	26888603	27537309 <i>CA10</i>
ENSECAG00000015470	11	37593800	37600820 <i>SLFN14</i>

ENSECAG00000005740	11	46912016	46912963
ENSECAG00000008405	11	50650758	50755571 <i>DNAH2</i>
ENSECAG00000020649	11	60516244	60550896 <i>COPS3</i>
ENSECAG00000007226	12	15216596	15217510
ENSECAG00000007333	12	15286039	15286686
ENSECAG00000007531	12	15454415	15455314 <i>OR4C16</i>
ENSECAG00000007832	12	15610206	15611063
ENSECAG00000013095	12	16261152	16289644
ENSECAG00000004140	12	16324193	16324567
ENSECAG00000024915	12	21861788	21872504 <i>FADS3</i>
ENSECAG00000004558	12	31345741	31384898 <i>CARS</i>
ENSECAG00000016004	13	7376813	7413151 <i>CYP3A97</i>
ENSECAG00000014383	13	18476085	18501261 <i>ITGAD</i>
ENSECAG00000020849	14	24748992	25035501 <i>GRIA1</i>
ENSECAG00000005337	14	33232464	33233150
ENSECAG00000022073	14	38639763	38740388 <i>KLHL3</i>
ENSECAG00000017148	14	43689392	43816195 <i>FNIP1</i>
ENSECAG00000004274	14	88274367	88497811 <i>IQGAP2</i>
ENSECAG00000015844	14	93067977	93103018 <i>ZNF496</i>
ENSECAG00000014657	15	89904427	90059240 <i>MYTIL</i>
ENSECAG00000018642	16	25094888	25176715 <i>PRICKLE2</i>
ENSECAG00000019144	16	67434752	67979351 <i>CPNE4</i>
ENSECAG00000024376	17	24024635	24214094 <i>LRCH1</i>
ENSECAG00000000691	18	2995767	3091726 <i>WDR33</i>
ENSECAG00000014391	19	24631534	24648943
ENSECAG00000015662	19	55037013	55066469 <i>CMSSI</i>
ENSECAG00000016973	19	55096972	55373784 <i>FILIP1L</i>
ENSECAG00000024989	20	19761828	19762210
ENSECAG00000009741	20	34964834	35039840 <i>UHRF1BPI</i>
ENSECAG00000020043	20	49908274	49909879 <i>IL17F</i>
ENSECAG00000023723	21	16920043	17035896 <i>SKIV2L2</i>
ENSECAG00000019217	22	22594366	22634766 <i>HM13</i>
ENSECAG00000015942	23	17800873	17921080 <i>TRPM6</i>
ENSECAG00000020120	23	44764250	44802755 <i>PLAA</i>
ENSECAG00000009492	25	6675249	6813088 <i>INVS</i>
ENSECAG00000019348	26	41361824	41366098 <i>YBEY</i>
ENSECAG00000000438	27	3070919	3081019 <i>VDAC3</i>
ENSECAG00000021346	27	3995445	4008535
ENSECAG00000015006	27	7326045	7364762 <i>FGFR1</i>
ENSECAG00000011627	27	25729556	25833246 <i>ENPP6</i>
ENSECAG00000000152	28	14702021	14741239 <i>KITLG</i>
ENSECAG00000000374	28	18286144	18313988 <i>PLEKHG7</i>
ENSECAG00000005111	28	29570191	29729790 <i>RFX4</i>

ENSECAG00000018973	28	36532728	36621416 <i>ENTHD1</i>
ENSECAG00000007986	29	16607198	16673011
ENSECAG00000020866	30	5988308	6024559 <i>SCCPDH</i>
ENSECAG00000010669	30	24999325	25195424 <i>CRB1</i>
ENSECAG00000018766	31	16476682	16493033 <i>IYD</i>

Table S17

WebGestalt analysis for protein-coding genes overlapping the X:40,375,001-53,375,001 region. Only functional categories with an adjusted p -value < 0.1 are shown. The yellow gradient scales with the adjusted p -value.

Analysis	Model organism	Term	Adjusted p -value
Wiki Pathways	human	AGE-RAGE pathway	0.0574
Wiki Pathways	human	mRNA processing	0.068
KEGG	human	Cytokine-cytokine receptor interaction	0.0678
Disease	human	Neurobehavioral Manifestations	0.0002
Disease	human	FG Syndrome	0.0061
Disease	human	Anovulation	0.0061
Disease	human	Abnormalities, Drug-Induced	0.0113
Disease	human	Malocclusion	0.0135
Disease	human	Robin sequence	0.018
Gene Ontology	human	long-chain-alcohol O-fatty-acyltransferase activity	0.0012
Gene Ontology	human	ionotropic glutamate receptor binding	0.0382
Gene Ontology	human	glutamate receptor binding	0.0638
Gene Ontology	human	transferase activity, transferring acyl groups other than amino-acyl groups	0.0638
Gene Ontology	mouse	long-chain-alcohol O-fatty-acyltransferase activity	0.0011
Gene Ontology	mouse	transferase activity, transferring acyl groups other than amino-acyl groups	0.0042
Gene Ontology	mouse	transferase activity, transferring acyl groups	0.0117
Gene Ontology	mouse	ionotropic glutamate receptor binding	0.055
Gene Ontology	mouse	histone methyltransferase complex	0.0296
Gene Ontology	mouse	methyltransferase complex	0.0296
Gene Ontology	mouse	MLL1 complex	0.0611
Gene Ontology	mouse	MLL1/2 complex	0.0611
Phenotype	human	X-linked dominant inheritance	0.0015
Phenotype	human	Long fingers	0.0016
Phenotype	human	Intellectual disability	0.0032
Phenotype	human	Low frustration tolerance	0.0032
Phenotype	human	Pectus excavatum	0.0032
Phenotype	human	Aplasia/Hypoplasia of the nipples	0.0154
Phenotype	human	Arachnodactyly	0.0154
Phenotype	mouse	increased thigmotaxis	0.0381
Phenotype	mouse	abnormal cranium size	0.0571
Phenotype	mouse	abnormal excitatory postsynaptic potential	0.0804
Phenotype	mouse	abnormal spatial learning	0.0953

Table S18

WebGestalt analysis for genes overlapping the top-1000 PBS regions in the Berel' population. Only functional categories with an adjusted p -value < 0.1 are shown. The yellow gradient scales with the adjusted p -value.

Statistic	Population	Analysis	Model organism	Term	Adjusted p -value
PBS	Berel'	Wiki Pathways	human	Striated Muscle Contraction	0.079
PBS	Berel'	Wiki Pathways	mouse	Striated Muscle Contraction	0.0772
PBS	Berel'	Wiki Pathways	mouse	Endochondral Ossification	0.0954
PBS	Berel'	KEGG	human	Olfactory transduction	0.0084
PBS	Berel'	KEGG	human	Other types of O-glycan biosynthesis	0.0192
PBS	Berel'	KEGG	mouse	Other types of O-glycan biosynthesis	0.0372
PBS	Berel'	KEGG	mouse	Olfactory transduction	0.057
PBS	Berel'	Disease	human	Autolysis	0.0357
PBS	Berel'	Gene Ontology	human	blastoderm segmentation	0.0073
PBS	Berel'	Gene Ontology	human	tripartite regional subdivision	0.0073
PBS	Berel'	Gene Ontology	human	anterior/posterior axis specification, embryo	0.0073
PBS	Berel'	Gene Ontology	human	embryonic axis specification	0.0721
PBS	Berel'	Gene Ontology	human	regulation of DNA recombination	0.0866
PBS	Berel'	Gene Ontology	human	regulation of isotype switching to IgG isotypes	0.0866
PBS	Berel'	Gene Ontology	human	anterior/posterior axis specification	0.0866
PBS	Berel'	Gene Ontology	human	segmentation	0.0866
PBS	Berel'	Gene Ontology	human	isotype switching to IgG isotypes	0.0898
PBS	Berel'	Gene Ontology	human	negative regulation of DNA recombination	0.0981
PBS	Berel'	Gene Ontology	human	endopeptidase activity	0.0073
PBS	Berel'	Gene Ontology	human	xylosyltransferase activity	0.0099
PBS	Berel'	Gene Ontology	human	metalloendopeptidase activity	0.0099
PBS	Berel'	Gene Ontology	human	UDP-xylosyltransferase activity	0.0099
PBS	Berel'	Gene Ontology	human	peptidase activity, acting on L-amino acid peptides	0.0264
PBS	Berel'	Gene Ontology	human	peptidase activity	0.0286
PBS	Berel'	Gene Ontology	human	olfactory receptor activity	0.0302
PBS	Berel'	Gene Ontology	human	metallopeptidase activity	0.0347
PBS	Berel'	Gene Ontology	mouse	O-glycan processing	0.0564
PBS	Berel'	Gene Ontology	mouse	xylosyltransferase activity	0.0186
PBS	Berel'	Gene Ontology	mouse	UDP-xylosyltransferase activity	0.0186
PBS	Berel'	Gene Ontology	mouse	endopeptidase activity	0.0319
PBS	Berel'	Gene Ontology	mouse	metallopeptidase activity	0.0319
PBS	Berel'	Gene Ontology	mouse	metalloendopeptidase activity	0.0319
PBS	Berel'	Gene Ontology	mouse	peptidase activity	0.0319
PBS	Berel'	Gene Ontology	mouse	peptidase activity, acting on L-amino acid peptides	0.0319

PBS	Berel'	Phenotype	human	Abnormality of the posterior pituitary	0.0444
PBS	Berel'	Phenotype	human	Abnormality of secondary sexual hair	0.0444
PBS	Berel'	Phenotype	human	Sparse axillary hair	0.0444
PBS	Berel'	Phenotype	human	Abnormality of the axillary hair	0.0444
PBS	Berel'	Phenotype	human	Aplasia/Hypoplasia of the breasts	0.0493
PBS	Berel'	Phenotype	human	Breast hypoplasia	0.0493
PBS	Berel'	Phenotype	human	Breast aplasia	0.0493
PBS	Berel'	Phenotype	human	Absent hand	0.0493
PBS	Berel'	Phenotype	human	Hypoplastic nipples	0.0493
PBS	Berel'	Phenotype	mouse	absent carpal bone	0.0304
PBS	Berel'	Phenotype	mouse	absent mammary gland	0.0348
PBS	Berel'	Phenotype	mouse	abnormal mammary gland pattern	0.0348
PBS	Berel'	Phenotype	mouse	abnormal clitoris morphology	0.0696
				abnormal external female genitalia morphology	
PBS	Berel'	Phenotype	mouse		0.0889
PBS	Berel'	Phenotype	mouse	abnormal T-helper 1 cell number	0.0889
PBS	Berel'	Phenotype	mouse	abnormal ulna morphology	0.0889
PBS	Berel'	Phenotype	mouse	increased airway responsiveness	0.0889
PBS	Berel'	Phenotype	mouse	increased interleukin-17 secretion	0.0889
				abnormal pancreatic acinar cell zymogen granule morphology	
PBS	Berel'	Phenotype	mouse		0.0905
PBS	Berel'	PheWas	human	Spinal stenosis	0.032
PBS	Berel'	PheWas	human	Symptoms involving digestive system	0.064
PBS	Berel'	PheWas	human	Reflux esophagitis	0.091
				Disturbances of sulphur-bearing amino-acid metabolism	
PBS	Berel'	PheWas	human		0.091
PBS	Berel'	PheWas	human	Decreased libido	0.091
PBS	Berel'	PheWas	human	Congenital anomalies of genital organs	0.091
PBS	Berel'	PheWas	human	Dermatophytosis of nail	0.091
PBS	Berel'	PheWas	human	Vascular disorders of penis	0.091
PBS	Berel'	PheWas	human	Other disorders of eye	0.091
PBS	Berel'	PheWas	human	Disturbances of amino-acid transport	0.091

Table S19

WebGestalt analysis for genes overlapping the top-1000 LSD_{norm} regions. Pre-Berel' refers to the ancestral population existing prior to the divergence between Berel' horses and present-day domesticates (Fig 3). Only functional categories with an adjusted p -value < 0.1 are shown. The yellow gradient scales with the adjusted p -value.

Statistic	Population	Analysis	Model organism	Term	Adjusted p -value
LSDnorm	Pre-Berel'	Wiki Pathway	human	Calcium Regulation in the Cardiac Cell	0.003
LSDnorm	Pre-Berel'	Wiki Pathway	human	IL12 SIGNALING PATHWAY	0.01
LSDnorm	Pre-Berel'	Wiki Pathway	mouse	Calcium Regulation in the Cardiac Cell	0.0021
LSDnorm	Pre-Berel'	Wiki Pathway	mouse	PluriNetWork	0.0203
LSDnorm	Pre-Berel'	Gene Ontology	human	hormone receptor binding	0.0145
LSDnorm	Pre-Berel'	Gene Ontology	human	steroid hormone receptor binding	0.0185
LSDnorm	Pre-Berel'	Gene Ontology	human	calcium ion transmembrane transporter activity	0.0185
LSDnorm	Pre-Berel'	Gene Ontology	human	nuclear hormone receptor binding	0.0185
LSDnorm	Pre-Berel'	Gene Ontology	human	glycerophosphodiester phosphodiesterase activity	0.0266
LSDnorm	Pre-Berel'	Gene Ontology	human	divalent inorganic cation transmembrane transporter activity	0.0296
LSDnorm	Pre-Berel'	Gene Ontology	human	passive transmembrane transporter activity	0.0296
LSDnorm	Pre-Berel'	Gene Ontology	human	androgen receptor binding	0.0296
LSDnorm	Pre-Berel'	Gene Ontology	human	channel activity	0.0296
LSDnorm	Pre-Berel'	Gene Ontology	human	pattern binding	0.0444
LSDnorm	Pre-Berel'	Gene Ontology	human	HOPS complex	0.0041
LSDnorm	Pre-Berel'	Gene Ontology	human	FHF complex	0.0252
LSDnorm	Pre-Berel'	Gene Ontology	human	vacuolar membrane	0.0936
LSDnorm	Pre-Berel'	Gene Ontology	human	synapse part	0.0936
LSDnorm	Pre-Berel'	Gene Ontology	human	sarcoplasmic reticulum	0.0936
LSDnorm	Pre-Berel'	Gene Ontology	human	sarcoplasm	0.0936
LSDnorm	Pre-Berel'	Gene Ontology	human	microtubule	0.0936
LSDnorm	Pre-Berel'	Gene Ontology	mouse	ion transmembrane transporter activity	0.0372
LSDnorm	Pre-Berel'	Gene Ontology	mouse	channel activity	0.0372
LSDnorm	Pre-Berel'	Gene Ontology	mouse	substrate-specific channel activity	0.0372
LSDnorm	Pre-Berel'	Gene Ontology	mouse	glycerophosphodiester phosphodiesterase activity	0.0372
LSDnorm	Pre-Berel'	Gene Ontology	mouse	ion channel activity	0.0372
LSDnorm	Pre-Berel'	Gene Ontology	mouse	divalent inorganic cation transmembrane transporter activity	0.0372
LSDnorm	Pre-Berel'	Gene Ontology	mouse	passive transmembrane transporter activity	0.0372
LSDnorm	Pre-Berel'	Gene Ontology	mouse	substrate-specific transmembrane transporter activity	0.0372
LSDnorm	Pre-Berel'	Gene Ontology	mouse	transmembrane transporter activity	0.0447
LSDnorm	Pre-Berel'	Gene Ontology	mouse	calcium ion transmembrane transporter activity	0.0447

LSDnorm Pre-Berel'	Gene Ontology	mouse	asymmetric synapse	0.0532
LSDnorm Pre-Berel'	Gene Ontology	mouse	FHF complex	0.0532
LSDnorm Pre-Berel'	Gene Ontology	mouse	dendritic shaft	0.0532
LSDnorm Pre-Berel'	Gene Ontology	mouse	synaptic membrane	0.0532
LSDnorm Pre-Berel'	Gene Ontology	mouse	microtubule	0.0532
LSDnorm Pre-Berel'	Gene Ontology	mouse	sarcoplasmic reticulum	0.0761
LSDnorm Pre-Berel'	Gene Ontology	mouse	postsynaptic membrane	0.0761
LSDnorm Pre-Berel'	Gene Ontology	mouse	sarcoplasm	0.0802
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal associative learning	0.0077
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal ear shape	0.0259
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal neural crest cell morphology	0.0311
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal synaptic transmission	0.0311
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal CNS synaptic transmission	0.0311
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal neural crest morphology	0.0432
LSDnorm Pre-Berel'	Phenotype	mouse	increased erythrocyte protoporphyrin level	0.0848
LSDnorm Pre-Berel'	Phenotype	mouse	lowered ear position	0.0848
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal action potential	0.0848
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal brain wave pattern	0.0848
LSDnorm Pre-Berel'	PheWas	human	Complications of gastrostomy, colostomy and enterostomy	0.0036
LSDnorm Pre-Berel'	PheWas	human	Cancer of oropharynx	0.0173
LSDnorm Pre-Berel'	PheWas	human	Salicylates causing adverse effects in therapeutic use	0.0173
LSDnorm Pre-Berel'	PheWas	human	Disorders of function of stomach	0.0308
LSDnorm Pre-Berel'	PheWas	human	Unspecified polyarthropathy or polyarthritis	0.0308
LSDnorm Pre-Berel'	PheWas	human	Sepsis and SIRS	0.0308
LSDnorm Pre-Berel'	PheWas	human	Poisoning by analgesics, antipyretics, and antirheumatics	0.0338
LSDnorm Pre-Berel'	PheWas	human	Ill-defined descriptions and complications of heart disease	0.0638
LSDnorm Pre-Berel'	PheWas	human	Adverse effects of antirheumatics	0.0914
LSDnorm Pre-Berel'	PheWas	human	Other specified erythematous conditions	0.0914

Table S20

Developmental stages and anatomical body parts where genes overlapping the X:40,375,001-53,375,001 region are transcribed, as revealed by Bgee. Only transcriptional categories with a *p*-value < 0.1 are shown. None of them remained significant after correcting for multiple testing.

Statistic	Model organism	Term	<i>P</i>-value
FST	Mouse	inner ear associated mesenchyme (mouse)	0.00557
FST	Mouse	mandible primordium (mouse)	0.01148
FST	Mouse	cervical ganglion	0.02151
FST	Mouse	vestibular organ	0.03048
FST	Mouse	spleen primordium	0.03079
FST	Mouse	Purkinje cell layer of cerebellar cortex	0.03465
FST	Mouse	pedal digit 5 phalanx	0.05062
FST	Mouse	thyroid gland	0.06396
FST	Mouse	diencephalon lateral wall marginal layer (mouse)	0.06984
FST	Mouse	anterior olfactory nucleus	0.06984
FST	Mouse	endometrium	0.07656
FST	Mouse	umbilical artery	0.07656
FST	Mouse	pedal digit 4 phalanx	0.08013
FST	Mouse	mesenchyme of submandibular gland primordium	0.08323
FST	Mouse	pedal digit 2 phalanx	0.08655
FST	Mouse	pedal digit 3 phalanx	0.08655
FST	Mouse	intrinsic muscle of tongue	0.08985
FST	Mouse	umbilical vein	0.08985
FST	Mouse	meninx	0.09441
FST	Mouse	definitive endoderm	0.09642
FST	Mouse	femur cartilage element	0.09969
FST	Human	muscle organ	0.0684
FST	Human	cardiac ventricle	0.0765

Table S21

Developmental stages and anatomical body parts where genes overlapping the top-1000 PBS windows are transcribed, as revealed by Bgee. Only transcriptional categories with a p -value < 0.1 are shown. None of them remained significant after correcting for multiple testing.

Statistic	Lineage	Model organism	Term	<i>P</i> -value
PBS	Berel'	Mouse	visceral epithelium of maturing renal corpuscle (mouse)	0.00138
PBS	Berel'	Mouse	lung	0.00321
PBS	Berel'	Mouse	parietal epithelium (mouse)	0.00517
PBS	Berel'	Mouse	brain	0.00582
PBS	Berel'	Mouse	sacral vertebra cartilage element	0.00741
PBS	Berel'	Mouse	glomerular tuft	0.00781
PBS	Berel'	Mouse	viscus	0.00818
PBS	Berel'	Mouse	spermatogonium	0.01071
PBS	Berel'	Mouse	visceral epithelium of capillary loop renal corpuscle (mouse)	0.011
PBS	Berel'	Mouse	pituitary gland	0.01224
PBS	Berel'	Mouse	ear epithelium	0.01258
PBS	Berel'	Mouse	nasal cavity respiratory epithelium	0.01803
PBS	Berel'	Mouse	vestibular organ	0.02081
PBS	Berel'	Mouse	pectoral appendage apical ectodermal ridge	0.0217
PBS	Berel'	Mouse	spermatocyte	0.02268
PBS	Berel'	Mouse	Leydig cell region of testis	0.03036
PBS	Berel'	Mouse	primitive bladder (mouse)	0.0306
PBS	Berel'	Mouse	liver	0.03159
PBS	Berel'	Mouse	clavicle	0.03334
PBS	Berel'	Mouse	post-embryonic organism	0.03345
PBS	Berel'	Mouse	nasal epithelium (mouse)	0.0346
PBS	Berel'	Mouse	adrenal cortex	0.0346
PBS	Berel'	Mouse	tooth of upper jaw	0.03578
PBS	Berel'	Mouse	basioccipital endochondral element	0.03687
PBS	Berel'	Mouse	renal system	0.043
PBS	Berel'	Mouse	glomerular visceral epithelium	0.04319
PBS	Berel'	Mouse	tibia	0.04341
PBS	Berel'	Mouse	ulna	0.04447
PBS	Berel'	Mouse	future brain	0.04513
PBS	Berel'	Mouse	tooth of lower jaw	0.04597
PBS	Berel'	Mouse	upper jaw incisor	0.04983
PBS	Berel'	Mouse	skeletal muscle tissue of tongue	0.0532
PBS	Berel'	Mouse	mesonephric mesenchyme of female mesonephros (mouse)	0.06391
PBS	Berel'	Mouse	mesonephric mesenchyme of male (mouse)	0.06663
PBS	Berel'	Mouse	pes connective tissue	0.06848
PBS	Berel'	Mouse	hindlimb stylopod muscle	0.06987

PBS	Berel'	Mouse	kidney vasculature	0.07001
PBS	Berel'	Mouse	lower jaw incisor	0.07101
PBS	Berel'	Mouse	diencephalon lateral wall	0.07341
PBS	Berel'	Mouse	Naris	0.07345
PBS	Berel'	Mouse	Stomach	0.07565
PBS	Berel'	Mouse	radius bone	0.08236
PBS	Berel'	Mouse	mesentery of urinary system	0.08275
PBS	Berel'	Mouse	egg cell	0.08407
PBS	Berel'	Mouse	simple squamous epithelium	0.08733
PBS	Berel'	Mouse	hindbrain	0.09167
PBS	Berel'	Mouse	urothelium	0.09219
PBS	Berel'	Mouse	mesonephric tubule (mouse)	0.09381
PBS	Berel'	Human	nasal cavity mucosa	0.00729
PBS	Berel'	Human	metanephros	0.0146
PBS	Berel'	Human	Sperm	0.02427
PBS	Berel'	Human	frontal cortex	0.02797
PBS	Berel'	Human	uterine cervix	0.0396
PBS	Berel'	Human	cortex of kidney	0.04379
PBS	Berel'	Human	germinal epithelium of ovary	0.04471
PBS	Berel'	Human	Pancreas	0.05
PBS	Berel'	Human	bone element	0.05241
PBS	Berel'	Human	peritoneum	0.06636
PBS	Berel'	Human	prostate gland	0.07292
PBS	Berel'	Human	Jejunum	0.08198
PBS	Berel'	Human	urinary bladder	0.08569
PBS	Berel'	Human	tibialis anterior	0.08615
PBS	Berel'	Human	Blood	0.09141
PBS	Berel'	Human	epithelium of bronchus	0.09404
PBS	Berel'	Human	epithelial cell of pancreas	0.09639

Table S22

Developmental stages and anatomical body parts where genes overlapping top-1000 LSD_{norm} windows are transcribed, as revealed by Bgee. Pre-Berel' refers to the ancestral population existing prior to the divergence between Berel' horses and present-day domesticates (Fig 3). Only transcriptional categories with a *p*-value < 0.1 are shown. None of them remained significant after correcting for multiple testing.

Statistic	Lineage	Model organism	Term	P-value
LSDnorm	Pre-Berel'	Mouse	larynx	0.00334
LSDnorm	Pre-Berel'	Mouse	suprachiasmatic nucleus	0.01247
LSDnorm	Pre-Berel'	Mouse	choroid plexus epithelium	0.01528
LSDnorm	Pre-Berel'	Mouse	tail dermomyotome group (mouse)	0.01614
LSDnorm	Pre-Berel'	Mouse	condyle	0.01687
LSDnorm	Pre-Berel'	Mouse	medulla oblongata	0.01758
LSDnorm	Pre-Berel'	Mouse	inferior colliculus	0.01847
LSDnorm	Pre-Berel'	Mouse	cortical layer II	0.01895
LSDnorm	Pre-Berel'	Mouse	lateral geniculate body	0.02119
LSDnorm	Pre-Berel'	Mouse	somatic nerve plexus	0.02163
LSDnorm	Pre-Berel'	Mouse	CA1 field of hippocampus	0.02329
LSDnorm	Pre-Berel'	Mouse	superior colliculus	0.0245
LSDnorm	Pre-Berel'	Mouse	associated mesenchyme of trachea	0.0282
LSDnorm	Pre-Berel'	Mouse	mesenchyme derived from head neural crest	0.0297
LSDnorm	Pre-Berel'	Mouse	inner cell mass	0.0298
LSDnorm	Pre-Berel'	Mouse	lateral hypothalamic area	0.03032
LSDnorm	Pre-Berel'	Mouse	subparaventricular zone	0.03116
LSDnorm	Pre-Berel'	Mouse	ventral tegmental area	0.03478
LSDnorm	Pre-Berel'	Mouse	physiological umbilical hernia (mouse)	0.03524
LSDnorm	Pre-Berel'	Mouse	trophectoderm	0.03538
LSDnorm	Pre-Berel'	Mouse	pretectal region	0.03738
LSDnorm	Pre-Berel'	Mouse	ganglionic eminence	0.0396
LSDnorm	Pre-Berel'	Mouse	medial geniculate body	0.03993
LSDnorm	Pre-Berel'	Mouse	facial nucleus	0.04081
LSDnorm	Pre-Berel'	Mouse	medial preoptic region	0.04229
LSDnorm	Pre-Berel'	Mouse	tongue	0.04276
LSDnorm	Pre-Berel'	Mouse	cerebellum lobe	0.04341
LSDnorm	Pre-Berel'	Mouse	pontine nuclear group	0.04357
LSDnorm	Pre-Berel'	Mouse	secondary oocyte	0.04534
LSDnorm	Pre-Berel'	Mouse	mature ovarian follicle	0.04868
LSDnorm	Pre-Berel'	Mouse	primary motor cortex	0.05156
LSDnorm	Pre-Berel'	Mouse	substantia nigra	0.05173
LSDnorm	Pre-Berel'	Mouse	medial dorsal nucleus of thalamus	0.05182
LSDnorm	Pre-Berel'	Mouse	ventral horn of spinal cord	0.05566
LSDnorm	Pre-Berel'	Mouse	cortical layer IV	0.05798

LSDnorm Pre-Berel' Mouse	epithelium of lens	0.05935
LSDnorm Pre-Berel' Mouse	mammillary body	0.05951
LSDnorm Pre-Berel' Mouse	undifferentiated genital tubercle	0.05956
LSDnorm Pre-Berel' Mouse	cortical layer VI	0.06156
LSDnorm Pre-Berel' Mouse	spleen primordium	0.06181
LSDnorm Pre-Berel' Mouse	chordate pharynx	0.06194
LSDnorm Pre-Berel' Mouse	dorsomedial nucleus of hypothalamus	0.06283
LSDnorm Pre-Berel' Mouse	spinal cord mantle layer	0.06336
LSDnorm Pre-Berel' Mouse	ulna	0.06521
LSDnorm Pre-Berel' Mouse	retrosplenial region	0.06564
LSDnorm Pre-Berel' Mouse	dentate gyrus of hippocampal formation	0.06569
LSDnorm Pre-Berel' Mouse	vibrissa follicle	0.06827
LSDnorm Pre-Berel' Mouse	ovarian follicle	0.06955
LSDnorm Pre-Berel' Mouse	dorsal striatum	0.06984
LSDnorm Pre-Berel' Mouse	tooth enamel organ	0.07161
LSDnorm Pre-Berel' Mouse	endolymphatic appendage	0.07657
LSDnorm Pre-Berel' Mouse	spinal cord ventricular layer (mouse)	0.07675
LSDnorm Pre-Berel' Mouse	granular layer of cerebellar cortex	0.07923
LSDnorm Pre-Berel' Mouse	arcuate nucleus of hypothalamus	0.08073
LSDnorm Pre-Berel' Mouse	ventromedial nucleus of hypothalamus	0.08322
LSDnorm Pre-Berel' Mouse	radius endochondral element	0.08978
LSDnorm Pre-Berel' Mouse	thoracic ganglion	0.0902
LSDnorm Pre-Berel' Mouse	diencephalon mantle layer	0.09027
LSDnorm Pre-Berel' Mouse	lateral septal nucleus	0.091
LSDnorm Pre-Berel' Mouse	nucleus accumbens	0.0913
LSDnorm Pre-Berel' Mouse	cerebellar nuclear complex	0.09365
LSDnorm Pre-Berel' Mouse	facial bone primordium	0.09453
LSDnorm Pre-Berel' Mouse	anterior amygdaloid area	0.09859
LSDnorm Pre-Berel' Mouse	olfactory bulb mitral cell layer	0.09885
LSDnorm Pre-Berel' Human	anterior cingulate cortex	0.00658
LSDnorm Pre-Berel' Human	sperm	0.03275
LSDnorm Pre-Berel' Human	endothelial cell	0.04192
LSDnorm Pre-Berel' Human	left ventricle myocardium	0.06135
LSDnorm Pre-Berel' Human	middle temporal gyrus	0.06373
LSDnorm Pre-Berel' Human	cochlea	0.08516
LSDnorm Pre-Berel' Human	nucleus accumbens	0.09849

Table S23

WebGestalt analysis for genes overlapping the top-1000 PBS regions. Segmental duplications (CNVs) are excluded from the candidate and background set. Only categories with an adjusted p -value < 0.1 are shown. The yellow gradient scales with the adjusted p -value.

Statistic	Population	Analysis	Model organism	Term	Adjusted p -value
PBS	Berel'	Wiki Pathways	human	Endochondral Ossification	0.0912
PBS	Berel'	Wiki Pathways	mouse	Endochondral Ossification	0.0662
PBS	Berel'	KEGG	human	Other types of O-glycan biosynthesis	0.0124
PBS	Berel'	KEGG	human	Olfactory transduction	0.0124
PBS	Berel'	KEGG	mouse	Other types of O-glycan biosynthesis	0.0252
PBS	Berel'	KEGG	mouse	Olfactory transduction	0.0294
PBS	Berel'	Gene Ontology	human	anterior/posterior axis specification, embryo	0.0029
PBS	Berel'	Gene Ontology	human	blastoderm segmentation	0.0029
PBS	Berel'	Gene Ontology	human	tripartite regional subdivision	0.0029
PBS	Berel'	Gene Ontology	human	embryonic axis specification	0.0263
PBS	Berel'	Gene Ontology	human	regulation of DNA recombination	0.0438
PBS	Berel'	Gene Ontology	human	anterior/posterior axis specification	0.0438
PBS	Berel'	Gene Ontology	human	isotype switching to IgG isotypes	0.0526
PBS	Berel'	Gene Ontology	human	regulation of isotype switching to IgG isotypes	0.0526
PBS	Berel'	Gene Ontology	human	negative regulation of DNA recombination	0.057
PBS	Berel'	Gene Ontology	human	negative regulation of nitrogen compound metabolic process	0.057
PBS	Berel'	Gene Ontology	human	UDP-xylosyltransferase activity	0.011
PBS	Berel'	Gene Ontology	human	xylosyltransferase activity	0.011
PBS	Berel'	Gene Ontology	mouse	O-glycan processing	0.0348
PBS	Berel'	Gene Ontology	mouse	regulation of DNA recombination	0.0623
PBS	Berel'	Gene Ontology	mouse	anterior/posterior axis specification, embryo	0.0623
PBS	Berel'	Gene Ontology	mouse	tripartite regional subdivision	0.0623
PBS	Berel'	Gene Ontology	mouse	blastoderm segmentation	0.0623
PBS	Berel'	Gene Ontology	mouse	regulation of isotype switching to IgG isotypes	0.0692
PBS	Berel'	Gene Ontology	mouse	isotype switching to IgG isotypes	0.0741
PBS	Berel'	Gene Ontology	mouse	negative regulation of DNA recombination	0.0908
PBS	Berel'	Gene Ontology	mouse	UDP-xylosyltransferase activity	0.0105
PBS	Berel'	Gene Ontology	mouse	xylosyltransferase activity	0.0105
PBS	Berel'	Phenotype	human	Absent hand	0.0279
PBS	Berel'	Phenotype	human	Sparse axillary hair	0.0279
PBS	Berel'	Phenotype	human	Abnormality of the posterior pituitary	0.0279
PBS	Berel'	Phenotype	human	Aplasia/Hypoplasia of the breasts	0.0279

PBS	Berel'	Phenotype	human	Hypoplastic nipples	0.0279
PBS	Berel'	Phenotype	human	Breast hypoplasia	0.0279
PBS	Berel'	Phenotype	human	Abnormality of the axillary hair	0.0279
PBS	Berel'	Phenotype	human	Abnormality of secondary sexual hair	0.0279
PBS	Berel'	Phenotype	human	Breast aplasia	0.0279
PBS	Berel'	Phenotype	human	Aplasia/Hypoplasia of the nipples	0.0614
PBS	Berel'	Phenotype	mouse	absent carpal bone	0.0158
PBS	Berel'	Phenotype	mouse	abnormal mammary gland pattern	0.0203
PBS	Berel'	Phenotype	mouse	absent mammary gland	0.0203
PBS	Berel'	Phenotype	mouse	abnormal clitoris morphology	0.0381
PBS	Berel'	Phenotype	mouse	increased airway responsiveness	0.0441
PBS	Berel'	Phenotype	mouse	abnormal external female genitalia morphology	0.0441
PBS	Berel'	Phenotype	mouse	abnormal ulna morphology	0.0441
PBS	Berel'	Phenotype	mouse	increased interleukin-17 secretion	0.0441
PBS	Berel'	Phenotype	mouse	abnormal T-helper 1 cell number	0.0441
PBS	Berel'	Phenotype	mouse	abnormal pancreatic acinar cell zymogen granule morphology	0.0457
PBS	Berel'	PheWas	human	Spinal stenosis	0.0009
PBS	Berel'	PheWas	human	Reflux esophagitis	0.0425
PBS	Berel'	PheWas	human	Congenital anomalies of genital organs	0.0425
PBS	Berel'	PheWas	human	Symptoms involving digestive system	0.0425
PBS	Berel'	PheWas	human	Vascular disorders of penis	0.0425
PBS	Berel'	PheWas	human	Decreased libido	0.0425
PBS	Berel'	PheWas	human	Spinal stenosis of lumbar region	0.0484
PBS	Berel'	PheWas	human	Disease of capillaries	0.0633
PBS	Berel'	PheWas	human	Cancer of other lymphoid, histiocytic tissue	0.0633
PBS	Berel'	PheWas	human	Sexual and gender identity disorders	0.0717

Table S24

WebGestalt analysis for genes overlapping the top-1000 LSD_{norm} regions. Pre-Berel' refers to the ancestral population existing prior to the divergence between Berel' horses and present-day domesticates (Fig 3). Segmental duplications (CNVs) are excluded from the candidate and background set. Only categories with an adjusted *p*-value < 0.1 are shown. The yellow gradient scales with the adjusted *p*-value.

Statistic	Lineage	Analysis	Model organism	Term	Adjusted <i>p</i> -value
LSDnorm	Pre-Berel'	Wiki Pathways	human	Calcium Regulation in the Cardiac Cell	0.003
LSDnorm	Pre-Berel'	Wiki Pathways	human	IL12 SIGNALING PATHWAY	0.0095
LSDnorm	Pre-Berel'	Wiki Pathways	mouse	Calcium Regulation in the Cardiac Cell	0.0014
LSDnorm	Pre-Berel'	Wiki Pathways	mouse	PluriNetWork	0.0175
LSDnorm	Pre-Berel'	Disease	human	Dysostoses	0.0952
LSDnorm	Pre-Berel'	Disease	human	Tooth, Supernumerary Central Nervous System	0.0952
LSDnorm	Pre-Berel'	Disease	human	Infections	0.0952
LSDnorm	Pre-Berel'	Disease	human	Sarcoma, Endometrial Stromal	0.0952
LSDnorm	Pre-Berel'	Gene Ontology	human	hormone receptor binding	0.0123
LSDnorm	Pre-Berel'	Gene Ontology	human	steroid hormone receptor binding	0.0242
LSDnorm	Pre-Berel'	Gene Ontology	human	nuclear hormone receptor binding	0.0242
LSDnorm	Pre-Berel'	Gene Ontology	human	channel activity	0.0311
LSDnorm	Pre-Berel'	Gene Ontology	human	glycerophosphodiester phosphodiesterase activity	0.0311
LSDnorm	Pre-Berel'	Gene Ontology	human	passive transmembrane transporter activity	0.0311
LSDnorm	Pre-Berel'	Gene Ontology	human	androgen receptor binding	0.0311
LSDnorm	Pre-Berel'	Gene Ontology	human	calcium ion transmembrane transporter activity	0.0488
LSDnorm	Pre-Berel'	Gene Ontology	human	polysaccharide binding	0.0488
LSDnorm	Pre-Berel'	Gene Ontology	human	pattern binding	0.0488
LSDnorm	Pre-Berel'	Gene Ontology	human	HOPS complex	0.0036
LSDnorm	Pre-Berel'	Gene Ontology	human	FHF complex	0.0238
LSDnorm	Pre-Berel'	Gene Ontology	human	synapse part	0.0952
LSDnorm	Pre-Berel'	Gene Ontology	human	sarcoplasm	0.0952
LSDnorm	Pre-Berel'	Gene Ontology	human	microtubule	0.0952
LSDnorm	Pre-Berel'	Gene Ontology	human	sarcoplasmic reticulum	0.0952
LSDnorm	Pre-Berel'	Gene Ontology	mouse	glycerophosphodiester phosphodiesterase activity	0.0462
LSDnorm	Pre-Berel'	Gene Ontology	mouse	passive transmembrane transporter activity	0.0462
LSDnorm	Pre-Berel'	Gene Ontology	mouse	ion transmembrane transporter activity	0.0462
LSDnorm	Pre-Berel'	Gene Ontology	mouse	substrate-specific channel activity	0.0462
LSDnorm	Pre-Berel'	Gene Ontology	mouse	channel activity	0.0462
LSDnorm	Pre-Berel'	Gene Ontology	mouse	ion channel activity	0.0462

LSDnorm Pre-Berel'	Gene Ontology	mouse	calcium ion transmembrane transporter activity	0.048
LSDnorm Pre-Berel'	Gene Ontology	mouse	hormone receptor binding	0.048
LSDnorm Pre-Berel'	Gene Ontology	mouse	ATPase activity	0.048
LSDnorm Pre-Berel'	Gene Ontology	mouse	pattern binding	0.048
LSDnorm Pre-Berel'	Gene Ontology	mouse	synaptic membrane	0.0433
LSDnorm Pre-Berel'	Gene Ontology	mouse	microtubule	0.0433
LSDnorm Pre-Berel'	Gene Ontology	mouse	dendritic shaft	0.0433
LSDnorm Pre-Berel'	Gene Ontology	mouse	FHF complex	0.0433
LSDnorm Pre-Berel'	Gene Ontology	mouse	asymmetric synapse	0.0433
LSDnorm Pre-Berel'	Gene Ontology	mouse	postsynaptic membrane	0.0635
LSDnorm Pre-Berel'	Gene Ontology	mouse	sarcoplasmic reticulum	0.0635
LSDnorm Pre-Berel'	Gene Ontology	mouse	sarcoplasm	0.0684
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal associative learning	0.0047
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal synaptic transmission	0.0163
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal ear shape	0.0163
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal CNS synaptic transmission	0.0244
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal neural crest cell morphology	0.0293
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal neural crest morphology	0.0325
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal brain wave pattern	0.0665
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal action potential	0.0665
LSDnorm Pre-Berel'	Phenotype	mouse	patent ductus arteriosus	0.0665
LSDnorm Pre-Berel'	Phenotype	mouse	abnormal learning/memory/conditioning	0.0665
LSDnorm Pre-Berel'	PheWas	human	Complications of gastrostomy, colostomy and enterostomy	0.0036
LSDnorm Pre-Berel'	PheWas	human	Cancer of oropharynx	0.0173
LSDnorm Pre-Berel'	PheWas	human	Salicylates causing adverse effects in therapeutic use	0.0173
LSDnorm Pre-Berel'	PheWas	human	Disorders of function of stomach	0.0308
LSDnorm Pre-Berel'	PheWas	human	Unspecified polyarthropathy or polyarthritis	0.0308
LSDnorm Pre-Berel'	PheWas	human	Sepsis and SIRS	0.0308
LSDnorm Pre-Berel'	PheWas	human	Poisoning by analgesics, antipyretics, and antirheumatics	0.0338
LSDnorm Pre-Berel'	PheWas	human	Ill-defined descriptions and complications of heart disease	0.0638
LSDnorm Pre-Berel'	PheWas	human	Adverse effects of antirheumatics	0.0914
LSDnorm Pre-Berel'	PheWas	human	Other specified erythematous conditions	0.0914

Table S25

Developmental stages and anatomical body parts where genes overlapping top-1000 PBS windows are transcribed, as revealed by Bgee. Segmental duplications (CNVs) are excluded from the candidate and background set. Only transcriptional categories with a p -value < 0.1 are shown. None of them remained significant after correcting for multiple testing.

Statistic	Lineage	Model organism	Term	<i>P</i> -value
PBS	Berel'	Mouse	visceral epithelium of maturing renal corpuscle (mouse)	0.000623
PBS	Berel'	Mouse	parietal epithelium (mouse)	0.003425
PBS	Berel'	Mouse	glomerular tuft	0.00435
PBS	Berel'	Mouse	sacral vertebra cartilage element	0.004919
PBS	Berel'	Mouse	brain	0.005821
PBS	Berel'	Mouse	visceral epithelium of capillary loop renal corpuscle (mouse)	0.006168
PBS	Berel'	Mouse	spermatogonium	0.007135
PBS	Berel'	Mouse	lung	0.008698
PBS	Berel'	Mouse	nasal cavity respiratory epithelium	0.00882
PBS	Berel'	Mouse	ear epithelium	0.01072
PBS	Berel'	Mouse	pectoral appendage apical ectodermal ridge	0.014588
PBS	Berel'	Mouse	spermatocyte	0.015257
PBS	Berel'	Mouse	upper jaw incisor	0.018238
PBS	Berel'	Mouse	pituitary gland	0.020817
PBS	Berel'	Mouse	nasal epithelium (mouse)	0.023445
PBS	Berel'	Mouse	adrenal cortex	0.023445
PBS	Berel'	Mouse	lower jaw incisor	0.027158
PBS	Berel'	Mouse	glomerular visceral epithelium	0.029403
PBS	Berel'	Mouse	basioccipital endochondral element	0.030255
PBS	Berel'	Mouse	ulna	0.030298
PBS	Berel'	Mouse	future brain	0.038793
PBS	Berel'	Mouse	pes connective tissue	0.040979
PBS	Berel'	Mouse	thoracic ganglion	0.051479
PBS	Berel'	Mouse	radius bone	0.057065
PBS	Berel'	Mouse	diencephalon lateral wall	0.059915
PBS	Berel'	Mouse	liver	0.060794
PBS	Berel'	Mouse	pharyngotympanic tube	0.064033
PBS	Berel'	Mouse	mesonephric tubule (mouse)	0.065287
PBS	Berel'	Mouse	cervicothoracic ganglion	0.069494
PBS	Berel'	Mouse	primitive bladder (mouse)	0.070756
PBS	Berel'	Mouse	inner cell mass	0.071389
PBS	Berel'	Mouse	simple squamous epithelium	0.079452
PBS	Berel'	Mouse	fundus of urinary bladder	0.081519
PBS	Berel'	Mouse	upper jaw molar	0.084267

PBS	Berel'	Mouse	primary ovarian follicle	0.084278
PBS	Berel'	Mouse	dorsal horn of spinal cord	0.085427
PBS	Berel'	Mouse	paramesonephric duct of female, mesonephric portion (mouse)	0.088063
PBS	Berel'	Mouse	thoracic vertebral cartilage condensation (mouse)	0.088303
PBS	Berel'	Mouse	pineal primordium (mouse)	0.088303
PBS	Berel'	Mouse	endometrium	0.088303
PBS	Berel'	Mouse	ectoderm of footplate	0.088303
PBS	Berel'	Mouse	posterior mesonephric tubule	0.088303
PBS	Berel'	Mouse	hindlimb skin	0.092311
PBS	Berel'	Mouse	anterior mesonephric tubule	0.092311
PBS	Berel'	Mouse	pelvic appendage apical ectodermal ridge	0.092311
PBS	Berel'	Mouse	hindlimb bud ectoderm (mouse)	0.096301
PBS	Berel'	Mouse	presumptive neural retina	0.096301
PBS	Berel'	Mouse	pallidum	0.097184
PBS	Berel'	Human	frontal cortex	0.0146
PBS	Berel'	Human	sperm	0.024
PBS	Berel'	Human	nasal cavity mucosa	0.0429
PBS	Berel'	Human	peritoneum	0.0444
PBS	Berel'	Human	bone element	0.0452
PBS	Berel'	Human	uterine cervix	0.0683
PBS	Berel'	Human	germinal epithelium of ovary	0.0701
PBS	Berel'	Human	epithelium of bronchus	0.0718
PBS	Berel'	Human	nasopharynx	0.073
PBS	Berel'	Human	sympathetic trunk	0.0967
PBS	Berel'	Human	metanephros	0.0988

Table S26

Developmental stages and anatomical body parts where genes overlapping top-1000 LSD_{norm} windows are transcribed, as revealed by Bgee. Pre-Berel' refers to the ancestral population existing prior to the divergence between Berel' horses and present-day domesticates (Fig. 3). Segmental duplications (CNVs) are excluded from the candidate and background set. Only transcriptional categories with a *p*-value < 0.1 are shown. None of them remained significant after correcting for multiple testing.

Statistic	Lineage	Model organism	Term	<i>P</i> -value
LSDnorm	Pre-Berel'	Mouse	larynx	0.00787
LSDnorm	Pre-Berel'	Mouse	condyle	0.00944
LSDnorm	Pre-Berel'	Mouse	suprachiasmatic nucleus	0.01178
LSDnorm	Pre-Berel'	Mouse	tail dermomyotome group (mouse)	0.01521
LSDnorm	Pre-Berel'	Mouse	cortical layer II	0.01746
LSDnorm	Pre-Berel'	Mouse	inferior colliculus	0.01931
LSDnorm	Pre-Berel'	Mouse	somatic nerve plexus	0.02112
LSDnorm	Pre-Berel'	Mouse	CA1 field of hippocampus	0.02384
LSDnorm	Pre-Berel'	Mouse	lateral geniculate body	0.02422
LSDnorm	Pre-Berel'	Mouse	superior colliculus	0.02466
LSDnorm	Pre-Berel'	Mouse	choroid plexus epithelium	0.02543
LSDnorm	Pre-Berel'	Mouse	associated mesenchyme of trachea	0.02663
LSDnorm	Pre-Berel'	Mouse	medulla oblongata	0.02683
LSDnorm	Pre-Berel'	Mouse	inner cell mass	0.02752
LSDnorm	Pre-Berel'	Mouse	mesenchyme derived from head neural crest	0.02802
LSDnorm	Pre-Berel'	Mouse	medial geniculate body	0.02855
LSDnorm	Pre-Berel'	Mouse	pontine nuclear group	0.03027
LSDnorm	Pre-Berel'	Mouse	subparaventricular zone	0.0327
LSDnorm	Pre-Berel'	Mouse	trophectoderm	0.03271
LSDnorm	Pre-Berel'	Mouse	lateral hypothalamic area	0.03279
LSDnorm	Pre-Berel'	Mouse	physiological umbilical hernia (mouse)	0.03329
LSDnorm	Pre-Berel'	Mouse	ventral tegmental area	0.0364
LSDnorm	Pre-Berel'	Mouse	medial dorsal nucleus of thalamus	0.03807
LSDnorm	Pre-Berel'	Mouse	facial nucleus	0.0395
LSDnorm	Pre-Berel'	Mouse	medial preoptic region	0.04467
LSDnorm	Pre-Berel'	Mouse	cerebellum lobe	0.04646
LSDnorm	Pre-Berel'	Mouse	retrosplenial region	0.04875
LSDnorm	Pre-Berel'	Mouse	substantia nigra	0.05071
LSDnorm	Pre-Berel'	Mouse	ganglionic eminence	0.05472
LSDnorm	Pre-Berel'	Mouse	cortical layer IV	0.05488
LSDnorm	Pre-Berel'	Mouse	secondary oocyte	0.05511
LSDnorm	Pre-Berel'	Mouse	primary motor cortex	0.05563
LSDnorm	Pre-Berel'	Mouse	epithelium of lens	0.0564
LSDnorm	Pre-Berel'	Mouse	mature ovarian follicle	0.05679

LSDnorm	Pre-Berel'	Mouse	spinal cord mantle layer	0.05769
LSDnorm	Pre-Berel'	Mouse	cortical layer VI	0.05828
LSDnorm	Pre-Berel'	Mouse	chordate pharynx	0.05841
LSDnorm	Pre-Berel'	Mouse	ulna	0.06175
LSDnorm	Pre-Berel'	Mouse	mammillary body	0.06393
LSDnorm	Pre-Berel'	Mouse	dorsomedial nucleus of hypothalamus	0.06571
LSDnorm	Pre-Berel'	Mouse	ovarian follicle	0.06786
LSDnorm	Pre-Berel'	Mouse	lateral septal nucleus	0.0687
LSDnorm	Pre-Berel'	Mouse	pretectal region	0.06877
LSDnorm	Pre-Berel'	Mouse	spinal cord ventricular layer (mouse)	0.06913
LSDnorm	Pre-Berel'	Mouse	tooth enamel organ	0.06947
LSDnorm	Pre-Berel'	Mouse	cerebellar nuclear complex	0.07014
LSDnorm	Pre-Berel'	Mouse	dentate gyrus of hippocampal formation	0.07209
LSDnorm	Pre-Berel'	Mouse	endolymphatic appendage	0.07256
LSDnorm	Pre-Berel'	Mouse	granular layer of cerebellar cortex	0.07506
LSDnorm	Pre-Berel'	Mouse	ventral horn of spinal cord	0.07641
LSDnorm	Pre-Berel'	Mouse	diencephalon mantle layer	0.07946
LSDnorm	Pre-Berel'	Mouse	thoracic ganglion	0.08039
LSDnorm	Pre-Berel'	Mouse	subiculum	0.0826
LSDnorm	Pre-Berel'	Mouse	olfactory bulb	0.0859
LSDnorm	Pre-Berel'	Mouse	arcuate nucleus of hypothalamus	0.08592
LSDnorm	Pre-Berel'	Mouse	radius endochondral element	0.08712
LSDnorm	Pre-Berel'	Mouse	ventromedial nucleus of hypothalamus	0.09261
LSDnorm	Pre-Berel'	Mouse	olfactory bulb mitral cell layer	0.09381
LSDnorm	Pre-Berel'	Mouse	undifferentiated genital tubercle	0.09452
LSDnorm	Pre-Berel'	Mouse	diencephalon mantle layer (mouse)	0.09685
LSDnorm	Pre-Berel'	Mouse	cortical layer III	0.0978
LSDnorm	Pre-Berel'	Human	anterior cingulate cortex	0.011
LSDnorm	Pre-Berel'	Human	sperm	0.0467
LSDnorm	Pre-Berel'	Human	endothelial cell	0.0682
LSDnorm	Pre-Berel'	Human	middle temporal gyrus	0.0843
LSDnorm	Pre-Berel'	Human	left ventricle myocardium	0.0934

Table S27. DNA extractions and trimming/rescaling statistics. DNA extractions and post-mortem damage. DNA extraction details, and adjustments applied to the sequenced reads. D, S, and Y extraction methods follow the nomenclature of Gamba and colleagues (2015) (25).

Table S28. Sequencing statistics, disaggregated by library. Details on sequencing runs per sample, and library. Sequencing runs are referred to as SE, for single end, PE for paired-end, followed by the number of sequencing cycles. nuDNA, nuclear DNA; mtDNA, mitochondrial DNA.

Table S29. Comparative genomes panel. ID, breed (or population), age and gender of the 54 horse samples comprising the comparative panel, including their Biosample accessions to SRA/ENA, and their respective nuclear coverage.

Table S30. Mendelian traits. Number of reads that pass all quality filters specified in the section Genotype probabilities vs. genotype calls, and support each nucleotide variant (A; C; G; T) for the 41 Mendelian traits. Cells highlighted in green were not genotyped; those in red have a depth-of-coverage lower than three-fold.

Table S31. ABBA-BABA test. Results for the ABBA-BABA test, in the form $\{[(H1,H2),H3], \text{outgroup}\}$, with the outgroup being *E. africanus somaliensis*.

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